TRANSFUSION MEDICINE



Erytra Eflexis® TAS training course

Erytra Eflexis® Configuration Guide

Diagnostic Commercial Technical Services (DxCTS)

September 2020

TYPING

GRIFOLS

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GRIFOLS

Importer configuration

GRIFOLS

Importer configuration

Introduction: Importer

- **What is the IMPORTER?**
  - Importer is the software that some IH instruments use **for loading the Reagents Pack** into the database.
  - Importer also loads the Reagents Pack information into **Diana System Connectivity (DSC).**
  - Importer allows to configure some instrument properties also accessible through the **erytra.properties.xml** file.

GRIFOLS

Importer configuration

Introduction: Reagents Pack

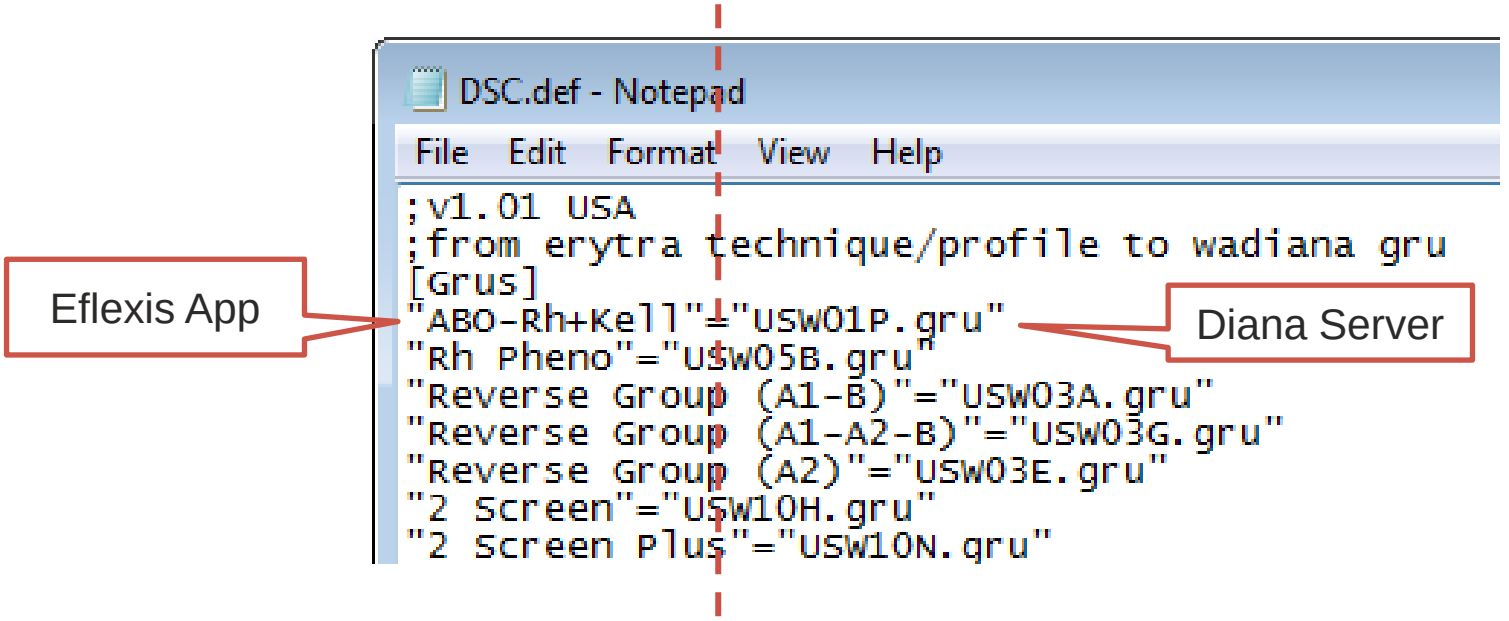
- **What is the Reagents Pack?**
  - The Reagent Pack is a **set of files that contains complete information for running different techniques and using related resources.**
  - Includes Defs files:
    - Tests definitions
    - Cards definitions (well, segments and meta-segments)
    - Reagents kits
    - QC Kits files.
  - Includes the final interpretation from the cards processed.
  - Includes the link between the Erytra Eflexis® profiles (techniques) and old Diana System connectivity based on *.gru and *.pln. These links are in the file **DSC.def.**
  - Include the templates (plantillas).

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Importer configuration

Introduction: DSC

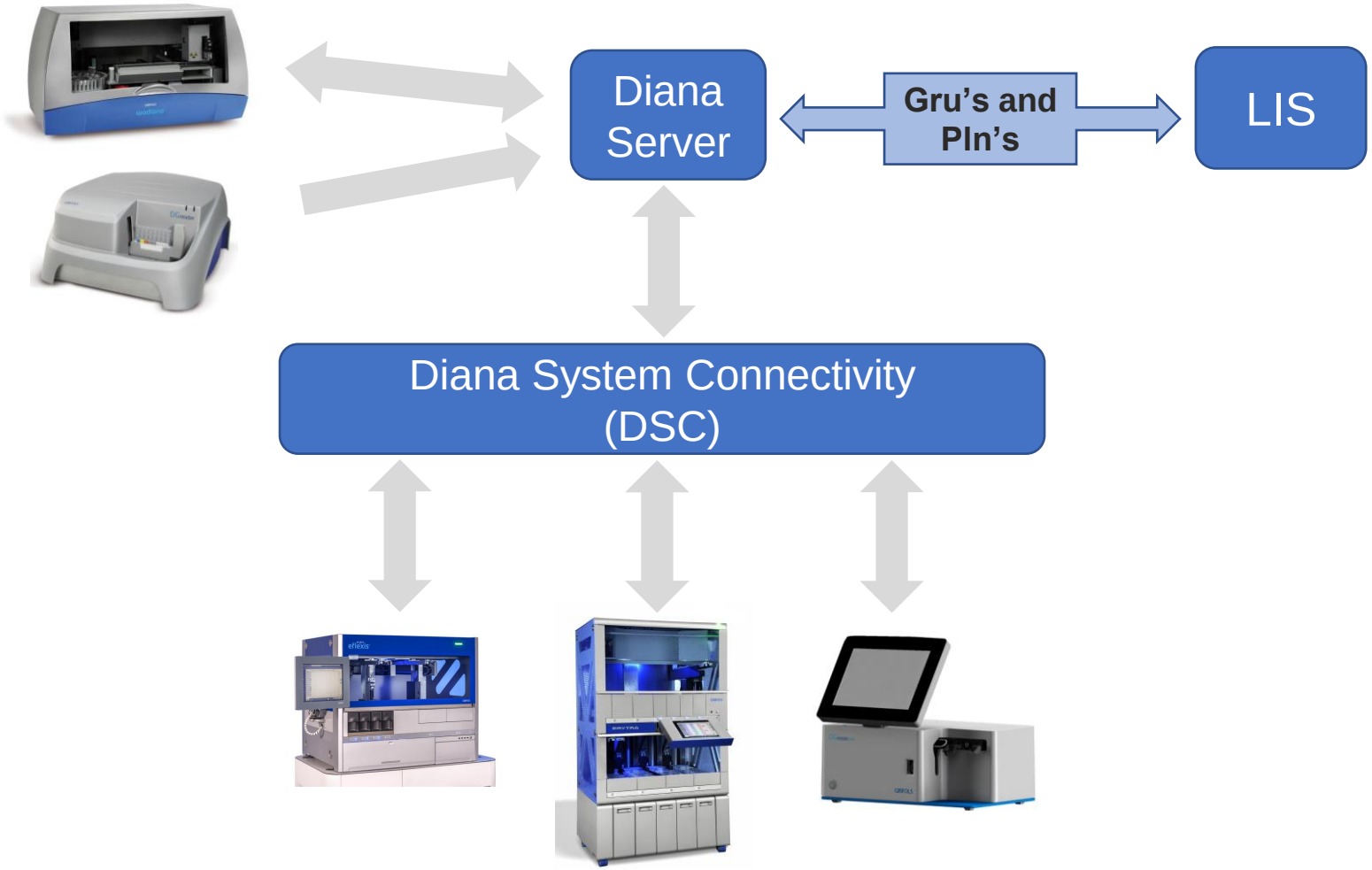
- **What is the DSC – Diana System Connectivity?**
  - **DSC** is the software section that translates/links between Erytra Eflexis® Software and **Diana Server.**
  - **Diana Server** receives the profile to execute from the LIS in format *.gru and send it to Erytra Eflexis® Software in profile format.



GRIFOLS

Importer configuration

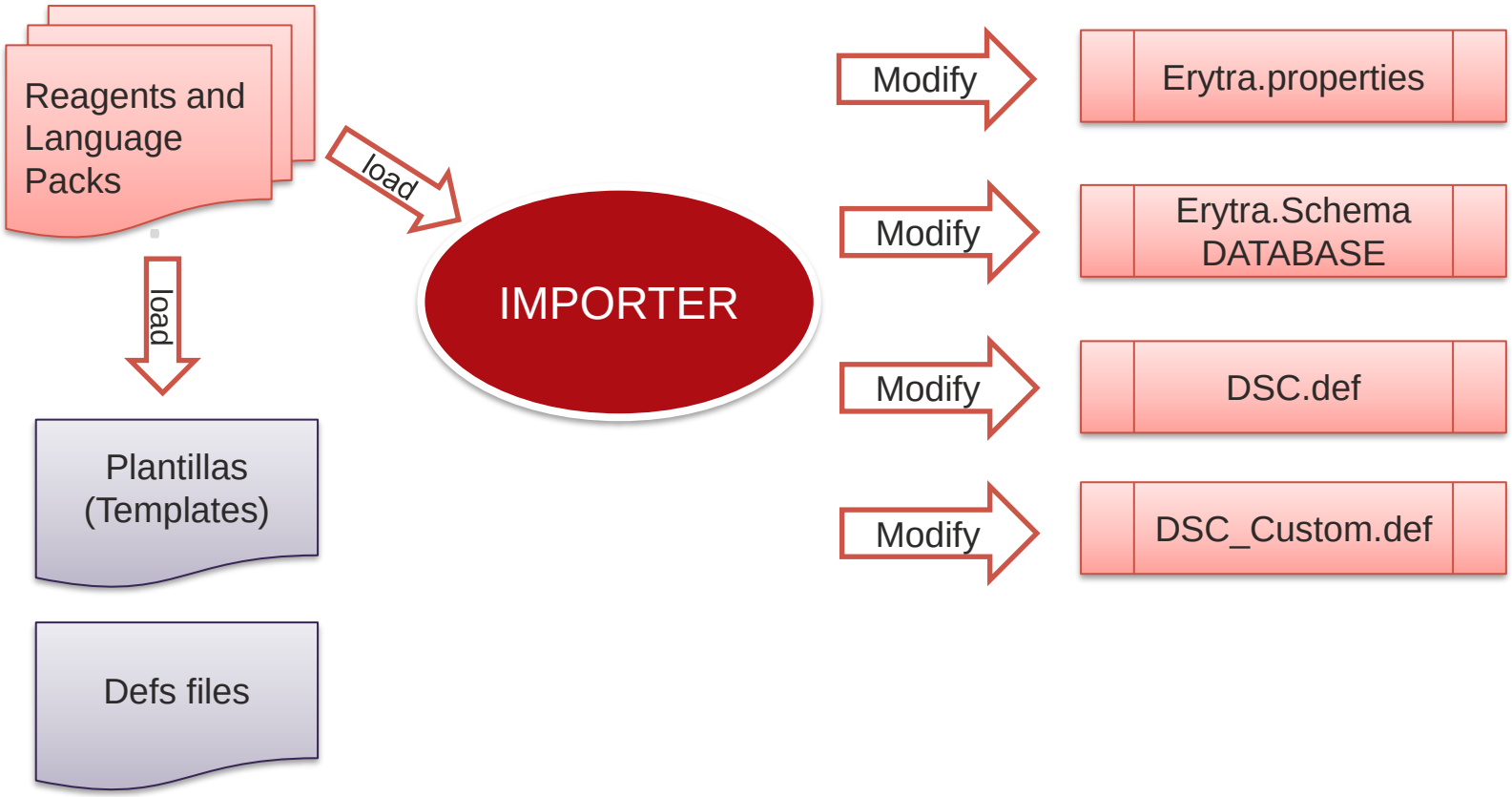
Introduction: DSC



GRIFOLS

Importer configuration

Introduction: Importer schema



GRIFOLS

Importer configuration

Reagents Pack

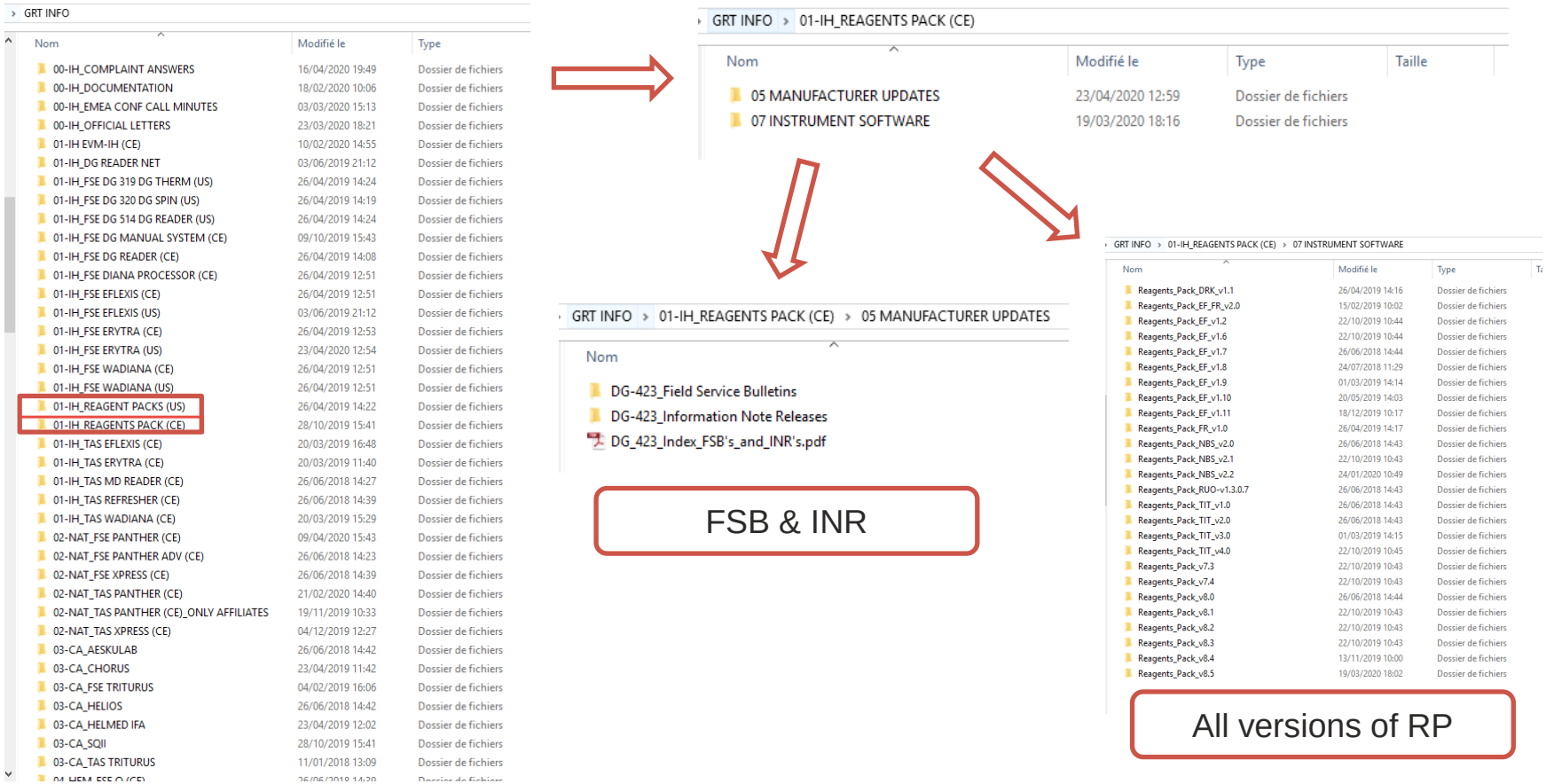
- The **Reagents Pack** (RP) files are separate from the software version and are released periodically with the newer or modified, validated techniques. However, to update a new RP, sometimes one specific software version might be mandatory.
- The manufacturer notifies the new version using the FSB or INR.
- There are unique RP for different regions due to differences in consumables used.
- There are RP by version.
- There are additional RP as Titration RP.
- You CANNOT downgrade to a previous version.
- ASK your MANAGER before you install a new RP.

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Importer configuration

Reagents Pack

- Reagents Packs can be downloaded from GRT FTP:

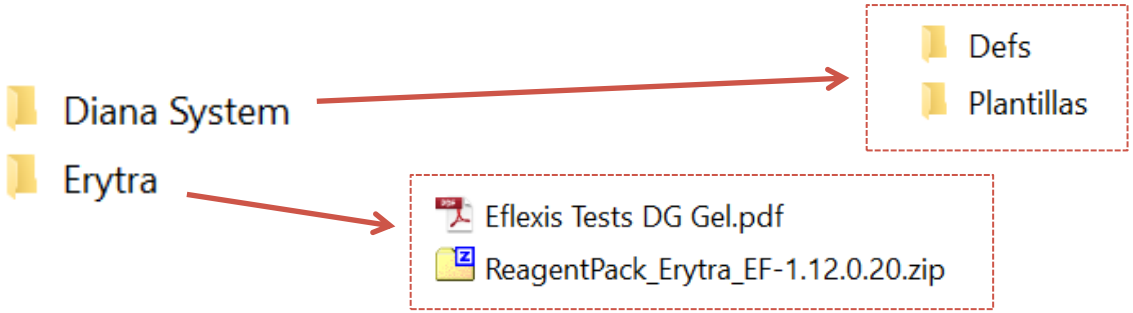


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Importer configuration

Reagents Pack

- Reagents Pack contains all information necessary for all IH products (Erytra®, Erytra Eflexis®, DG Reader, WADiana and DG Reader Net).
- Contains the following folders:



  - **Diana System** includes Definitions files (Defs) and Plantillas (templates)
  - **Erytra folder** includes the “Reagents Pack” in zip format and a pdf which explains in detail all the reagents pack content in terms of available tests and their composition.

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Importer configuration

Reagents Pack

- Test table

REAGENTS PACK EF v1.12

Name of the test

Table: Erytra Eflexis Tests for DG Gel cards

Version Date: 07/10/2020

Name of the cards and references

List of reagents and references

Reagent manufacturer	Description	Version	Template wells	Card	Reagent	Equivalent Gru
DIAGNOSTIC GRIFOLS	2 Screen (AHG)(50uL)	1.00	I.IAC,II.IAC	DG Gel Coombs (210342)	Serascan Diana 2 (210204)	DGW10Z.gru
DIAGNOSTIC GRIFOLS	2 Screen (AHG)	1.01	I.IAC,II.IAC	DG Gel Coombs (210342)	Serascan Diana 2 (210204)	DGW10A.gru
DIAGNOSTIC GRIFOLS	2 Screen (IgG)	1.01	I.IGG,II.IGG	DG Gel Anti-IgG (210344)	Serascan Diana 2 (210204)	DGW10H.gru
DIAGNOSTIC GRIFOLS	2 Screen (Neutral)	1.01	Is.IAS,IIs.IAS	DG Gel Neutral (210343)	Serascan Diana 2 (210204)	DGW10B.gru
DIAGNOSTIC GRIFOLS	2 Screen +AC (AHG)	3.00	I.IAC,II.IAC,AC.ACC	DG Gel Coombs (210342)	DG Gel Sol (210354),Serascan Diana 2 (210204)	DGW10D.gru
DIAGNOSTIC GRIFOLS	2 Screen+AC (Neutral)	3.00	Is.IAS,IIs.IAS,ACs.ACS	DG Gel Neutral (210343)	DG Gel Sol (210354),Serascan Diana 2 (210204)	DGW10L.gru
DIAGNOSTIC GRIFOLS	2 Screen+Dia (AHG)	1.01	I.IAC,II.IAC,DiaC.IAC	DG Gel Coombs (210342)	Serascan Diana 2 (210204),Serascan Diana Dia (210203)	DGW10E.gru
DIAGNOSTIC GRIFOLS	2 Screen+Dia (IgG)	1.01	I.IGG,II.IGG,DiaG.IGG	DG Gel Anti-IgG (210344)	Serascan Diana 2 (210204),Serascan Diana Dia (210203)	DGW10J.gru
DIAGNOSTIC GRIFOLS	2 Screen+Mia (AHG)	1.00	I.IAC,II.IAC,MiaC.IAC	DG Gel Coombs (210342)	Mi(a)-Cyte 0.8% (213386),Serascan Diana 2 (210204)	DGW13M.gru
DIAGNOSTIC GRIFOLS	2 Screen+Mia (IgG)	1.00	I.IGG,II.IGG,MiaG.IGG	DG Gel Anti-IgG (210344)	Mi(a)-Cyte 0.8% (213386),Serascan Diana 2 (210204)	DGW13N.gru
DIAGNOSTIC GRIFOLS	2 Screen+Mia+Dia (AHG)	1.00	I.IAC,II.IAC,MiaC.IAC,DiaC.IAC	DG Gel Coombs (210342)	Mi(a)-Cyte 0.8% (213386),Serascan Diana 2 (210204),Serascan Diana Dia (210203)	DGW13O.gru
DIAGNOSTIC GRIFOLS	2 Screen+Mia+Dia (IgG)	1.00	I.IGG,II.IGG,MiaG.IGG,DiaG.IGG	DG Gel Anti-IgG (210344)	Mi(a)-Cyte 0.8% (213386),Serascan Diana 2 (210204),Serascan Diana Dia (210203)	DGW13P.gru
DIAGNOSTIC GRIFOLS	2P Screen (Neutral)	1.01	IP.IAP,IIP.IAP	DG Gel Neutral (210343)	Serascan Diana 2P (210205)	DGW10C.gru

Define the type of reagent (Grifols/Medion)

Version of the test (will be automatically updated)

Name of the wells (useful to create templates)

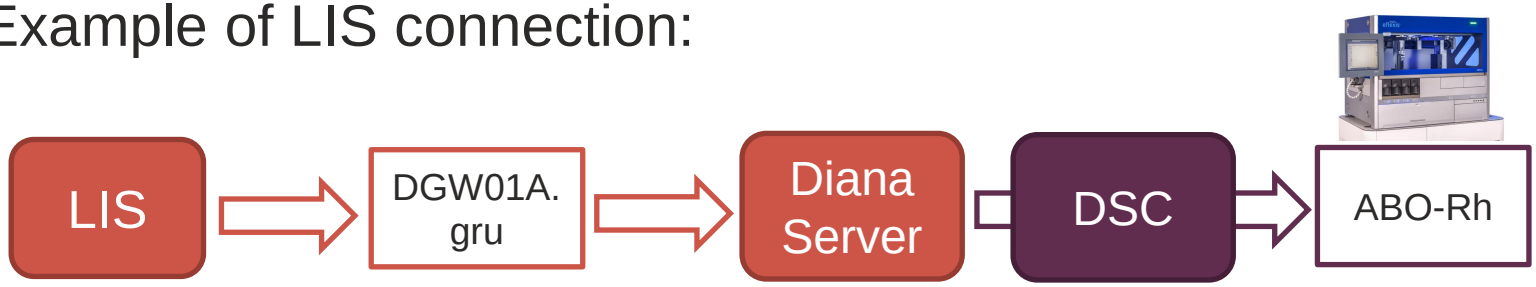
Equivalence of *.gru names (used for the DSC and LIS)

GRIFOLS

Importer configuration

Reagents Pack

- Example of LIS connection:

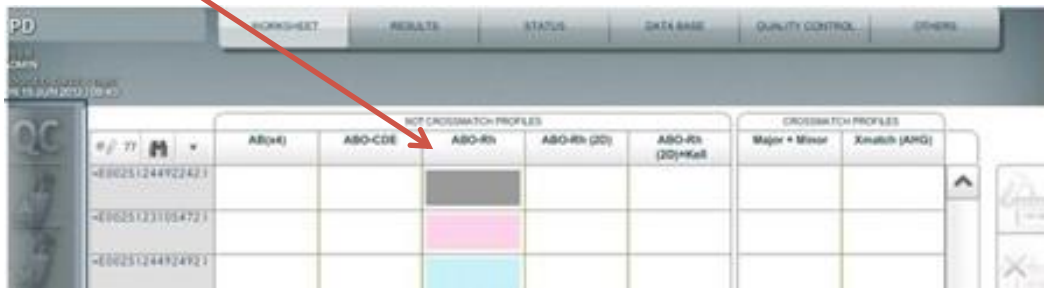


REAGENTS PACK EF v1.12

Table: Erytra Eflexis Tests for DG Gel cards

Version Date: 07/10/2020

Reagent manufacturer	Description	Version	Template wells	Card	Reagent	Equivalent Gru
DIAGNOSTIC GRIFOLS	ABO-Rh (CR)(A1-A2-B-O)	1.00	A.GD,B.GD,1_D.RH,2_D.RH,A1.GI,A2.GI,B.GI,0.GI	DG Gel ABO/Rh (CR) (210378)	Gerigrup Diana A1/B (213659),Gerigrup Diana 4 (210215),DG Gel Sol (210354)	DGW2SD.gru
DIAGNOSTIC GRIFOLS	ABO-Rh (CR)(A1-B-O)	1.01	A.GD,B.GD,1_D.RH,2_D.RH,CTL.HM,A1.GI,B.GI,0.GI	DG Gel ABO/Rh (CR) (210378)	DG Gel Sol (210354),Gerigrup Diana A1/B (213659),Gerigrup Diana 4 (210215)	DGW2SC.gru
DIAGNOSTIC GRIFOLS	ABO-Rh (2D)RT	1.00	A.GD,B.GD,AB.GD,D.RH,D'.RH,CTL.HM,A1.GI,B.GI	DG Gel ABO/Rh (2D)(RT) (210126)	DG Gel Sol (210354),Gerigrup Diana A1/B (213659)	DGW09P.gru
DIAGNOSTIC GRIFOLS	ABO-Rh	1.00	A.GD,B.GD,AB.GD,D.RH,CDE.RH,CTL.HM,A1.GI,B.GI	DG Gel ABO-Rh (210355)	DG Gel Sol (210354),Gerigrup Diana A1/B (213659)	DGW01A.gru



GRIFOLS

Importer configuration

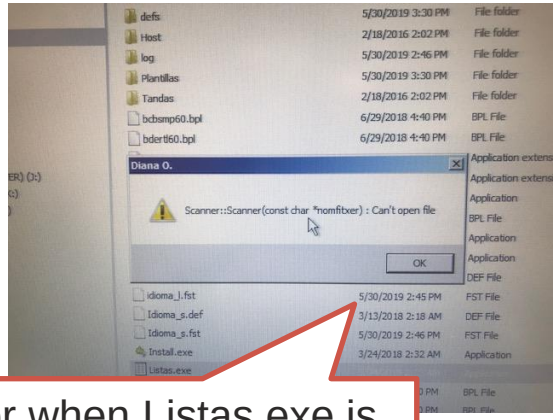
Reagents Pack: Installation notes

- Carefully read all INR/FSB documentation, because a bad installation can cause problems.

FIELD SERVICE BULLETIN

FSB-0423#0006: **Version 01**	**IH REAGENTS PACK ERYTRA EFLEXIS V1.12 (Build 1.12.0.20)**	Date: 30/10/20

REASON	- Erytra eflexis® Reagents Pack v1.12 (Build 1.12.0.20) has been released. - Provide installation procedure. - Describe improvements and modifications implemented.	
SCOPE	- All erytra eflexis® units running under CE configuration and software version 1.2.0 or higher.	
CHARACTER	☒Mandatory For units using Anti-N (213182, Antitoxin) reagent.	☐Recommended
PRIORITY	☐Immediate implantation	☒Asap



Error when Listas.exe is executed.

Missing DEFS files.

GRIFOLS

Importer configuration

Reagents Pack: Installation notes

- Before installing a new Reagents Pack, **do not forget to validate and export all samples and QC samples result**, there maybe some in the results list that cannot be possible to validate nor export after reloading the new RP.
- It is not possible to return to a previous RP.

2. Installation procedure

To install erytra eflexis® Reagents Pack v1.12 in units with software version 1.2.0 or 1.2.1, proceed as follows:

- Copy IH Reagents Pack EF v1.12 files to an USB Drive.

> **CAUTION!**
> Check that USB drive does not contain any virus before using it.

- Validate and export all samples and QC samples results and shut down erytra eflexis®.
- Switch on erytra eflexis® PC.
- Perform T shape on the touchscreen when erytra eflexis® logo is displayed on screen.
- Connect the USB drive to the PC

GRIFOLS

Importer configuration

Reagents Pack: Installation notes

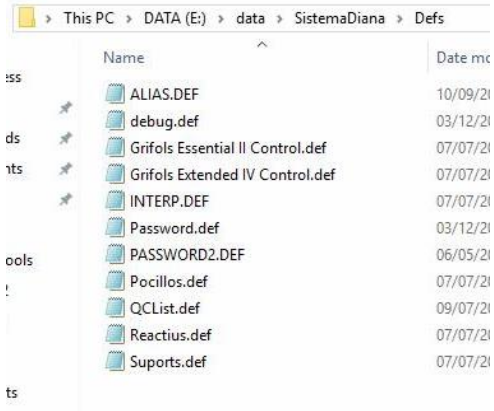
- The **most common problem** during the Reagents Pack installation is to **forget to delete existing and copy new DEF files and Templates to the appropriate folder**.
- Pay attention to the comment (all instruments). It means that applies to all IH instruments.

Diana System definition files installation (all instruments)

- *Delete ALIAS.DEF*, *INTERP.DEF*, *Pocillos.def*, *Reactius.def* and *Suports.def* in *Defs* subdirectory of Diana System working directory, if found.
- Copy *ALIAS.DEF*, *DG Gel Essential Control.def*, *DG Gel Extended Control.def*, *INTERP.DEF*, *Pocillos.def*, *Reactius.def* and *Suports.def* from the Reagents Pack

Diana System template files installation (all instruments)

- Delete all the files in *Plantillas* subdirectory of Diana System working directory.
- Copy all files from the Reagents Pack *Plantillas* subdirectory of *Diana System* directory to *Plantillas* subdirectory of Diana System working directory.
- Select all copied files in the destination directory and select *Properties*.
- Remove, if it exists, the check mark from the *Read only* box and click *Apply*.



Be careful!! Do NOT delete if there are custom templates. Copy, paste and overwrite is safest approach.

Be careful!! When you manage templates. New version can include new definitions not compatibles with the current LIS

GRIFOLS

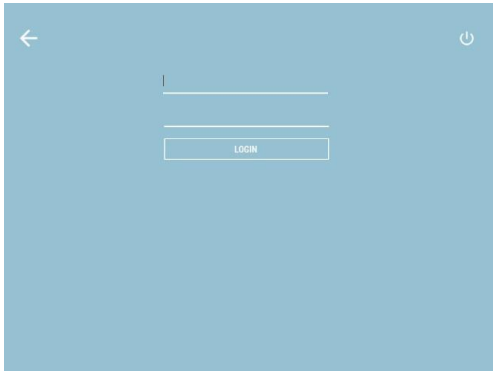
Importer configuration

Importer software

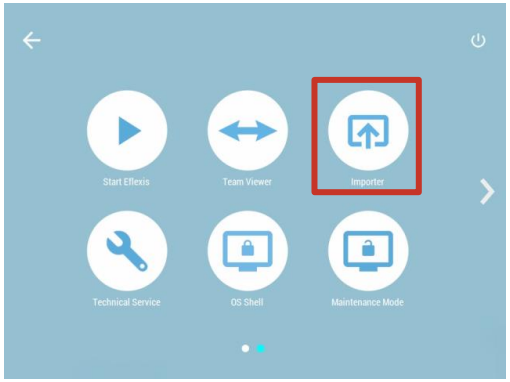
- To Access to the Importer software:
  - Access to Boot Menu by pressing the left-bottom icon and select Importer



Boot Application
Displayed only during 5 seconds



Username: adminUser
Password: admin

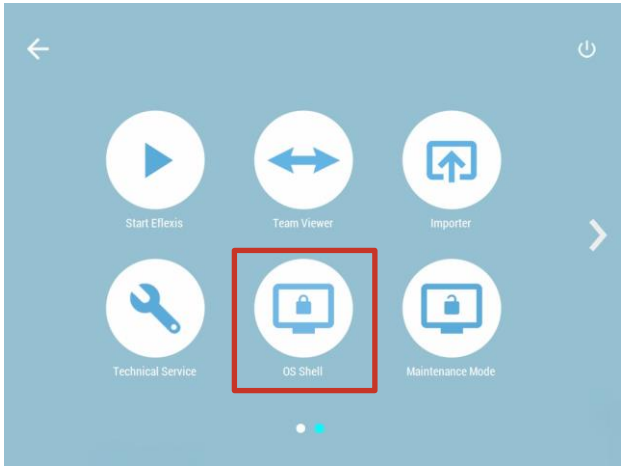


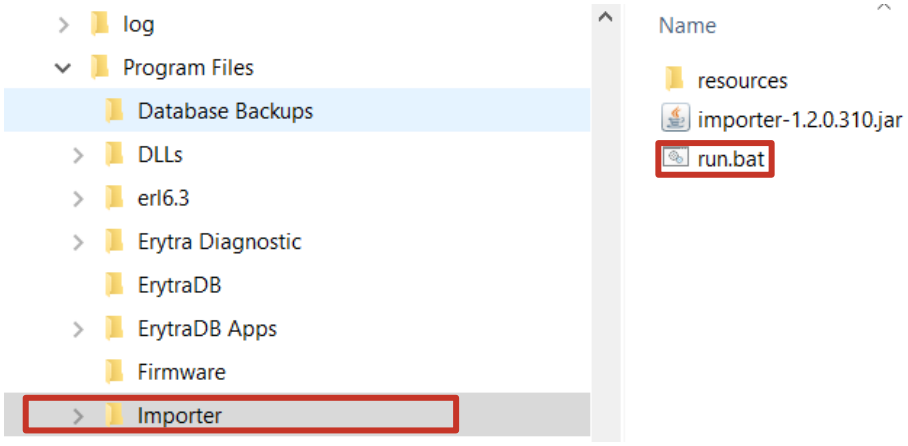
GRIFOLS

Importer configuration

Importer software

– Alternatively, **Importer** can be run from OS Shell.
– To do that, open a File Explorer and go to Program Files.
– Select the folder Importer and execute “run.bat”.





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Importer configuration

Importer software

- When opened, Importer tab is selected by default. There are three **3 different Pack files for loading**.

Reagent Pack file

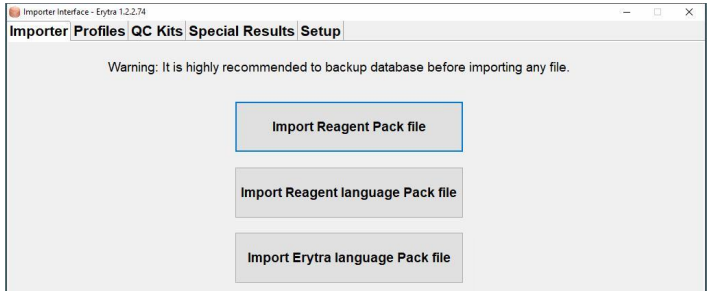
- Contains the different **techniques** and **resources**.
- Contains several folders where all the **information is password-protected**. It can not be opened nor modified.
- Includes: Tests, cards, reagents, QC Kits, Special results, DSC.def file.

Reagent language Pack file

- Contains the **information of the software** and its **translations** to several languages.

Erytra language Pack file

- Contains the **information of the reports** and its **translations** to several languages.

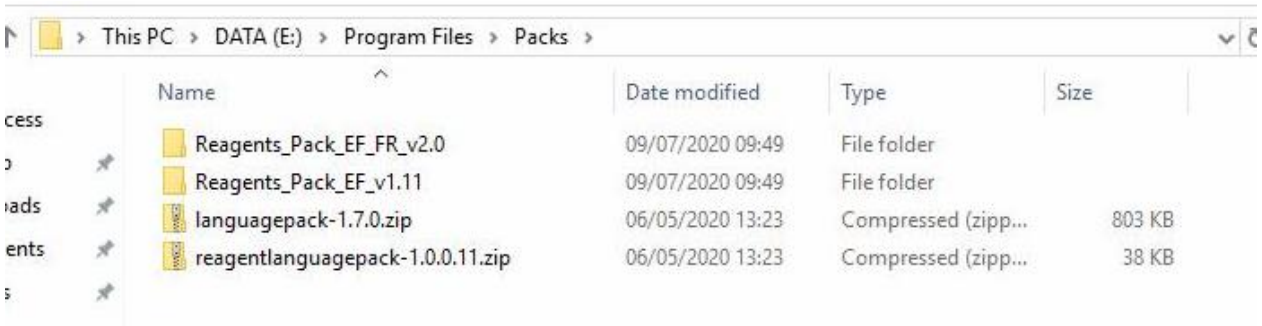


GRIFOLS

Importer configuration

Importer software

- **Reagent language Pack** and **Erytra language Pack** are stored and pre-installed by the software in E:\Program Files\Packs during the software installation.
- It’s highly recommended that you copy the latest **ReagentPack_XXX.zip** file version in this folder. This action helps other FSE and TAS in recognizing which version is loaded

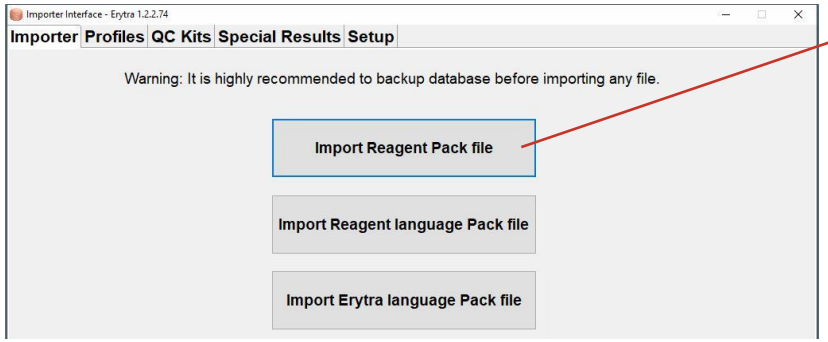


GRIFOLS

Importer configuration

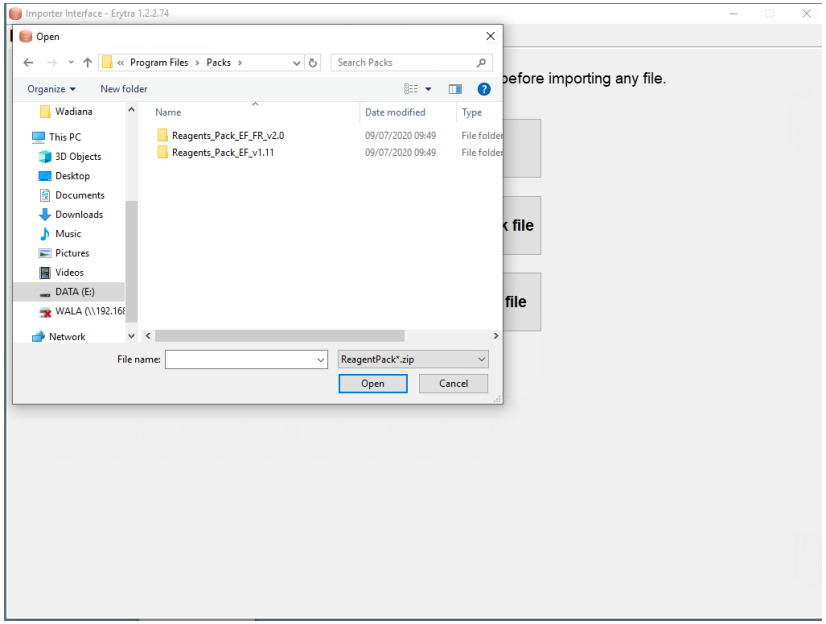
Importer software: Loading the Reagents Pack

- Select Importer tab for loading all Reagents Pack





Loading the RP can take several minutes!!



IF THE INSTALLATION IS BRAND NEW, LOAD ALL THREE RP FILES.
For US version, absolutely necessary!!!

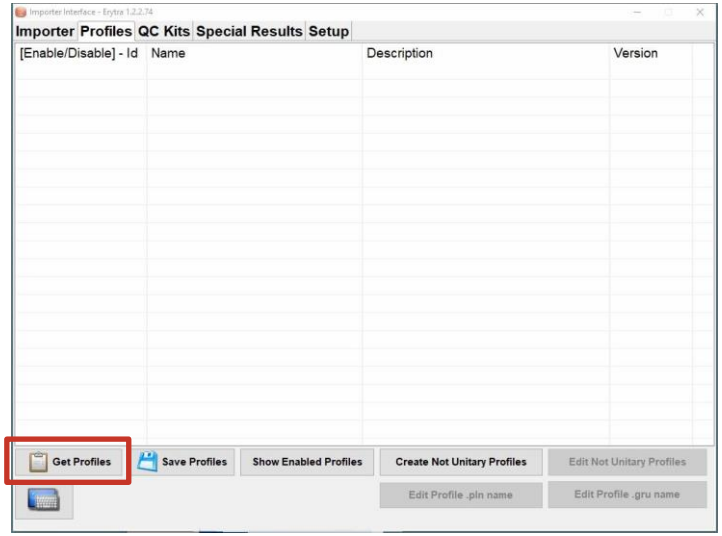
Note: Reagents Pack for titration should be imported after the Reagents Pack.

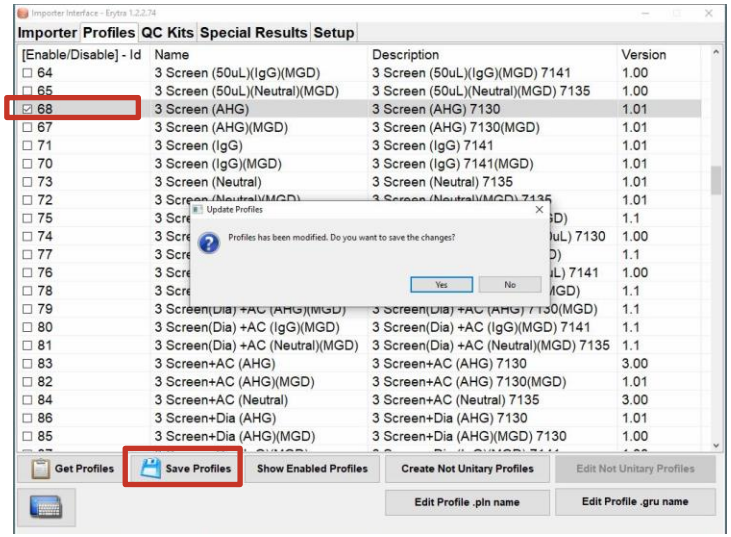
GRIFOLS

Importer configuration

Importer software: Profiles tab

- This tab allows **enabling the techniques** that will be **displayed on the worksheet**.
  - Press **Get Profiles** button and the full list of available techniques will be displayed in alphanumerical order.
  - Add a check in the boxes on the left of the technique names and then select the **Save Profiles** button to keep them enabled.



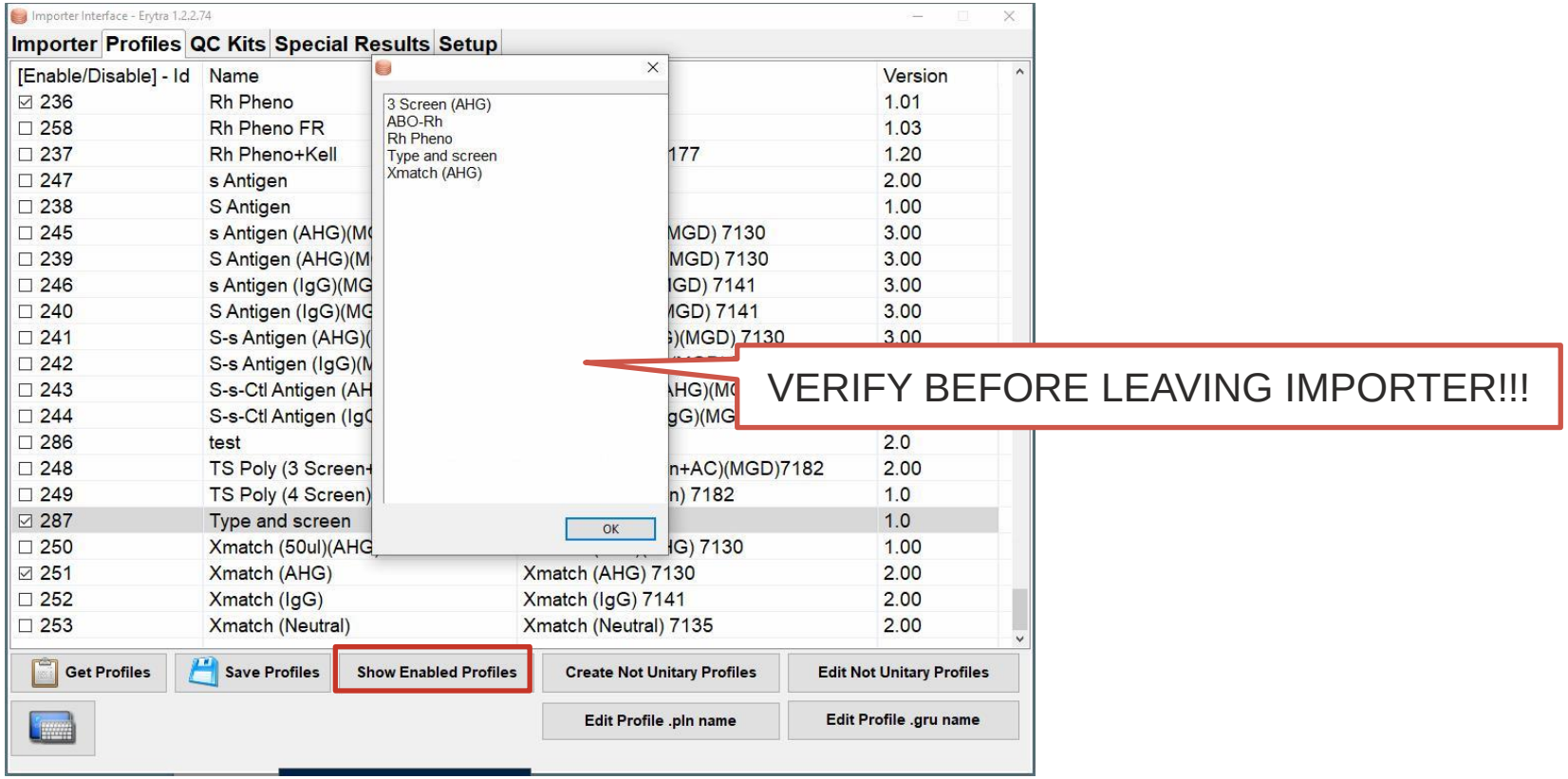


GRIFOLS

Importer configuration

Importer software: Profiles tab

- The list of only currently enabled profiles can be viewed by selecting the **Show Enabled Profiles** button.

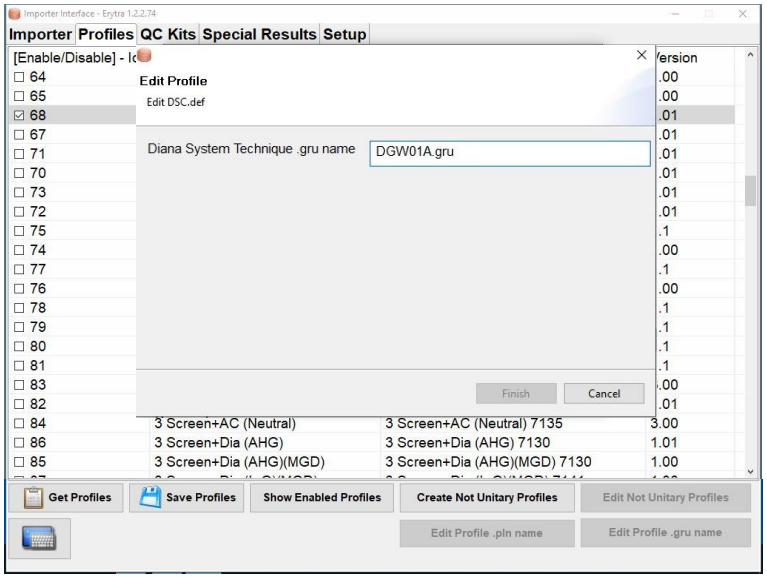


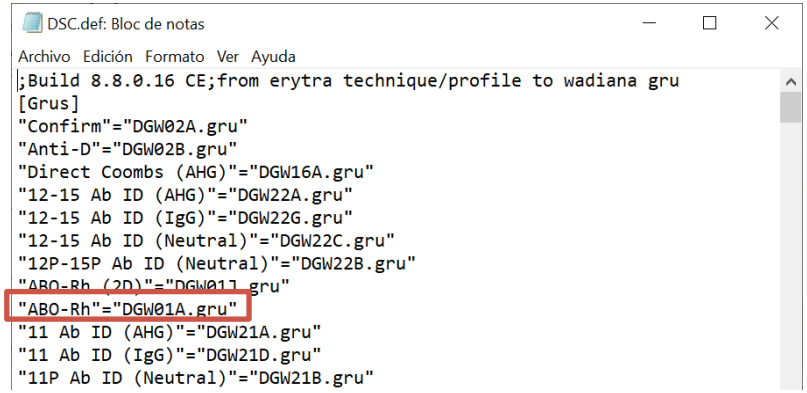
GRIFOLS

Importer configuration

Importer software: Profiles tab

- Diana System technique .gru names are automatically populated when the RP is uploaded. To modify it:
  - Select **Edit Profile .gru name** button.
  - The new *.gru has to be different to a *.gru existing (see Eflexis Test DG Gel.pdf or E:\data\Erytra\Wadiana\DSC.def).





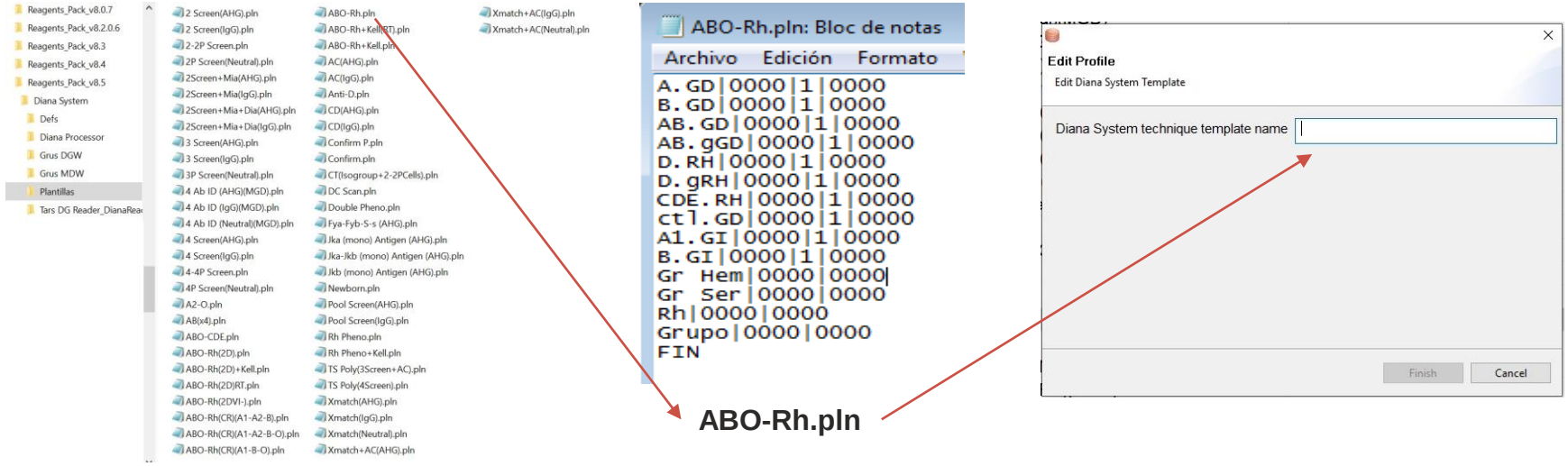
Do not modify the DSC.def file. This is different for every Reagents Pack version!!!

GRIFOLS

Importer configuration

Importer software: Profiles tab

- A Template (plantilla) is a file used by Sistema Diana for sending the results from the Erytra Eflexis® to LIS. To do this, each profile should have a *.pln filename template associated.
- It is possible to choose one of the default templates that can be found in E:\data\SistemaDiana\Plantillas. If none is entered, the system chooses Erytra.pln. This is empty by default.



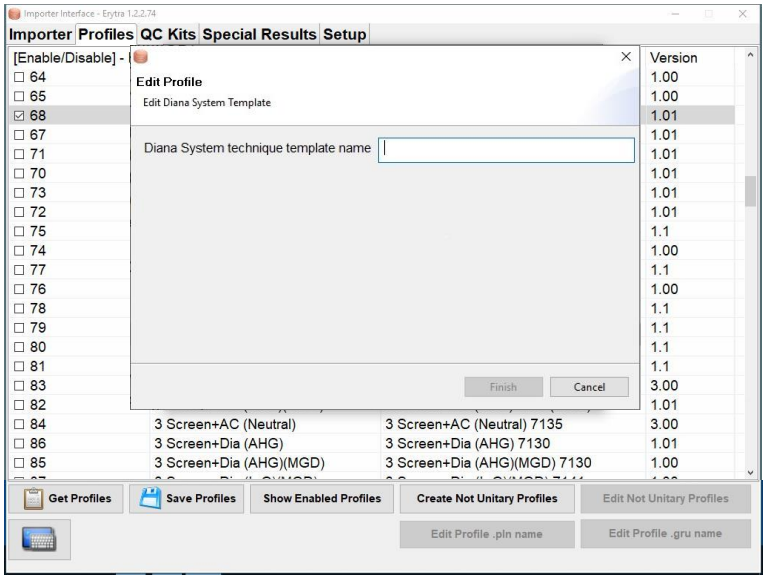
Note: It is possible to create a new template or modify an existing one. For additional information see “Creating Template Files” chapter in LIS module.

GRIFOLS

Importer configuration

Importer software: Profiles tab

- To assign a Template (plantilla) to a profile:
  - Select **Edit Profile .pln name** button.
  - The name of the *.pln has to be the same to an existing *.pln contained in E:\data\SistemaDiana\Plantillas. It can be one of the default templates or a new created template.



Note: It is possible to create a new template or modify an existing one. For additional information see “Creating Template Files” chapter in LIS module.

GRIFOLS

Importer configuration

Importer software: Not Unitary Profiles

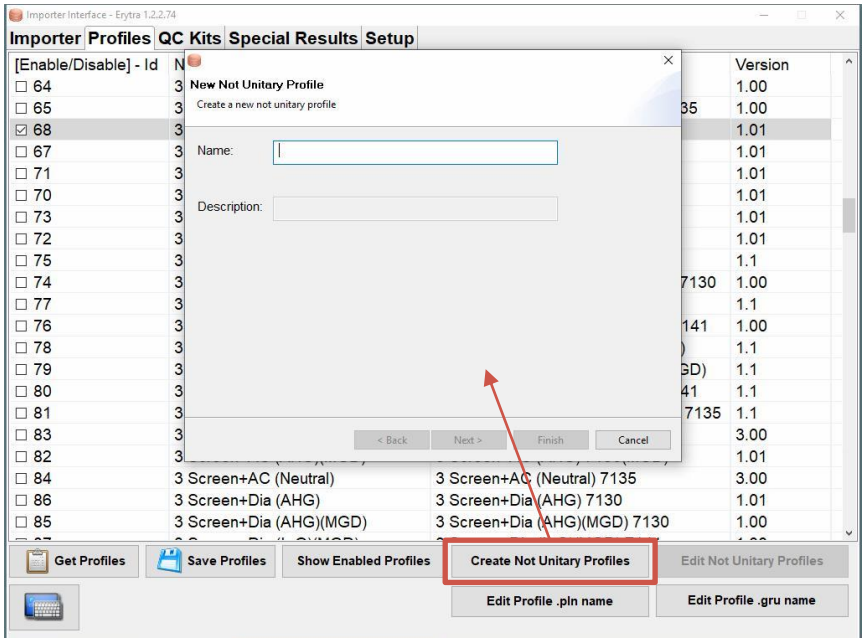
- A not unitary profile (NUP) is a customized profile that can be used to combine several unitary profiles (except for Crossmatch techniques).
- Using only one not unitary profile, the operator can run several techniques at the same time. However, until all associated profiles have been completed, the results cannot be validated and exported. Furthermore, when an individual result is rejected, all results obtained for the NUP are automatically rejected.
- For Erytra Eflexis®, a NUP can be also used to Rename a Unitary Profile.

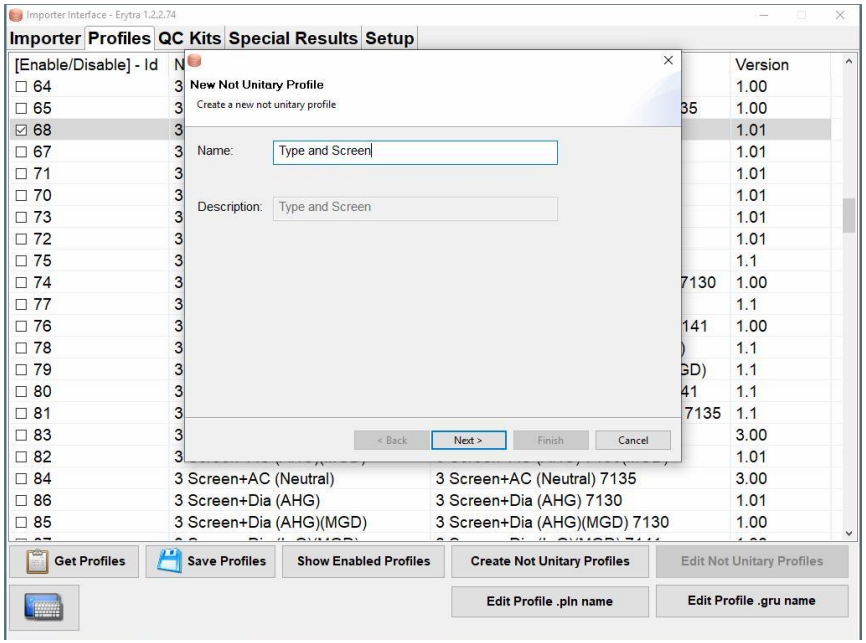
GRIFOLS

Importer configuration

Importer software: Not Unitary Profiles

- To create a not unitary profile:
  - Select **Create Not Unitary Profile** button.
  - Write the desired name for this Not Unitary Profile.



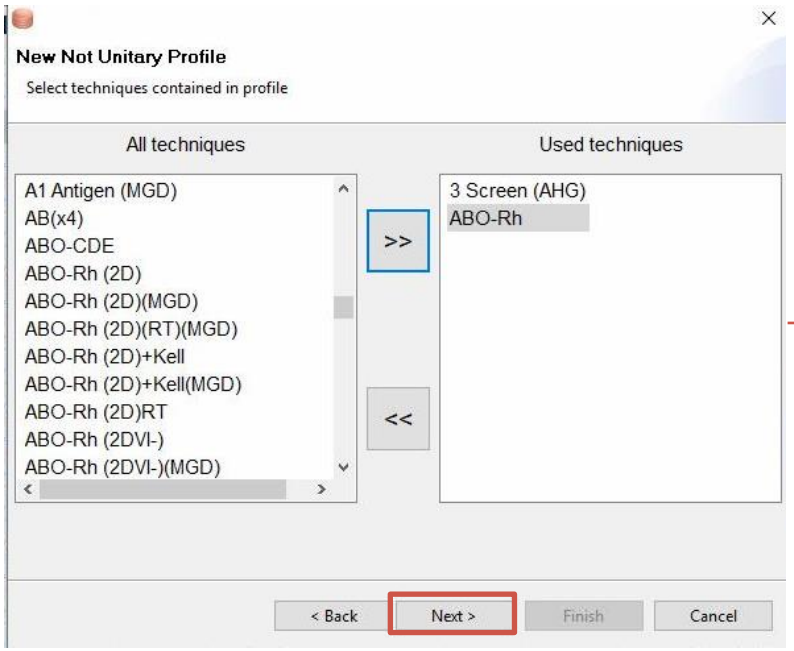


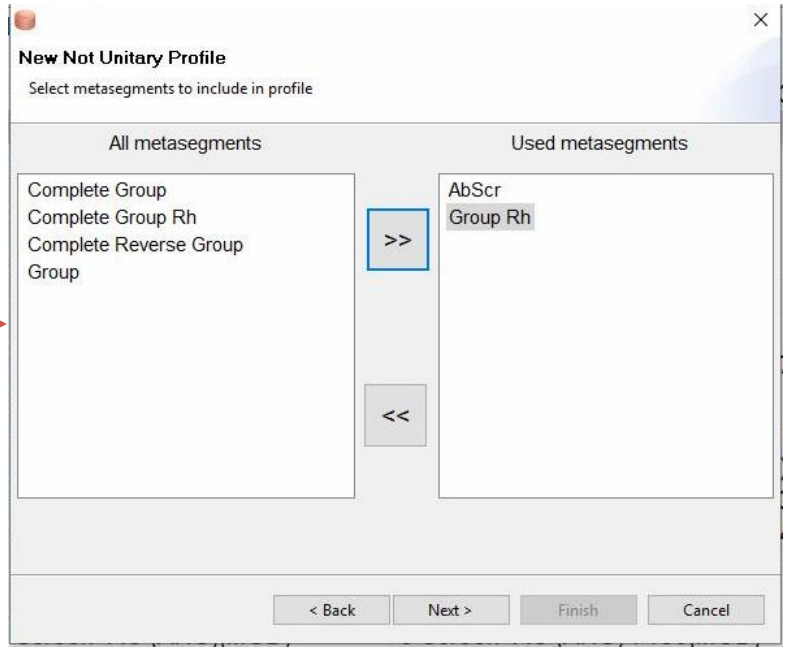
GRIFOLS

Importer configuration

Importer software: Not Unitary Profiles

- Select the techniques to include in the profile and press **Next >**.
- Select the meta-segments to include in the profile and that will appear on the Results List. They will be shown by default.



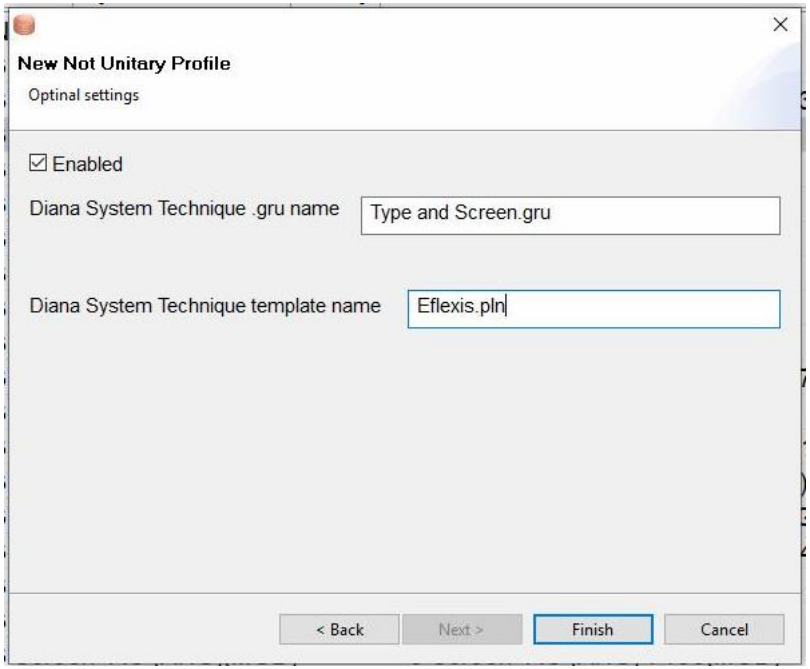


GRIFOLS

Importer configuration

Importer software: Not Unitary Profiles

- In case to be connected to HOST, assign the *.gru name and the template to the profile.
- Type *.gru and *.pln always in lowercase letters.
- Press Finish and Save the changes.



GRIFOLS

Importer configuration

Importer software: Not Unitary Profiles

- The introduced *.gru and *.pln names are saved in DSC_Custom.def that can be found in E:\data\Erytra\Wadiana.
- The template *.pln name has to be found in E:\data\Sistema Diana\Plantillas. If it is not there, the default template is Erytra.pln and this needs to be created using Listas.exe or Notepad++

DSC_Custom.def

```
[Grus]
"Titre 3 Screen (Cell I)(AHG)(MGD)" = "MDW33A.gru"
"Titre 3 Screen (Cell I)(AHG)" = "DGW33A.gru"
"Titre 3 Screen (Cell I)(IgG)(MGD)" = "MDW33D.gru"
"Titre 3 Screen (Cell I)(IgG)" = "DGW33D.gru"
"Titre 3 Screen (Cell II)(AHG)(MGD)" = "MDW33B.gru"
"Titre 3 Screen (Cell II)(AHG)" = "DGW33B.gru"
"Titre 3 Screen (Cell II)(IgG)(MGD)" = "MDW33E.gru"
"Titre 3 Screen (Cell II)(IgG)" = "DGW33E.gru"
"Titre 3 Screen (Cell III)(AHG)(MGD)" = "MDW33C.gru"
"Titre 3 Screen (Cell III)(AHG)" = "DGW33C.gru"
"Titre 3 Screen (Cell III)(IgG)(MGD)" = "MDW33F.gru"
"Titre 3 Screen (Cell III)(IgG)" = "DGW33F.gru"
"Titre 2 Screen (Cell I)(Neutral)" = "DGW33G.gru"
"Titre 2 Screen (Cell I)(Neutral)(MGD)" = "MDW33G.gru"
"Titre 2 Screen (Cell II)(Neutral)" = "DGW33H.gru"
"Titre 2 Screen (Cell II)(Neutral)(MGD)" = "MDW33H.gru"
"Titre 4 Screen (Cell I)(Neutral)" = "DGW33I.gru"
"Titre 4 Screen (Cell II)(Neutral)" = "DGW33J.gru"
"Titre 4 Screen (Cell III)(Neutral)" = "DGW33K.gru"
"Titre 4 Screen (Cell IV)(Neutral)" = "DGW33L.gru"
"Titre A1 (Neutral)" = "DGW33M.gru"
"Titre A1 (Neutral)(MGD)" = "MDW33M.gru"
"Titre B (Neutral)" = "DGW33O.gru"
"Titre B (Neutral)(MGD)" = "MDW33O.gru"
"Titre 2 Screen (Cell I)(AHG)" = "DGW33Q.gru"
"Titre 2 Screen (Cell I)(AHG)(MGD)" = "MDW33Q.gru"
"Titre 2 Screen (Cell II)(AHG)" = "DGW33R.gru"
"Titre 2 Screen (Cell II)(AHG)(MGD)" = "MDW33R.gru"
"Titre 4 Screen (Cell I)(AHG)" = "DGW33S.gru"
"Titre 4 Screen (Cell II)(AHG)" = "DGW33T.gru"
"Titre 4 Screen (Cell III)(AHG)" = "DGW33U.gru"
"Titre 4 Screen (Cell IV)(AHG)" = "DGW33V.gru"
"Titre A1 (AHG)(MGD)" = "MDW33W.gru"
"Titre A1 (AHG)" = "DGW33W.gru"
"Titre A1 (IgG)(MGD)" = "MDW33Y.gru"
"Titre A1 (IgG)" = "DGW33Y.gru"
"Titre B (AHG)(MGD)" = "MDW33X.gru"
"Titre B (AHG)" = "DGW33X.gru"
"Titre B (IgG)(MGD)" = "MDW33Z.gru"
"Titre B (IgG)" = "DGW33Z.gru"
"Titre Dia Screen (AHG)" = "DGW35A.gru"
"TypeandScreen" = "TypeandScreen.gru"

[Templates]
"TypeandScreen" = "Erytra.pln"
```

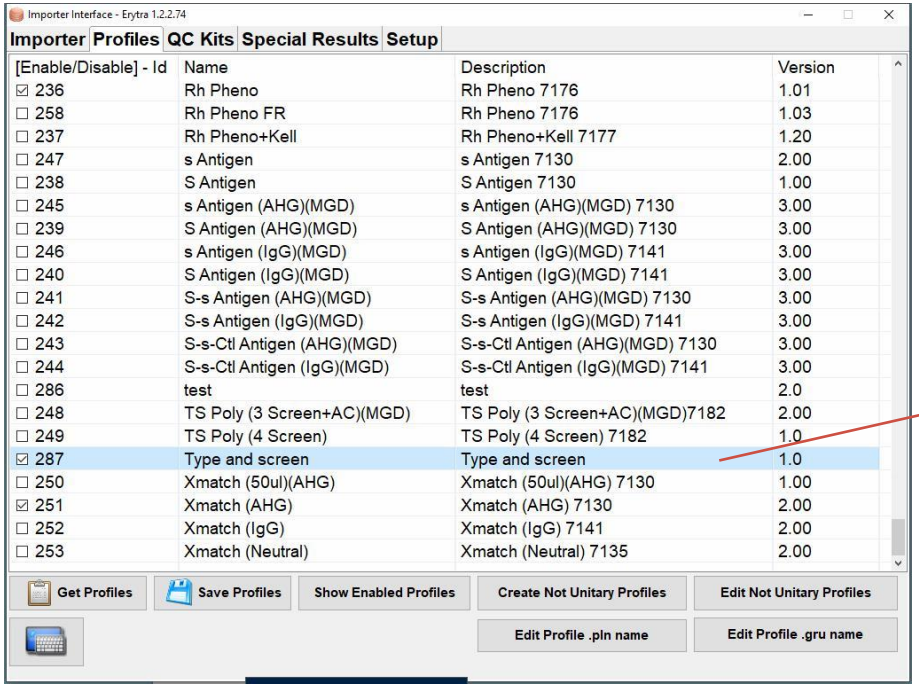
Note: It is possible to create a new template or modify an existing one. For additional information see “Creating Template Files” chapter in LIS module.

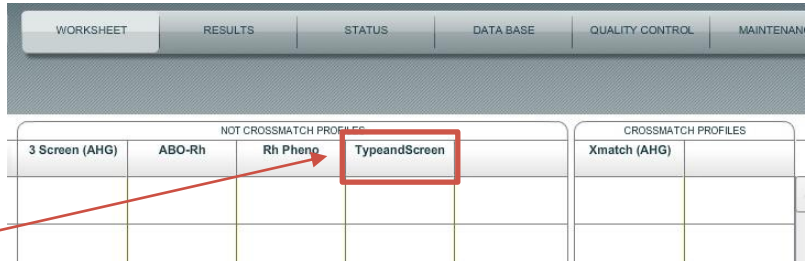
GRIFOLS

Importer configuration

Importer software: Not Unitary Profiles

- Note that NUP are enabled by default when they are first created.
- The new profile will appear in the Worksheet if enabled.



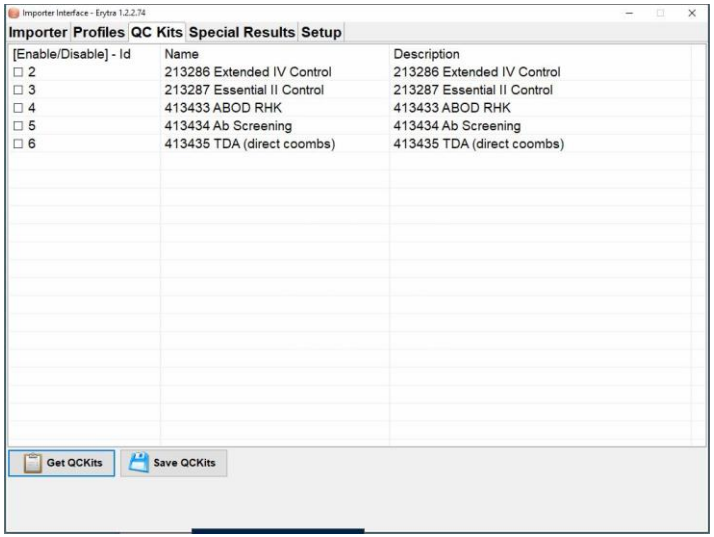


GRIFOLS

Importer configuration

Importer software: QC Kits tab

- This tab allows importing the QC kit from the Reagents Pack.
  - Select **Get QC kits** to display the list with the available QC kits.
  - Add a check in the corresponding checkbox and select **Save QC kits**.
  - More than one QC Kit can be enabled at the same time.



- Activate QC policy and define its frequency from the Setup tab or the erytra.properties.xml file is also necessary.

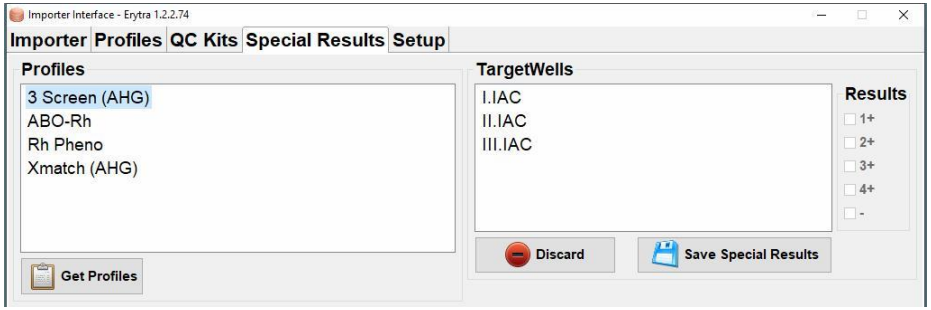
Note: The QC Kits available in QC Kits tab might vary between different regions and countries.

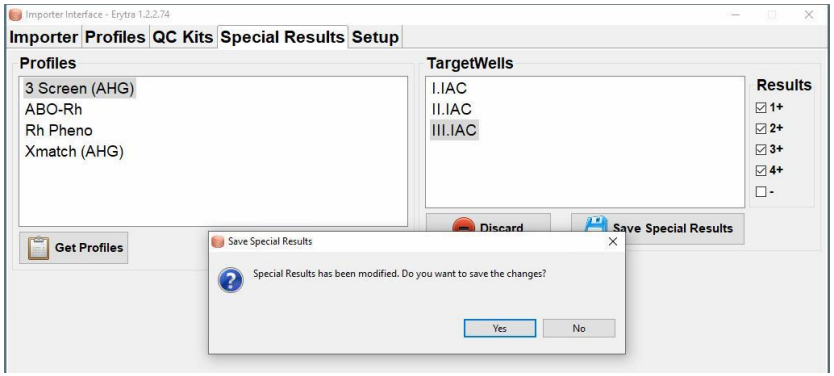
GRIFOLS

Importer configuration

Importer software: Special Results tab

- It is possible to configure **special results for each individual technique**.
- These results will appear on the RESULTS LIST marked with the red background on the **eye icon**, which **disables** the possibility to **automatically validate** those **results**.
- It will force the operator to go to the PROFILE INFORMATION screen and **accept these special results before validate** and **export** them.





GRIFOLS

Importer configuration

Importer software: Special Results tab

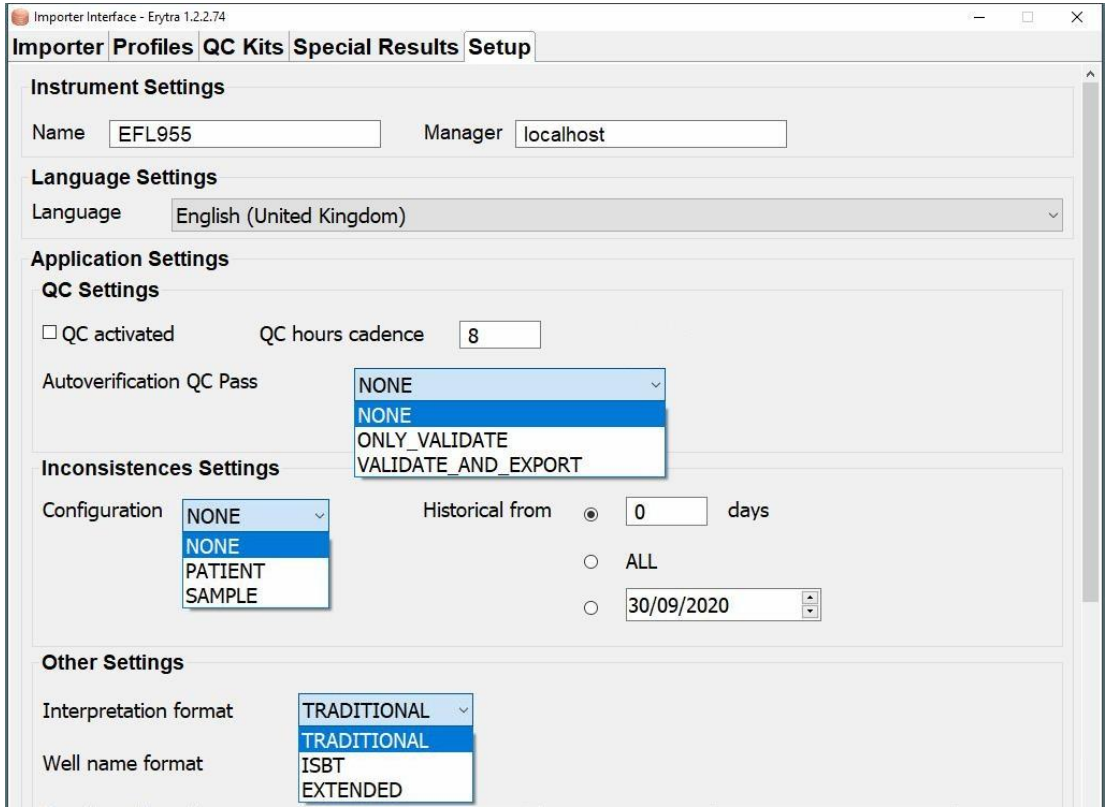
- **Some results** are considered, **by default**, **as special results**:
  - **Special Microtube Grade**:
    - Double Population (DP); US: Mixed Field (MF).
    - Doubtful reaction grade (?).
    - Doubtful positive reaction grade (+/-); US: (W+/-) only under user selection.
    - No Results Determined (NR); US: not applicable.
    - Any result previously configured as a special results in the Importer.
  - **Special Interpretations**:
    - Not Interpretable Results (NI).
    - Only US: NRD Results (No Result Determined).
    - Positive Autocontrol.
    - Positive Direct Antiglobulin Test.
  - **Discrepant Interpretations (Dis)**

GRIFOLS

Importer configuration

Importer software: Setup tab

- There are several **configurable parameters** that can be modified from the Setup tab.



Note: These parameters can also be defined by means of erytra.properties.xml file.

GRIFOLS

Importer configuration

Importer software: Setup tab

- These parameters are:
  - **Name of the instrument**: free text (max 10 characters).
  - **Language** of the interface of the Erytra Eflexis® program
  - **QC configuration**: activation of the QC Policy, QC hours cadence and autovalidation of the QC.
  - **Inconsistencies settings**:
    - None: no Inconsistencies are checked.
    - Sample: Inconsistencies are checked on Sample tube barcode basis.
    - Patient: Inconsistencies are checked on Patient ID basis.
  - **Interpretation format** and **Well name format:**
    - Traditional: it uses the traditional nomenclature.
    - ISBT: it uses the ISBT nomenclature
    - Extended (only for Interpretation): currently the same as Traditional.

Note: These parameters can also be defined by means of erytra.properties.xml file.

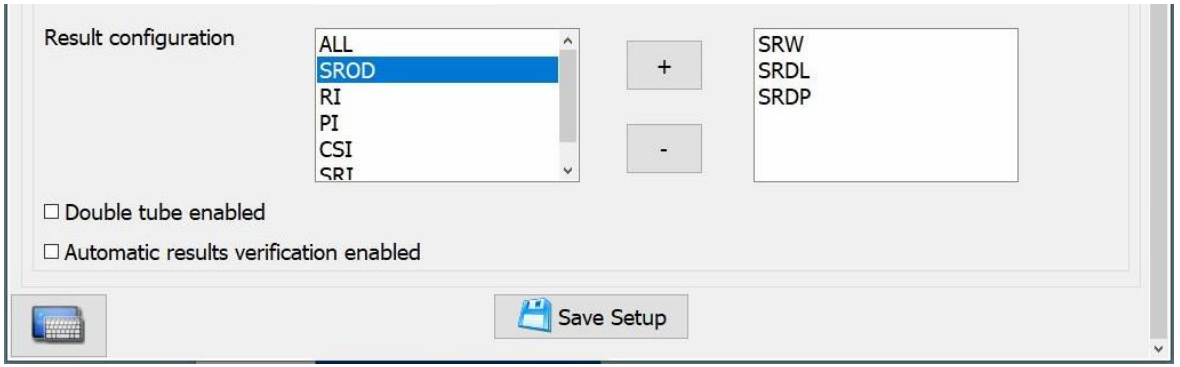
GRIFOLS

Importer configuration

Importer software: Setup tab

- **Result configuration**: refers to the cards with results that are placed in the service rack and defines the well results that shall be checked by the operator before validation.
- **Double tube enabled:** enables the possibility to work with two tubes with the same sample ID (RBCs and serum/plasma separated).
- **Automatic results verification**: if enabled, the not exceptional results will be automatically validated and exported.

- To save the changed done in the Importer Setup, it is necessary to select the Save Setup button before closed the Importer.



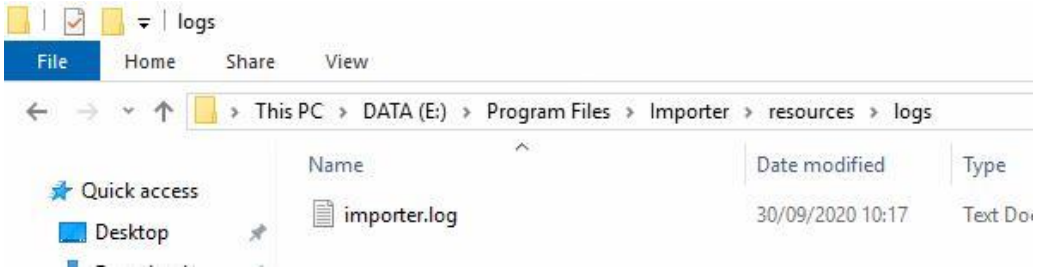
Note: These parameters can also be defined by means of erytra.properties.xml file.
Note 2: to learn more about Result configuration, see the erytra.properties chapter.

GRIFOLS

Importer configuration

Importer log files

- The Importer.log file traces information of the actions performed by the importer software.
- Has a cycle topology. So, when it is full, the new information renews the first information.
- The size of this file is short.
- Is located at E:/Program Files/Importer/resources/logs
- The system stores one backup of the importer.log



- To see when Importer was last modified can be useful when there are sudden, unexpected LIS and export issues.

GRIFOLS

Erytra properties

GRIFOLS

Erytra.properties

Introduction

- Erytra.properties is a configuration file.
- The present module is based on the **Service Manual** section **Software Configuration** for Erytra Eflexis®.
- To edit configuration use any .txt file editor as WordPad, Notepad, Notepad ++, etc.



NOTE:

Check that spelling of the written options is correct. Otherwise an error will appear when software is started.



NOTE:

Before modifying any parameter, check that software is not running.

Erytra.properties

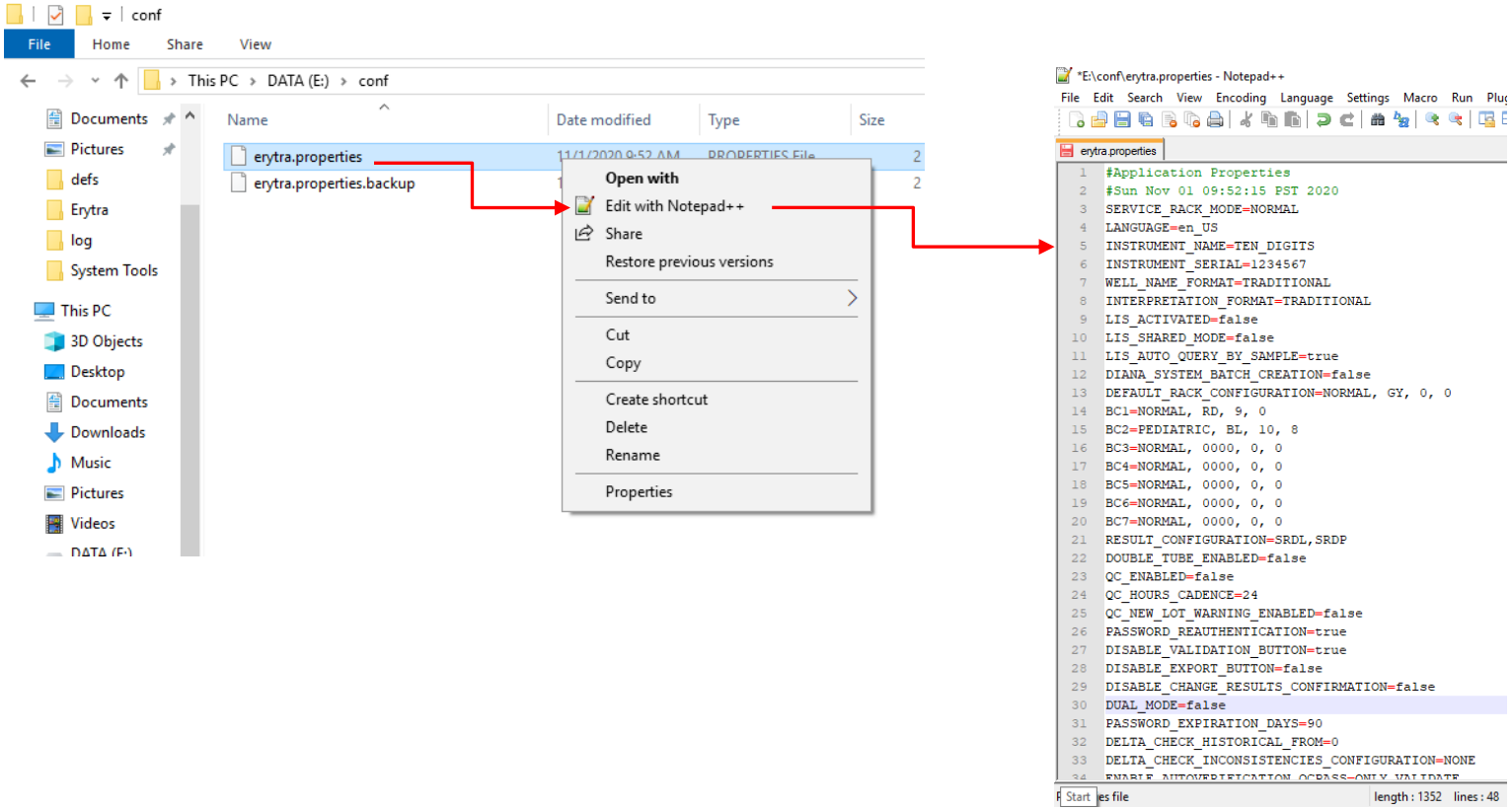
Scope

- The present document cover the erytra.properties configuration files under operating system Windows 10. That means:
  - **Erytra Eflexis® version from 1.2.2**
- Now a backup file named erytra.properties.backup is created during (each) initialization process (after log in). It is stored in *E:\conf*.
- If a corrupted erytra.properties file is found during instrument initialization, system will look for erytra.properties.backup file and, if it exists, it will use the information of this file to recreate erytra.properties file. If the file does not exist, a new erytra.properties file with default values will be recreated.
- For the parameters manageable in both Importer and erytra.properties is recommended to always use Importer to configure them.

Erytra.properties

Configuration

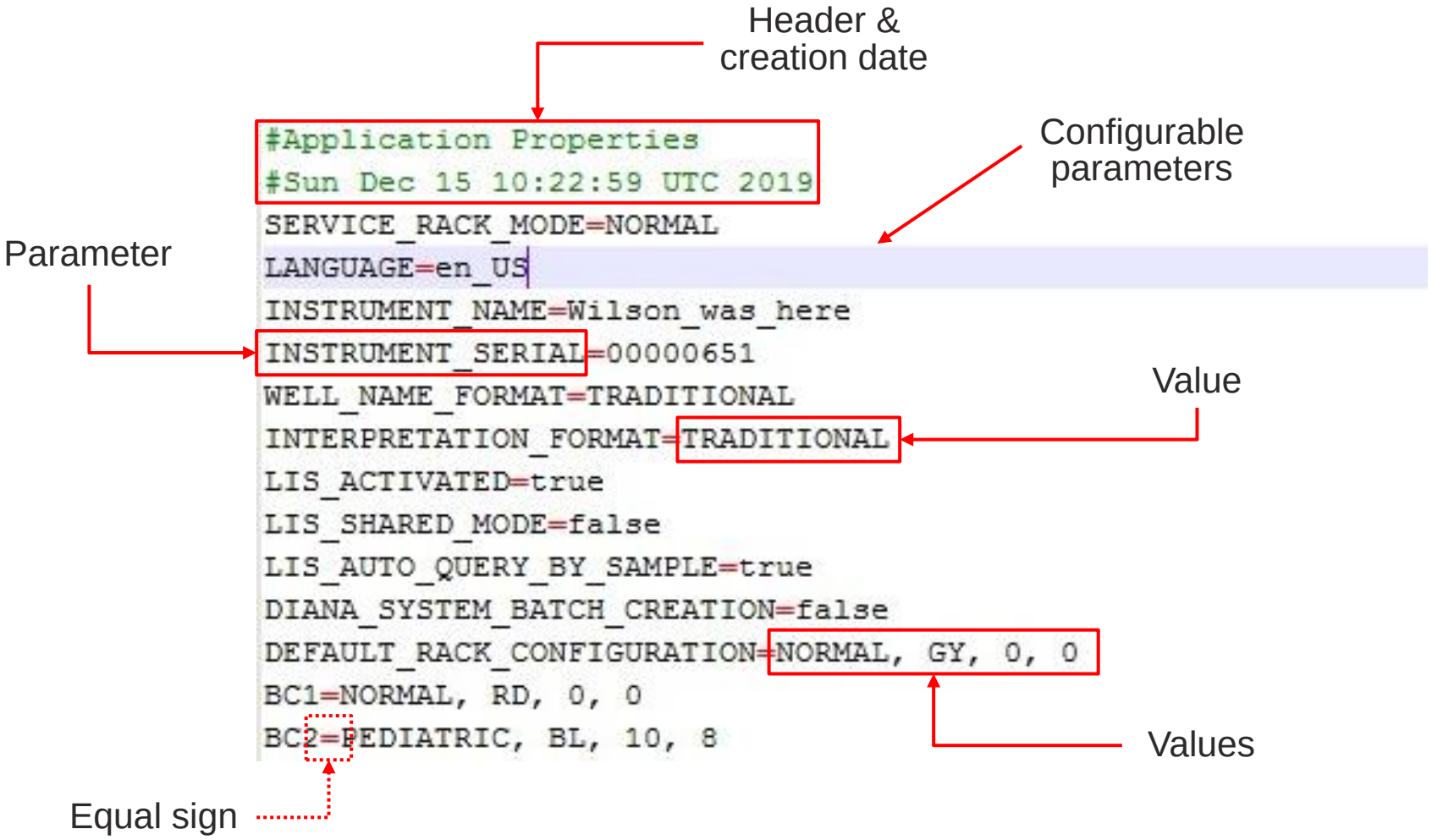
- To change configurations go to **E:\conf**, right click to display menu and open with Notepad ++.



GRIFOLS

Erytra.properties

Configuration



GRIFOLS

Erytra.properties

Configuration

- **LANGUAGE =** en_GB (by default)

=cs_CZ	: Czech.	=it_IT	: Italian.
=de_DE	: Deusth.	=ja_JP	: Japanese.
=en_GB	: England English.	=nb_NO	: Norwegian.
=en_US	: American English.	=pl_PL	: Polish.
=es_ES	: Spanish.	=pt_PT	: Portugal Portuguese.
=fr_FR	: French.	=sv_SE	: Swedish.
		=zh_SG	: Chinese.

  - Language parameter shows customer software language available in language pack according to country or region .

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
```

Note: This parameter can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **INSTRUMENT_NAME=**xxxxxxxxxx
  - Instrument name used for particular custom identification.
  - Maximum 10 characters.
  - Missing this detail can cause instability in the software.
- **INSTRUMENT_SERIAL**=(xxxxxxx format)
  - Serial number used to match physical serial number with traceability.
  - Maximum 7 numerical characters. E.g. SN0000730 is only 0000730
  - Missing this detail can cause instability in the software (critical in Erytra Eflexis®)

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
```

Note: Instrument name can be defined also by means of Importer application.
Note 2: Instrument serial number can also be defined by means of Technical Service Software.

GRIFOLS

Erytra.properties

Configuration

- **RESULTS_CONFIGURATION=**SRW, SRDP, SRDL (by default for CE) and SRDL, SRDP (by default for US)
  - Refers to the cards with results that are placed in the service rack and defines the well results that shall be checked by the operator before validation. The options to configure are:
    - **SRW** – Special Results that are **weak** (+/-) (Not available in US)
    - **SRDL** – Special Results that are **doubtful** (?)
    - **SRDP** – Special Results with **double population** (mixed field)
    - **SROD** – Special Results **defined by the operator**, using Importer.
    - **RI** – Results with **reading incidences.**
    - **PI** – Results with **process incidences.**
    - **CSI** – **Card or Sample Incidences**.
    - **ALL** – **All the process cards** are placed in the service rack.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA=false
DIANA_SYSTEM_BATCH_CREATION=false
DEFAULT_RACK_CONFIGURATION=NORMAL, GY, 0, 0
BC1=NORMAL, RD, 0, 0
BC2=PEDIATRIC, BL, 10, 8
BC3=NORMAL, 0000, 0, 0
BC4=NORMAL, 0000, 0, 0
BC5=NORMAL, 0000, 0, 0
BC6=NORMAL, 0000, 0, 0
BC7=NORMAL, 0000, 0, 0
RESULT_CONFIGURATION=SRW,SRDL,SRDP
```

  - All the above results can be combined. In this case, they should be written separated by a coma.

Note: These parameters can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **WELL_NAME_FORMAT=**TRADITIONAL (default)
  - Defines the standard used to designate well names.
    - **TRADITIONAL:** it uses the traditional nomenclature.
    - **ISBT:** it uses the ISBT nomenclature.
- **INTERPRETATION_FORMAT=**TRADITIONAL (default)
  - Defines the standard used to designate interpretation results.
    - **TRADITIONAL:** it uses the traditional nomenclature.
    - **ISBT:** it uses the ISBT nomenclature.
    - **EXTENDED:** currently the same as Traditional.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
```

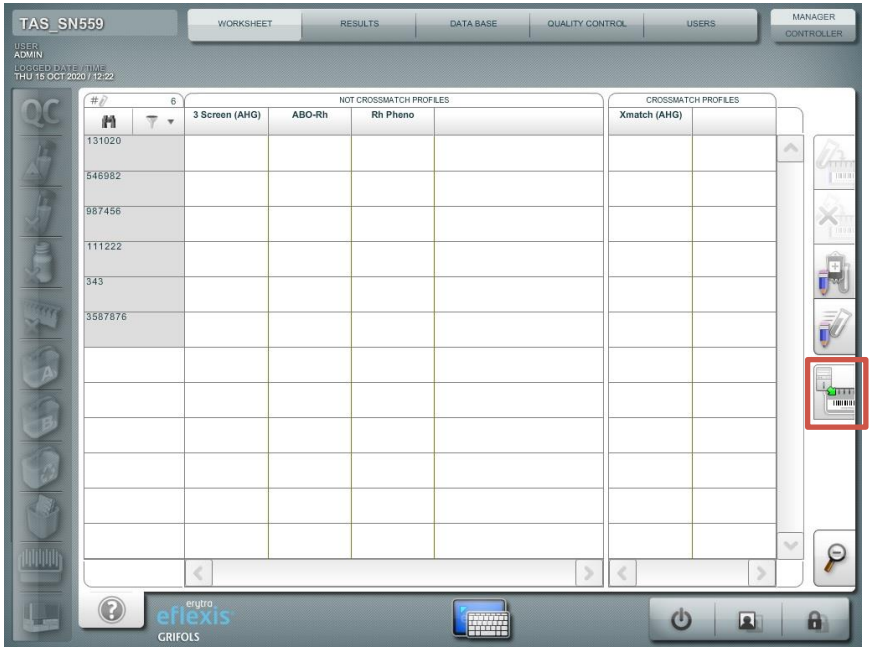
Note: These parameters can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **LIS_ACTIVATED=**false (default)
  - Enable or disable the software with DianaServer.
    - **true**: the LIS is activated and is allowed to communicate with the LIS. The **LIS Request** button will be enabled in the Worksheet.
    - **false**: the LIS connection is not activated.




```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA=false
```


GRIFOLS

Erytra.properties

Configuration

- **LIS_SHARED_MODE=**false (default)
  - Defines whether samples received from LIS and that are not inside the instrument are sent to LIS as cancelled.
    - **true:** the samples that are inside the instrument will be processed whereas those samples that are not inside will be sent back to LIS as cancelled. The missing alerts of the cancelled samples will not be activated.
    - **false:** the missing alerts of the samples that are not inside the instrument will be activated and will appear in the Worksheet.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA=false
```

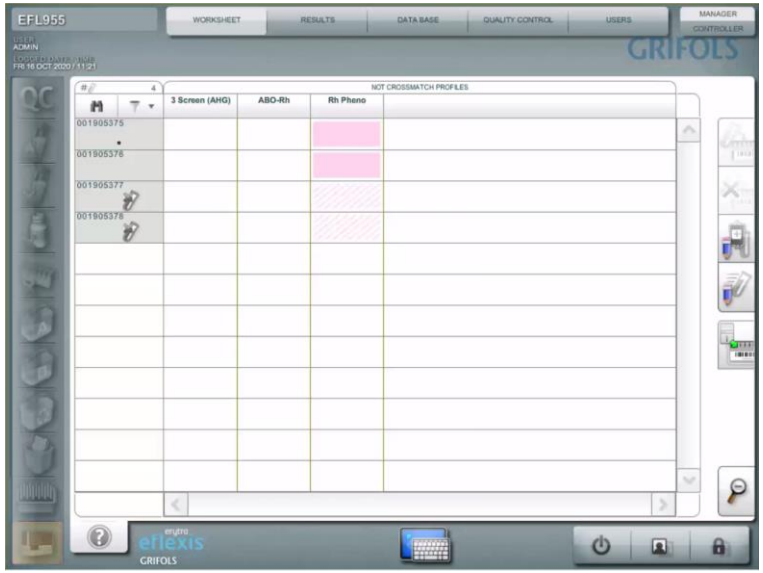
Note: For more detailed information, see LIS configuration chapter.

GRIFOLS

Erytra.properties

Configuration

- When several instruments share the same LIS connection, LIS_SHARE_MODE parameter must be set to **true**. Otherwise, all the workload would be sent to only one instrument and not received by the others.
- In case there is only instrument connected to LIS, setting the LIS_SHARE_MODE parameter to **false** allows all pending workload to be displayed in the Worksheet window.



Note: For more detailed information, see LIS configuration chapter.

GRIFOLS

Erytra.properties

Configuration

- **LIS_AUTO_QUERY_BY_SAMPLE=**false (default)
  - Request sent automatically to LIS for each sample when a sample rack has been identified.
    - **true:** automatic request.
    - **false:** no automatic request.
  - All samples in the instrument are included in the request.
  - Diana Server ensures that samples are not repeated unless previous results have already been exported to LIS.
  - **LIS Request** button is maintained in the Worksheet and active.

```
[6/14/2017 6:09:20 PM]-> Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170614180920
[6/14/2017 6:09:20 PM]-> Send: Q|1|ALL||ALL||||||||I|
[6/14/2017 6:09:20 PM]-> Send: L|1|N

[6/15/2017 2:07:56 PM]-> Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170615140756
[6/15/2017 2:07:56 PM]-> Send: Q|1|^=E00251711037721^||ALL||||||||I|
[6/15/2017 2:07:56 PM]-> Send: L|1|N
```

```
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
...
DEFAULT_BC_IS_ISBT=false
LIS_AUTO_QUERY_BY_SAMPLE=false
FLUIDIC_SYSTEM_ALWAYS_READY=true
AUTOMATIC_RESULTS_VERIFICATION=false
EXTERNAL_CARD_WASTE=false
```

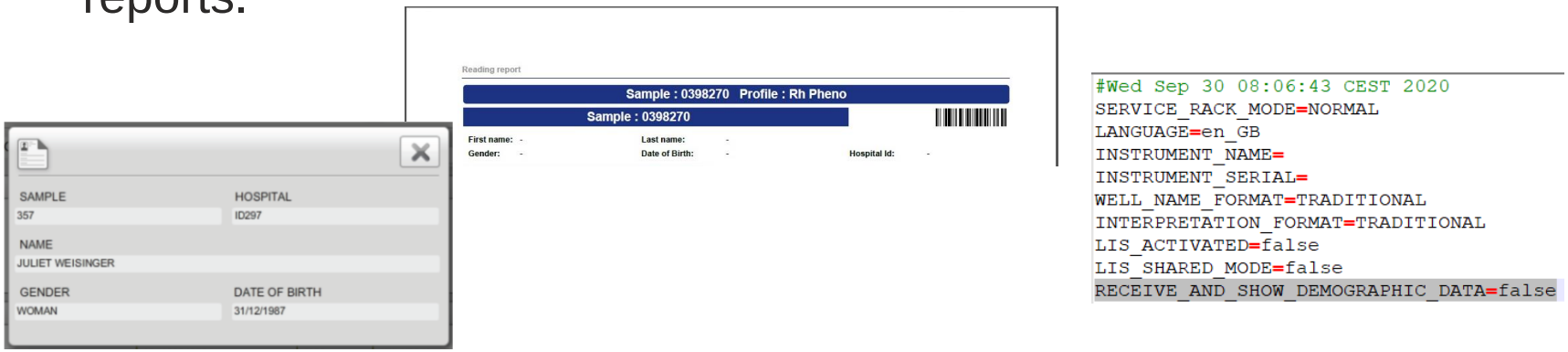
Note: For more detailed information, see LIS configuration chapter.

GRIFOLS

Erytra.properties

Configuration

- **RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA=**false (default)
  - Allows defining if sample demographic data sent from LIS is to be stored in the system and shown to the operator. The available options are:
    - **true**: to receive and show the demographic data.
    - **false**: to not show the demographic data.
  - If the parameter is set on true, demographic data will be displayed in **Sample Related Data** window, in **Traceability information** in the detailed Profile Information and in the reading and traceability reports.




```
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA=false
```

Note: This register is not shown in US version software, it is only for CE instrument configuration.

GRIFOLS

Erytra.properties

Configuration

- In Erytra Eflexis®, if new demographic data is received from LIS for a certain Patient ID which had already different demographic data associated in the analyser, a warning window will appear to indicate the discrepancy. In this window, current demographic data is shown and there is the option to maintain it and cancel the execution or to replace it for the new one.
- This checking is not configurable and will appear only if receive and show demographic data is activated.



Note: This register is not shown in US version software, it is only for CE instrument configuration.

GRIFOLS

Erytra.properties

Configuration

- **DIANA_SYSTEM_BATCH_CREATION=**false (default)
  - Defines whether the system creates Diana System batches or not. It can be configured as:
    - **true**: the tan file creation is activated.
    - **false**: the tan file creation is not activated.
  - The files are created in E:\data\SistemaDiana\Tandas
  - This option is used whenever the LIS must capture test images, since the ASTM protocol only sends text.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA=false
DIANA_SYSTEM_BATCH_CREATION=false
DEFAULT_RACK_CONFIGURATION=NORMAL, GY, 0, 0
```

Note: For more detailed information, see LIS configuration chapter.

GRIFOLS

Erytra.properties

Configuration

- **RACK_BC_ENABLED=**true (default)
  - It defines if a barcode is required in each samples holder
    - **true**: the barcode in the sample holders is required.
    - **false**: the barcode in the samples holders is not required.
  - To reduce a potential problem detecting the barcode in special holders, a barcode has been introduced in all holders. If a barcode is not detected, this holder will not appear in the Samples screen.



Yellow barcode label code: 213798



Blue barcode label code: 213672



Gray barcode label code: 213791



Red barcode label code: 213723

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL_NAME_FORMAT=TRADITIONAL
INTERPRETATION_FORMAT=TRADITIONAL
LIS_ACTIVATED=false
LIS_SHARED_MODE=false
RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA=false
DIANA_SYSTEM_BATCH_CREATION=false
DEFAULT_RACK_CONFIGURATION=NORMAL, GY, 0, 0
BC1=NORMAL, RD, 0, 0
BC2=PEDIATRIC, BL, 10, 8
BC3=NORMAL, 0000, 0, 0
BC4=NORMAL, 0000, 0, 0
BC5=NORMAL, 0000, 0, 0
BC6=NORMAL, 0000, 0, 0
BC7=NORMAL, 0000, 0, 0
...
ARCHIVE_PATH=NONE
ARCHIVE_INTERVAL=NONE
ARCHIVE_ENABLED=false
RACK_BC_ENABLED=true
```


GRIFOLS

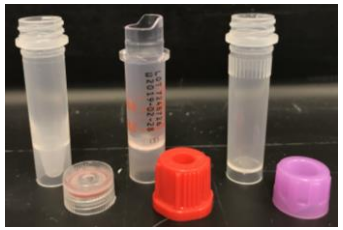
Erytra.properties

Configuration

- **BCx=**1st FIELD, 2nd FIELD, 3rd FIELD, 4th FIELD
  - In order to configure the **sample tubes** contained in a **samples holder**, different configurations are used depending on **barcode labels** attached to the samples holder. It is possible to configure 7 different barcode labels.
  - There are 4 configurable parameters for each barcode label:
    - **1st field**: (Samples tube type) identifies the samples tubes type: NORMAL, PLUNGER, PEDIATRIC.



NORMAL
Use for standard plastic or glass tubes



PEDIATRIC
From 500µl to 1ml pediatric tubes



PLUNGER
For tubes with plunger as a syringe

```
BC1=NORMAL, RD, 0, 0
BC2=PEDIATRIC, BL, 10, 8
BC3=NORMAL, 0000, 0, 0
BC4=NORMAL, 0000, 0, 0
BC5=NORMAL, 0000, 0, 0
BC6=NORMAL, 0000, 0, 0
BC7=NORMAL, 0000, 0, 0
```

Note: x is a number from 1 to 7

GRIFOLS

Erytra.properties

Configuration

- **BCx=**1st FIELD, 2nd FIELD, 3rd FIELD, 4th FIELD
  - **2nd field**: (Holder type) identifies the barcode of each samples holder type. Special barcodes that are already barcoded (GY, RD, BL, YL) can provide. It is a free text field and the default option is 0000.

GY: Grey (as default)



BL: Blue - Pediatric tubes (Ref. 210714)



YL: Yellow



RD: Red



SAMPLES HOLDER >16 mm (Ref. 210715)



To hold sample tubes from 9 to 16,92 mm

Note: Refer to INR-0511#0062 for its configuration.

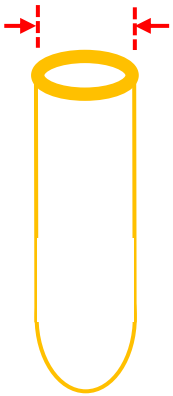
Note: x is a number from 1 to 7

GRIFOLS

Erytra.properties

Configuration

- **BCx=**1st FIELD, 2nd FIELD, 3rd FIELD, 4th FIELD
  - **3rd field**: (Sample tube diameter) external diameter of the samples' tubes.
    - If it is 0 (default option), the system use the presence detector for measuring the external diameter.
    - In any other case, the system will use any value between 9 and 16 mm introduced in this field. Values out of the permitted range can produce unresponsive behaviour in the software.



```
BC1=NORMAL, RD, 0, 0
BC2=PEDIATRIC, BL, 10, 8
BC3=NORMAL, 0000, 0, 0
BC4=NORMAL, 0000, 0, 0
BC5=NORMAL, 0000, 0, 0
BC6=NORMAL, 0000, 0, 0
BC7=NORMAL, 0000, 0, 0
```

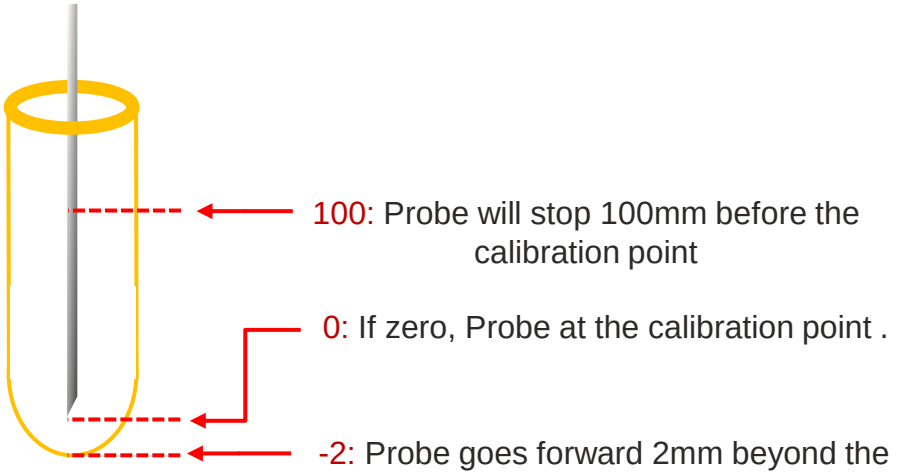
Note: x is a number from 1 to 7

GRIFOLS

Erytra.properties

Configuration

- **BCx=**1st FIELD, 2nd FIELD, 3rd FIELD, 4th FIELD
  - **4th field**: (Samples tube bottom height) indicates the samples' tubes bottom height.
    - It shall indicate the offset in mm from the bottom tube, considering the probe adjustment. Then, the probe does not reach the bottom, as it considers the indicated offset.
    - The default option is 0 and it can be configured as any number between -2 and 100.



Technical manual:

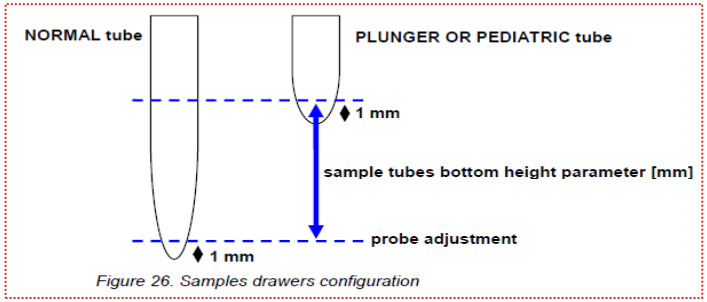


Figure 26. Samples drawers configuration

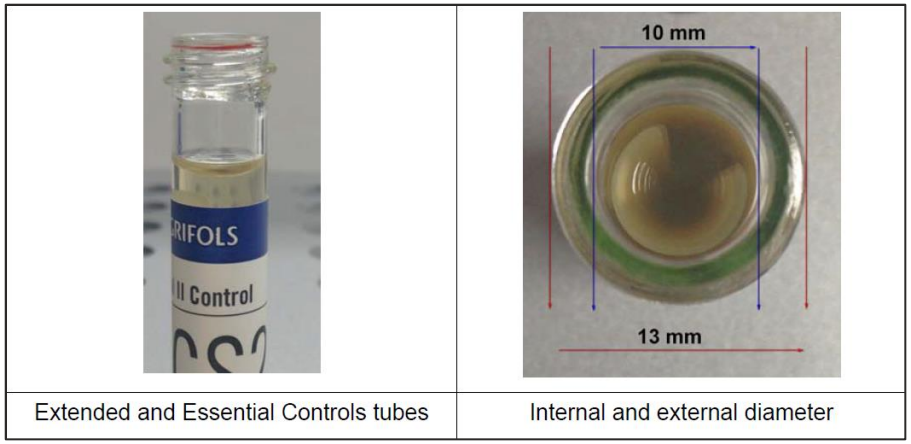
Note: x is a number from 1 to 7

GRIFOLS

Erytra.properties

Configuration

- **BCx=**1st FIELD, 2nd FIELD, 3rd FIELD, 4th FIELD
  - Configuration of Extended IV and Essential II Quality Control in Grifols Analyzers:



Extended and Essential Controls tubes | Internal and external diameter

Example: BC 3 = NORMAL, **YL**, **10**, 0

or

BC 3 = NORMAL, **RD**, **10**, 0

YL = For Yellow Labels

RD = For Red Labels





```
BC1=NORMAL, RD, 0, 0
BC2=PEDIATRIC, BL, 10, 8
BC3=NORMAL, 0000, 0, 0
BC4=NORMAL, 0000, 0, 0
BC5=NORMAL, 0000, 0, 0
BC6=NORMAL, 0000, 0, 0
BC7=NORMAL, 0000, 0, 0
```

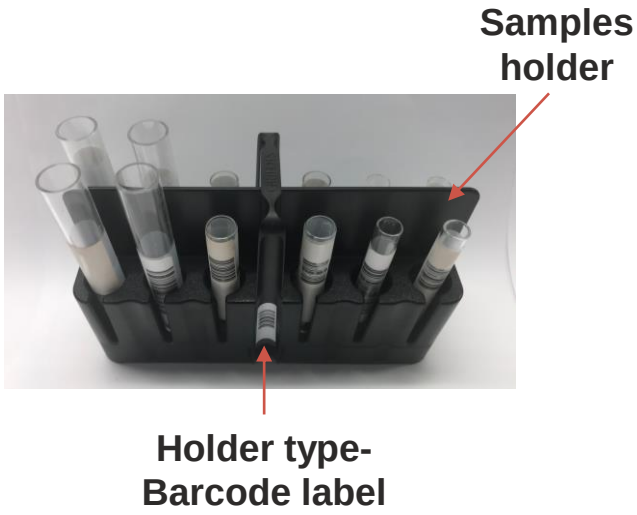
Note: x is a number from 1 to 7

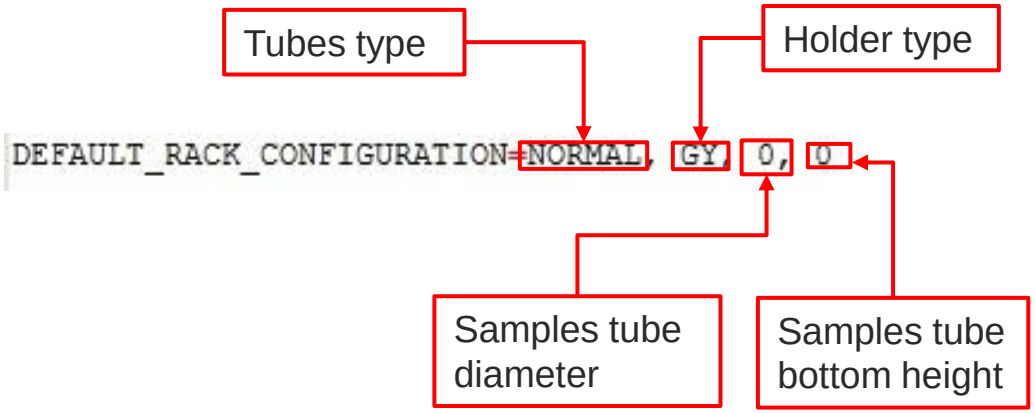
GRIFOLS

Erytra.properties

Configuration

- **DEFAULT_RACK_CONFIGURATION=**NORMAL, GY, 0, 0 (default)
  - Applies to those sample tubes racks that are not codified in any of the BC entrances, despite they have or not barcode.
  - The default option is for normal samples tubes racks, considering the diameter given by the presence detector and the samples tubes' bottom height as 0 mm.





GRIFOLS

Erytra.properties

Configuration

- **ISBT_ENABLED=**false (default)
  - **true:** ISBT 128 DIN barcodes are detected and interpreted.
  - **false:** ISBT 128 DIN barcodes are not detected nor interpreted
- **DEFAULT_BC_IS_ISBT=**false (default)
  - In case of manual entry for ISBT, system will control the number of digits and checksum.
    - **true:** any manually introduced barcode is considered as ISBT type (16 characters expected).
    - **false:** any manually introduced barcode is considered as non-ISBT type.
  - When manually identifying a sample tube barcode, the manual identification window, prompted by the software, includes a checkbox allowing to indicate whether the barcode is ISBT type or not (ticked by default when the DEFAULT_BC_IS_ISBT parameter is enabled and vice versa).

GRIFOLS

Erytra.properties

Configuration

- **ISBT_LIS_DIGITS=**FULL (default)
  - **DIN:** system communication to/from LIS consists of 13-character ISBT codes.
  - **FULL:** system communication to/from LIS consists of 16-character ISBT codes.
  - By enabling the corresponding parameters, ISBT 128 Donation Identification Number (DIN) barcodes can be interpreted by the system and then sent to LIS using the necessary number of characters.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
MANAGER_IP=localhost
ISBT_ENABLED=false
ISBT_LIS_DIGITS=FULL
DEFAULT_BC_IS_ISBT=false
LIS_AUTO_QUERY_BY_SAMPLE=false
FLUIDIC_SYSTEM_ALWAYS_READY=true
AUTOMATIC_RESULTS_VERIFICATION=false
EXTERNAL_CARD_WASTE=false
```

Note: There are special considerations with the barcode reader. Refer to the Erytra Eflexis® Service Manual, “Barcodes reader configuration”.

GRIFOLS

Erytra.properties

Configuration

- **RESERVE_INCUBATOR_FOR_STAT=**true (default)
  - Parameter to configure whether an incubator is reserved for STAT or not.
    - **true:** an incubator is reserved for STAT.
    - **false:** no incubators are reserved for STAT.
  - The software manages automatically incubators availability and that one incubator will be always be reserved and not used for routine samples, although temperature and incubator number can vary.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
IMPACT_DETECTOR=true
RESERVE_INCUBATOR_FOR_STAT=true
MANAGER_IP=localhost
ISBT_ENABLED=false
ISBT_LIS_DIGITS=FULL
DEFAULT_BC_IS_ISBT=false
LIS_AUTO_QUERY_BY_SAMPLE=false
FLUIDIC_SYSTEM_ALWAYS_READY=true
AUTOMATIC_RESULTS_VERIFICATION=false
EXTERNAL_CARD_WASTE=false
```


GRIFOLS

Erytra.properties

Configuration

- **ARCHIVE_ENABLED=**false (default)
  - This option moves the contents of E:\data\Erytra\Microtubes and E:\Log folders to the defined folder in ARCHIVE_PATH parameter.
    - **true:** to enable the archive process.
    - **false:** to disable the archive process.
- **ARCHIVE_PATH=**x\:\\xxx\\xxxx
  - It allows defining the destination folder where logs and microtubes are transferred. The options are any valid path. Note that any "\" symbol shall be introduced as "\\". (java compatibility)

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
ARCHIVE_PATH=NONE
ARCHIVE_INTERVAL=NONE
ARCHIVE_ENABLED=false
RACK_BC_ENABLED=true
IMPACT_DETECTOR=true
RESERVE_INCUBATOR_FOR_STAT=true
```

Note: For more detailed information, refer to the Erytra Eflexis® Service Manual.

GRIFOLS

Erytra.properties

Configuration

- **ARCHIVE_INTERVAL=**NONE (default)
  - Parameter to define the number of days that files to be archived remain in the corresponding folder of the instrument (recommended value ≤ 14).
  - After the number of days defined (checked by the software every hour), the files are moved to the folder defined in archive_path parameter. If the parameter is defined as 0, the files are continuously moved to the corresponding folder. Negative values are not allowed.
  - It is a free text field.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
ARCHIVE_PATH=NONE
ARCHIVE_INTERVAL=NONE
ARCHIVE_ENABLED=false
RACK_BC_ENABLED=true
IMPACT_DETECTOR=true
RESERVE_INCUBATOR_FOR_STAT=true
```

Note: For more detailed information, refer to the Erytra Eflexis® Service Manual.

GRIFOLS

Erytra.properties

Configuration

- **QC_ENABLED=**false (default)
  - Activation of QC policy
    - **true:** the QC is activated.
    - **false:** the QC is not activated
- **QC_HOURS_CADENCE=**8 (default time)
  - The periodicity in hours to request the Quality Control.
  - Once the QC is validated, it is ok to process samples for the number of hours defined in the value. After this time, if the operator tries to process some routine samples, they can face different situations:
    - If QC tubes are on board, the instrument will stop the routine samples and start automatically the QC.
    - If QC tubes are not on board, the QC warning will be activated and the operator has to decide if they want to process the QC or force the system to process samples (with QC not passed icon displayed).
  - Setting the value to 0 is not allowed.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
QC_ENABLED=false
QC_HOURS_CADENCE=8
QC_NEW_LOT_WARNING_ENABLED=false
...
ENABLE_AUTOVERIFICATION_QCPASS=NONE
```

Note: These parameters can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **QC_NEW_LOT_WARNING_ENABLED=**false (default)
  - Display a popup warning when a new reagent lot is introduced in the instrument with no QC processed for this reagent. Valid also for card lots.
    - **true:** popup warning is displayed
    - **false:** no popup warning




```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
QC_ENABLED=false
QC_HOURS_CADENCE=8
QC_NEW_LOT_WARNING_ENABLED=false
...
ENABLE_AUTOVERIFICATION_QCPASS=NONE
```

  - The message is only a warning, QC will not be processed automatically by the instrument.

GRIFOLS

Erytra.properties

Configuration

- **ENABLE_AUTOVERIFICATION_QCPASS=**NONE (default)
  - Auto-validation and export of QC when passed.
    - **NONE:** no automatic validation so no automatic export. The operator has to validate QC results and then export them.
    - **ONLY_VALIDATE:** All the QC results are auto-validated if they are passed and there are no special results or incidences. The operator has to export them.
    - **VALIDATE_AND_EXPORT:** All the QC results are auto-validated and automatically exported (if QC results are all passed and with no special results or incidences).

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
QC_ENABLED=false
QC_HOURS_CADENCE=8
QC_NEW_LOT_WARNING_ENABLED=false
...
ENABLE_AUTOVERIFICATION_QCPASS=NONE
```

Note: This parameter can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **DISABLE_VALIDATION_BUTTON=**false (default)
  - Enables or disables the **Validation** button in the Results List.
    - **true:** Validation button is disabled. This option forces the operator to validate the samples one by one (inside Profile information screen).
    - **false:** to have this button enabled on the Results List.
- **DISABLE_EXPORT_BUTTON=**false (default)
  - Enables or disables the **Export** button in the Results List
    - **true:** Export button is disabled. This option forces the operator to export the samples one by one (inside Profile information screen).
    - **false:** to have this button enabled.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
PASSWORD_REAUTHENTICATION=true
DISABLE_VALIDATION_BUTTON=false
DISABLE_EXPORT_BUTTON=false
DISABLE_CHANGE_RESULTS_CONFIRMATION=true
```


GRIFOLS

Erytra.properties

Configuration

- **DISABLE_CHANGE_RESULTS_CONFIRMATION=**true (default)
  - Configures the system to display a confirmation message whenever the user modifies any result.
    - **true:** do not request the confirmation messages.
    - **false:** request the confirmation messages (no password required).

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
PASSWORD_REAUTHENTICATION=true
DISABLE_VALIDATION_BUTTON=false
DISABLE_EXPORT_BUTTON=false
DISABLE_CHANGE_RESULTS_CONFIRMATION=true
```


GRIFOLS

Erytra.properties

Configuration

- **PASSWORD_REAUTHENTICATION=**true (default)
  - It allows enabling or disabling of the **identification request** (user/password) when any of the relative **actions configured** are performed.
    - **true:** to enable this password re-identification.
    - **false:** to disable this password re-identification.
  - It applies to following actions:
    - Acceptance of special results
    - Results' modification
    - Ignore incidences
    - Processing samples with a QC FAIL
    - Validating results with a QC FAIL

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
PASSWORD_REAUTHENTICATION=true
DISABLE_VALIDATION_BUTTON=false
DISABLE_EXPORT_BUTTON=false
```


GRIFOLS

Erytra.properties

Configuration

- **PASSWORD_EXPIRATION_DAYS**=90 (default)
  - It allows defining the password expiration interval in days. It is a free integer number and the default option is 90 days.
  - This parameter only applies to the new users created and not to those provided as default.
  - There is no limitation in the number (ex: 10000 → more than 27 years).
  - Expiration Date = Current date + number of days configured in the properties.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
DUAL_MODE=false
PASSWORD_EXPIRATION_DAYS=90
DELTA_CHECK_HISTORICAL_FROM=0
```


GRIFOLS

Erytra.properties

Configuration

- **SHOW_CONFIRMATION_WHEN_VALIDATE_MODIFIED_SAMPLEPROFILES=**true (default)
  - Whenever the user validates a modified result, the system shall request a username and password of the user.
    - **true:** to enable this confirmation and identification message. This is the default option.
    - **false:** to disable this confirmation and identification message.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
ENABLE_AUTOVERIFICATION_QCPASS=NONE
SHOW_CONFIRMATION_WHEN_VALIDATE_MODIFIED_SAMPLEPROFILES=true
```


GRIFOLS

Erytra.properties

Configuration

- **AUTOMATIC_RESULTS_VERIFICATION=**false (default)
  - Auto-validation and auto-export for samples.
    - **true:** auto-validation and auto-export is activated**.**
    - **false:** No auto-validation and auto-export, results have to be validated and exported by the operator.
  - Auto validation and export is done only if:
    - Related QC has been exported
    - There is no incidences
    - There is no special results
  - Samples will be exported in groups, not sample by sample.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
LIS_AUTO_QUERY_BY_SAMPLE=false
FLUIDIC_SYSTEM_ALWAYS_READY=true
AUTOMATIC_RESULTS_VERIFICATION=false
EXTERNAL_CARD_WASTE=false
```

Note: This parameter can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **DELTA_CHECK_INCONSISTENCIES_CONFIGURATION=** NONE (default)
  - Allows enabling or not historical checking for previous results and warning in case of discrepancies.
    - **NONE:** no inconsistencies are checked.
    - **SAMPLE:** Inconsistencies are checked on the basis of the sample tube barcode.
    - **PATIENT:** Inconsistencies are checked on the basis of the Patient ID (PID)

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
PASSWORD_EXPIRATION_DAYS=90
DELTA_CHECK_HISTORICAL_FROM=0
DELTA_CHECK_INCONSISTENCIES_CONFIGURATION=NONE
ENABLE_AUTOVERIFICATION_QCPASS=NONE
```

Note: This parameter can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **DELTA_CHECK_HISTORICAL_FROM**=0 (default)
  - Defines the interval of time within inconsistencies are checked in the exported results (database).
    - **ALL:** Inconsistencies are checked within all the data stored in the database.
    - **YYYY/MM/DD:** Inconsistencies are checked within all results from the defined date.
    - **Any number:** Inconsistencies are checked within the defined number of days prior the current date.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
PASSWORD_EXPIRATION_DAYS=90
DELTA_CHECK_HISTORICAL_FROM=0
DELTA_CHECK_INCONSISTENCIES_CONFIGURATION=NONE
ENABLE_AUTOVERIFICATION_QCPASS=NONE
```

Note: This parameter can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **FLUIDIC_SYSTEM_ALWAYS_READY=**true (default)
  - Inactivity wash performed instead of inactivity rinse.
    - **true:** inactivity wash will be performed.
    - **false:** inactivity rinse will be performed.
  - Allows keeping instrument primed at every hour of inactivity.
  - Inactivity wash will be performed every hour if instrument is not working and then, after 24 hours of inactivity, a rinse will be performed.
  - After inactivity wash, the instrument performs a small probe prime (1’55) as part of the wash process.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
DEFAULT_BC_IS_ISBT=false
LIS_AUTO_QUERY_BY_SAMPLE=false
FLUIDIC_SYSTEM_ALWAYS_READY=true
AUTOMATIC_RESULTS_VERIFICATION=false
EXTERNAL_CARD_WASTE=false
```


GRIFOLS

Erytra.properties

Configuration

- **DUAL_MODE=**false (default)
  - It allows working with external drain for the liquid waste and using Waste 1 and Waste 2 positions as additional Solution A and Solution B positions (DUAL_MODE).
    - **true:** to work in a dual mode, with the external drain and doubling the capacity of System Solutions A & B. This configuration requires a change in the manifold of the fluidic system.
    - **false:** to work in a mono mode.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
DISABLE_CHANGE_RESULTS_CONFIRMATION=true
DUAL_MODE=false
PASSWORD_EXPIRATION_DAYS=90
DELTA_CHECK_HISTORICAL_FROM=0
```

Note: Do not forget to modify the tab position in the load cells supports for waste 1 and waste 2 positions.
Note 2: Fluidic decontamination must be performed prior to changing from mono-mode to dual mode.

GRIFOLS

Erytra.properties

Configuration

- **Direct waste liquid drainage and dual mode**
  - It is possible to drain the liquid waste directly to the lab drain by adapting the fluidic circuit of the instrument (requires modifying the fluidic circuit).
  - This configuration allows working in a **dual mode** for System Solutions, doubling the capacity and allowing random access for System Solutions A & B, by using the waste containers position as a position for System Solutions.

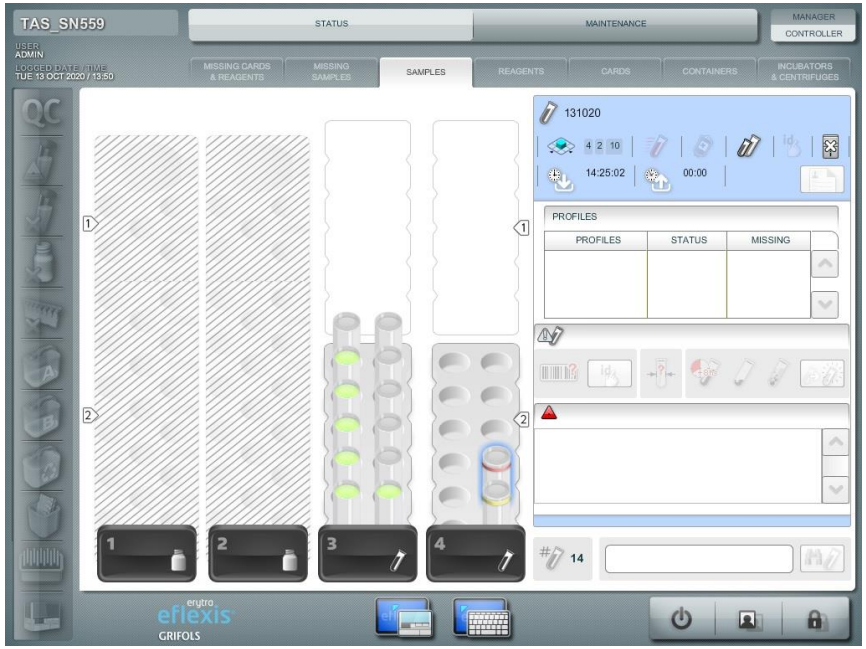


GRIFOLS

Erytra.properties

Configuration

- **DOUBLE_TUBE_ENABLED=**false (default)
  - Allows defining if double tube can be used in the system or not.
    - **true:** to enable the use of double tube. One tube for plasma/serum and one for RBCs.
    - **false:** to disable the use of double tube.




```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
RESULT_CONFIGURATION=SRW,SRDL,SRDP
DOUBLE_TUBE_ENABLED=false
QC_ENABLED=false
QC_HOURS_CADENCE=8
QC_NEW_LOT_WARNING_ENABLED=false
```

Note: This parameter can also be defined by means of Importer application.

GRIFOLS

Erytra.properties

Configuration

- **IMPACT_DETECTOR=**true (default)
  - It defines if the system works with impact detector or not (IMPACT_DETECTOR).
    - **true:** to work with the impact detector installed. It requires a change in the hardware of the PIP arm.
    - **false:** to not work with the impact detector installed. This option is not for Erytra Eflexis® as all units already contain this impact detector.
  - Erytra Eflexis® always has included this device. This functionally is included for compatibility with old Erytra® units.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
...
RACK_BC_ENABLED=true
IMPACT_DETECTOR=true
RESERVE_INCUBATOR_FOR_STAT=true
MANAGER_IP=localhost
ISBT_ENABLED=false
ISBT_LIS_DIGITS=FULL
DEFAULT_BC_IS_ISBT=false
LIS_AUTO_QUERY_BY_SAMPLE=false
FLUIDIC_SYSTEM_ALWAYS_READY=true
AUTOMATIC_RESULTS_VERIFICATION=false
EXTERNAL_CARD_WASTE=false
```


GRIFOLS

Erytra.properties

Configuration

- **SERVICE_RACK_MODE=**NORMAL (default)
  - Parameter to configure the card drawer rack position for use as service racks.
  - Service rack is a special card holder that contains cards with special results, incidences and partially used cards.
    - **NORMAL:** There are 2 service racks in position rack 4 of the card drawer 1 and rack 4 of the card drawer 2.
    - **HALF:** All racks in card drawer 2 are used as service rack.
    - **NONE:** No service rack is defined: all the cards are discarded directly in the waste cards container (also partially used cards!).
    - **SINGLE:** There is only 1 service rack in position rack 4 Card Drawer 2.

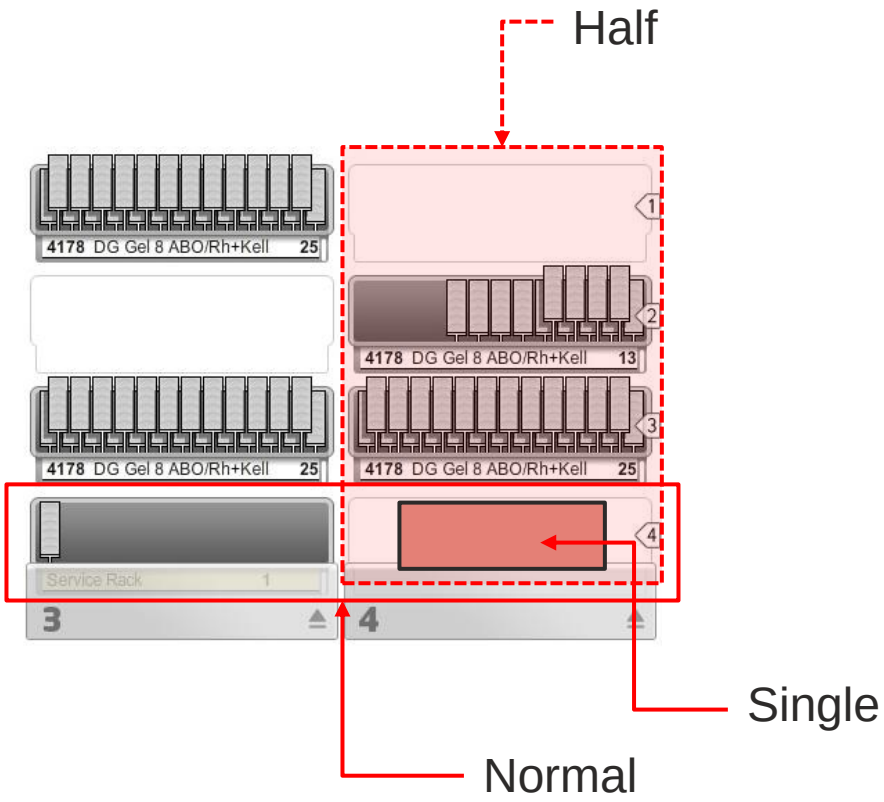
```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
INSTRUMENT_NAME=
INSTRUMENT_SERIAL=
WELL NAME FORMAT=TRADITIONAL
```


GRIFOLS

Erytra.properties

Configuration

- **SERVICE_RACK_MODE=**NORMAL (default)



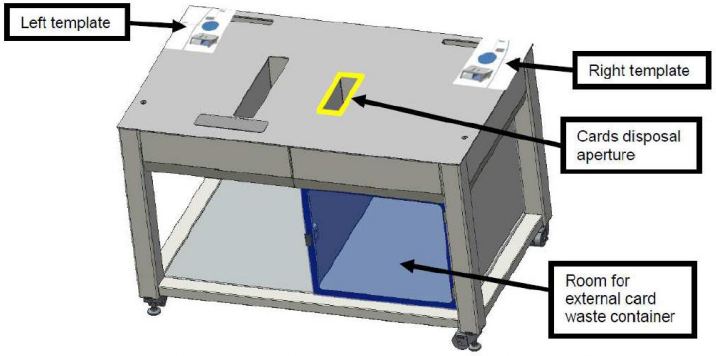
GRIFOLS

Erytra.properties

Configuration

- **EXTERNAL_CARD_WASTE=**false (default)
  - Dispose cards in an external card waste container. This feature requires software and hardware configuration.
    - **true**: external card waste is used. Card waste tray is not displayed in user interface and card waste level warning and card waste container presence warning are disabled.
    - **false**: Card waste tray/internal card waste is used.





Installation templates and cards disposal system in Grifols Bench.

```
#Application Properties
#Wed Sep 30 08:06:43 CEST 2020
SERVICE_RACK_MODE=NORMAL
LANGUAGE=en_GB
...
LIS_AUTO_QUERY_BY_SAMPLE=false
FLUIDIC_SYSTEM_ALWAYS_READY=true
AUTOMATIC_RESULTS_VERIFICATION=false
EXTERNAL_CARD_WASTE=false
```


GRIFOLS

Erytra.properties

Configuration

- **MANAGER_IP=**localhost (default)
  - Parameter to configure the IP address of the PC with the Manager software. Default value: localhost.

Note: This functionality is not available. It is not possible to install it on a different computer at the moment.

Note: This parameter can also be defined by means of Importer application.

GRIFOLS

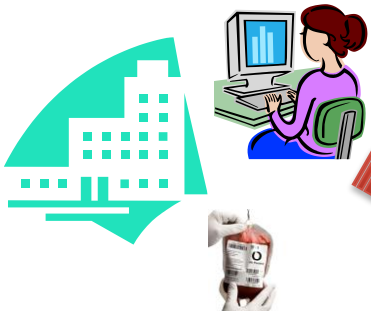
LIS Connection

GRIFOLS

LIS

Introduction

- LIS is 3rd party software that receives, stores, and manages data for laboratories



Test order

Test orders are logged into the hospital LIS

LIS

The operator queries the LIS to receive the pending work. Once downloaded, the worksheet is automatically populated

worksheet

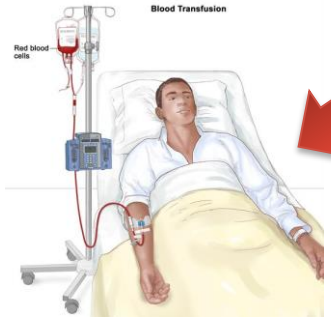


Test Results



After testing, the operator sends the results to the LIS

Transfusion order

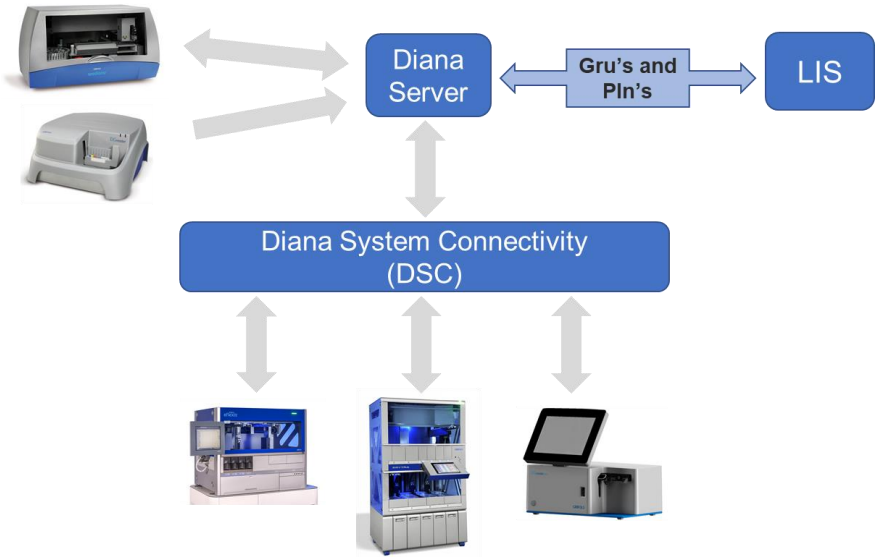


GRIFOLS

LIS

Introduction

- The **Diana Server** software **manages the information between LIS and all IH instruments.**
- The communication between the **Diana Server** and **Erytra Eflexis®** is **not direct**, it requires another software called **DSC (Diana System Connectivity).**

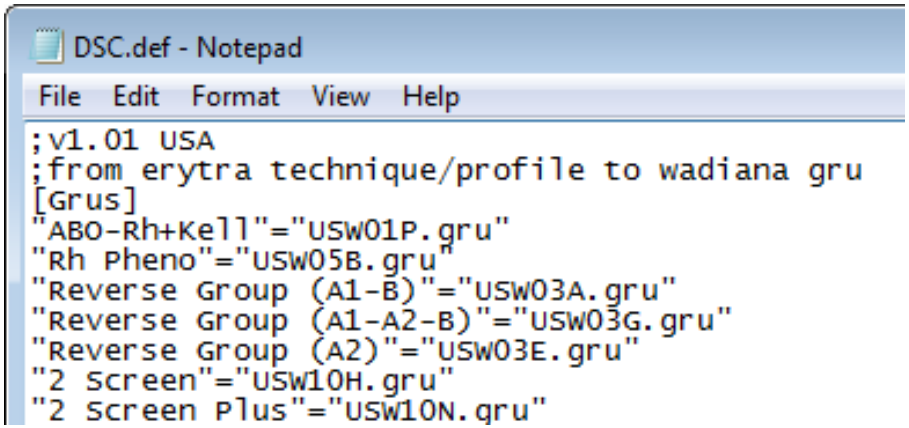


GRIFOLS

LIS

Introduction

- **What is the DSC – Diana System Connectivity?**
  - **DSC** Erytra Eflexis® software section that translates/links between them and **Diana Server** (*.gru and *.pln).
  - **Diana Server** uses (*.gru and *.pln) for communicating with **LIS**.



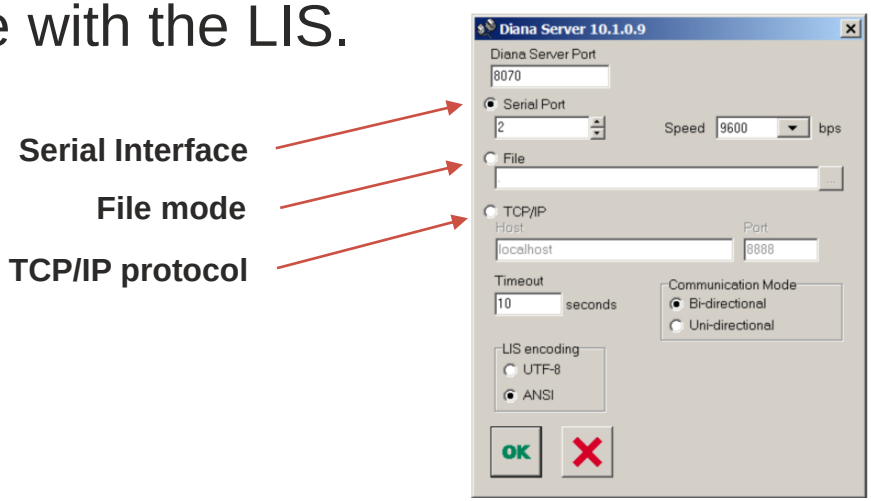
LIS

Diana Server Software

- LIS communicates with the Diana Server software using an international protocol called ASTM.



- All ASTM information is collected in a log file called: DianaASTM.log located in E:\data\Sistema Diana\Log
- The instruments have three different Physical channels to communicate with the LIS.

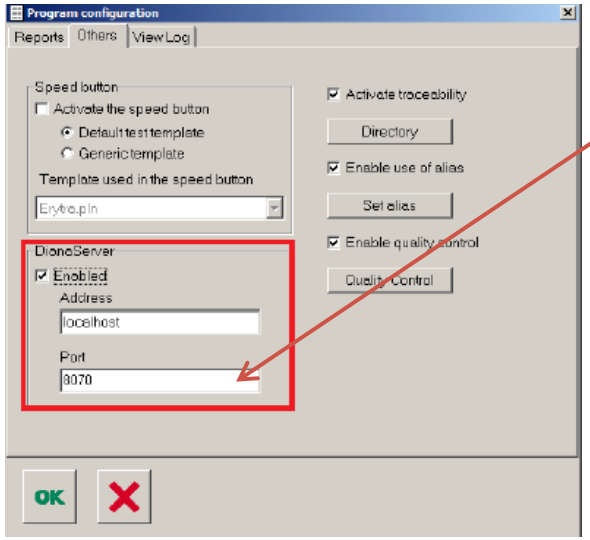


GRIFOLS

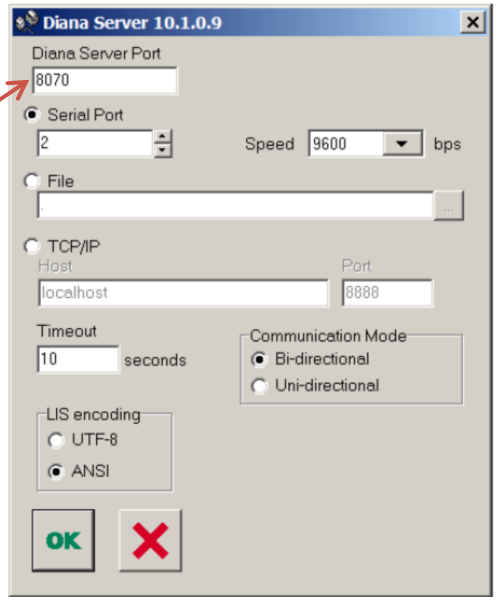
LIS

Diana Server Software

- Diana server has been designed as the Erytra Eflexis® software based on web technology.
- The way to communicate each other is using the protocol https.
- It is possible to configure the IP address and port of the Diana Server modifying these parameter in Diana O (Listas.exe) and Diana Server software.



Diana O (Listas.exe)



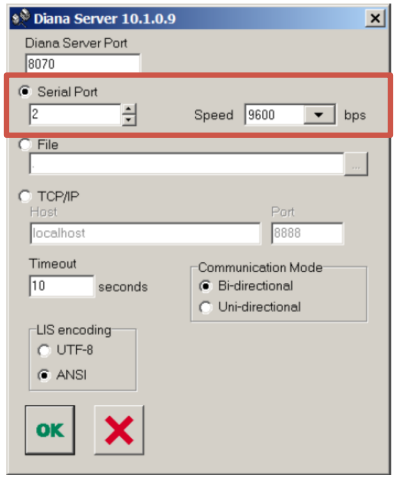
Diana Server

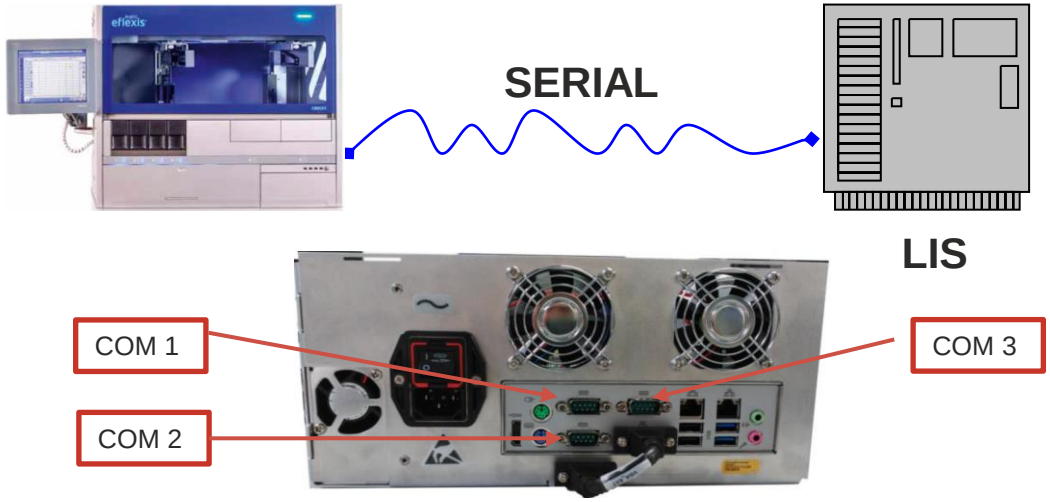
GRIFOLS

LIS

Serial port/cable connection

- Connects Erytra Eflexis® to the LIS/LIS Middleware trough the Serial Port of Erytra Eflexis® PC that is connected to the LIS.
- Serial Port is 1,2,3 depending of the physical port used.
- Serial to Ethernet or Serial to USB converter can be used.
- Messages are chopped into numbered frames to keep flow control and the acknowledgements (handshaking) are used to control transmissions.

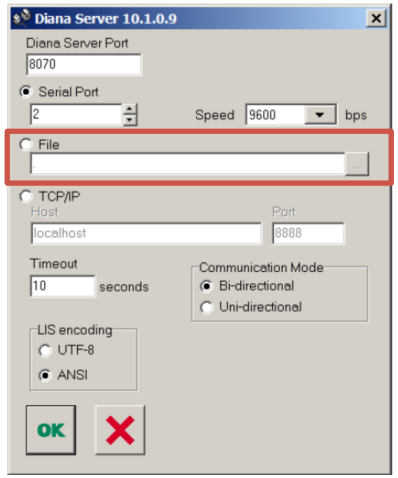


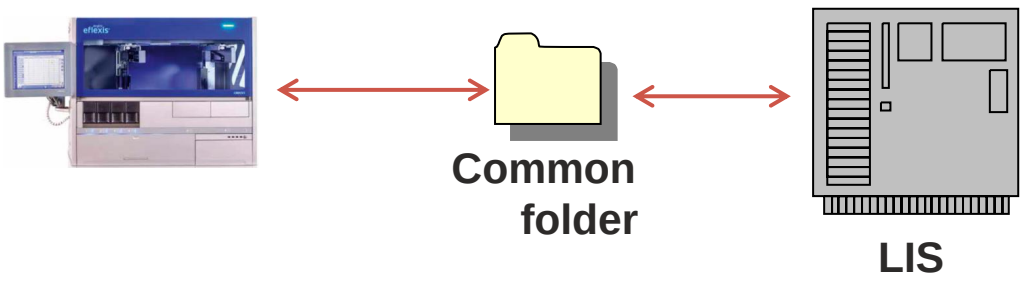


LIS

File sharing connection

- Files are written to/from a specified common folder.
- The exchange common folder should be unique.
  - **Diana Server** starts writing the transmission to a temporary file named TX_TMP.dat. Once all the information is written TX_TMP.dat, it is renamed to **OUT.dat**
  - **LIS** transmits data to the common folder in a file called **IN.dat**



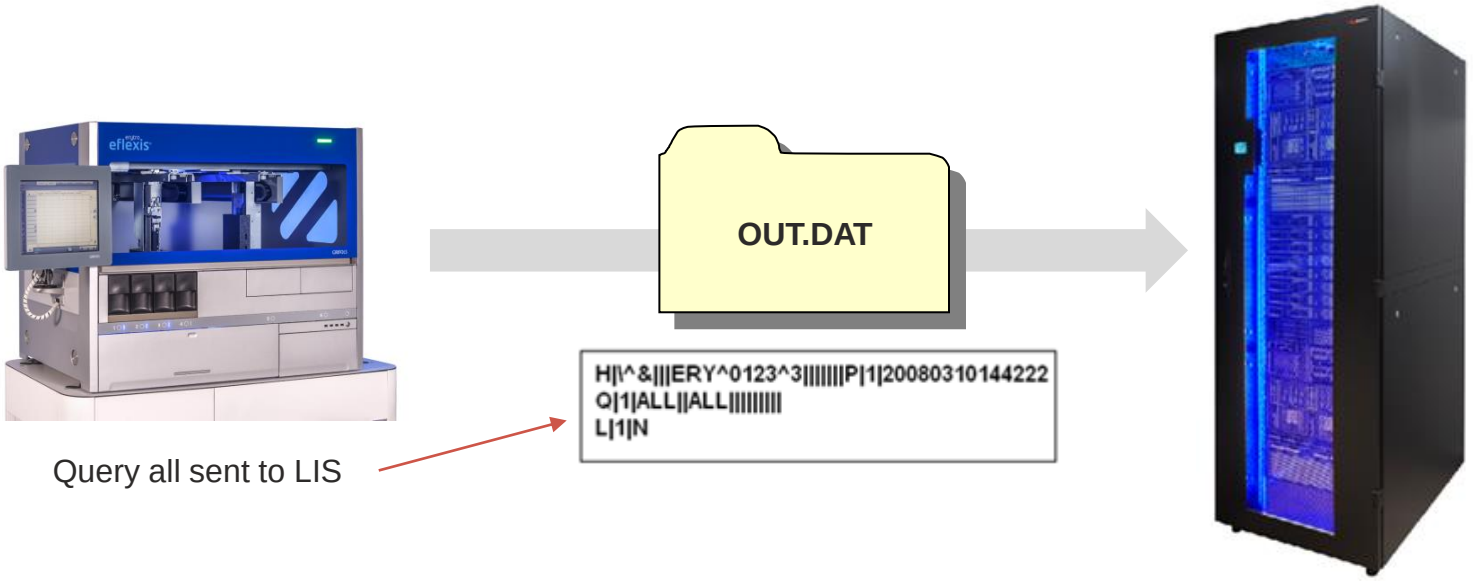


GRIFOLS

LIS

File sharing connection

– The analyser starts a communication process through DianaServer.
– File **OUT.DAT** is generated in the shared folder.



LIS

File sharing connection

– LIS deletes **OUT.DAT** when received. If not deleted within a defined timeout period, error messages are generated.



GRIFOLS

LIS

File sharing connection

– Requests Download (Query Worklist from LIS). LIS generates a new file called IN.DAT in the shared folder.





IN.DAT

Query worklist from LIS

```
H|\^&|||ERY^0123^3|||||||P|1|20080310144222
P|1|123456|||Dummy^Patient||20001231|M|
O|1|123456||^^^TEST1.GRU^TEMPLATE.PLN
L|1|N
```


GRIFOLS

LIS

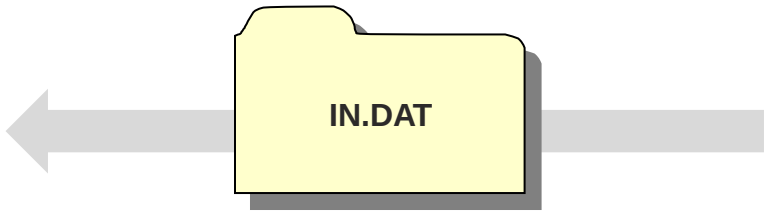
File sharing connection

- DianaServer detects the IN.DAT file and deletes it.
- DianaServer send the information of the IN.DAT to the instrument (DSC).
- If the file is not deleted within a TimeOut time, the file is deleted and an Error message is generated.



(DSC)

ABO-Rh = TEST1.gru




```
H|\^&|||ERY^0123^3||||||P|1|20080310144222
P|1|123456|||Dummy^Patient||20001231|M|
O|1|123456||^^^TEST1.GRU^TEMPLATE.PLN
L|1|N
```



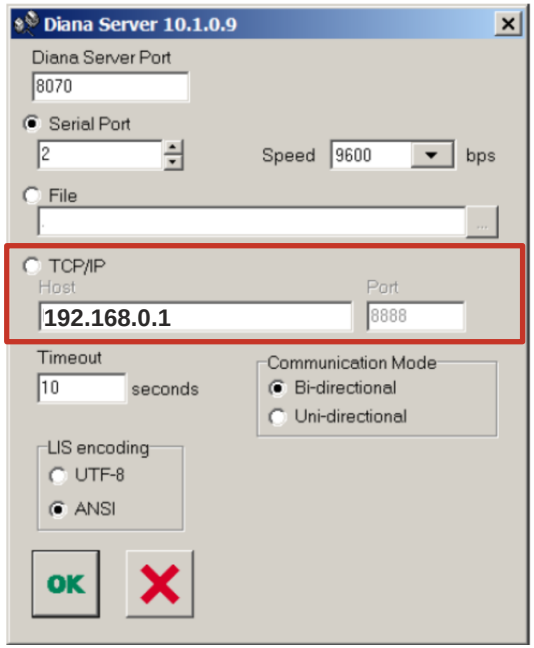
GRIFOLS

LIS

TCP/IP connection

- DianaServer sends data to LIS using TCP/IP protocol
- LIS IP Address is configured in DianaServer.
  - For example:
    - IP LIS is 192.168.0.1
    - Port: By default uses 8888





GRIFOLS

LIS

LIS configuration

- Sequence of enabling and configuring the LIS system:
  1. Switch on and select MAINTENANCE MODE.
  2. Unprotected the system by using Applocker bu.
  3. Edit the erytra.properties
  4. Set **LIS_ACTIVATED** parameter to True.
  5. Set **LIS_SHARE_MODE** according with customer preferences.
  6. Set **RECEIVED_AND_SHOW_DEMOGRAPHIC_DATA** (CE version) according with customer preferences.
  7. Set **LIS_AUTO_QUERY_BY_SAMPLE** according with customer preferences.
  8. To transmit the card barcodes and reagents to the LIS, modify Grifols Windows registry in **HKEY_LOCAL_MACHINE**, parameter SendBCReacSupp = 1.

Note: For further information, refer to the Erytra Eflexis® Service Manual or INR/FSB.

GRIFOLS

LIS

LIS configuration

9. Enable Diana Server.
10. **Configure the Diana Server** according with the installation requirements
11. Configure the **Diana Server** to **automatically start** at PC startup.
12. **Enable Server and TCP/IP Netbios Helper service** at “windows services”. To enable File and printer sharing feature and the access to a shared folder in a remote computer feature.
13. Protected the system by using Applocker eu.
14. **Reboot** the instrument.

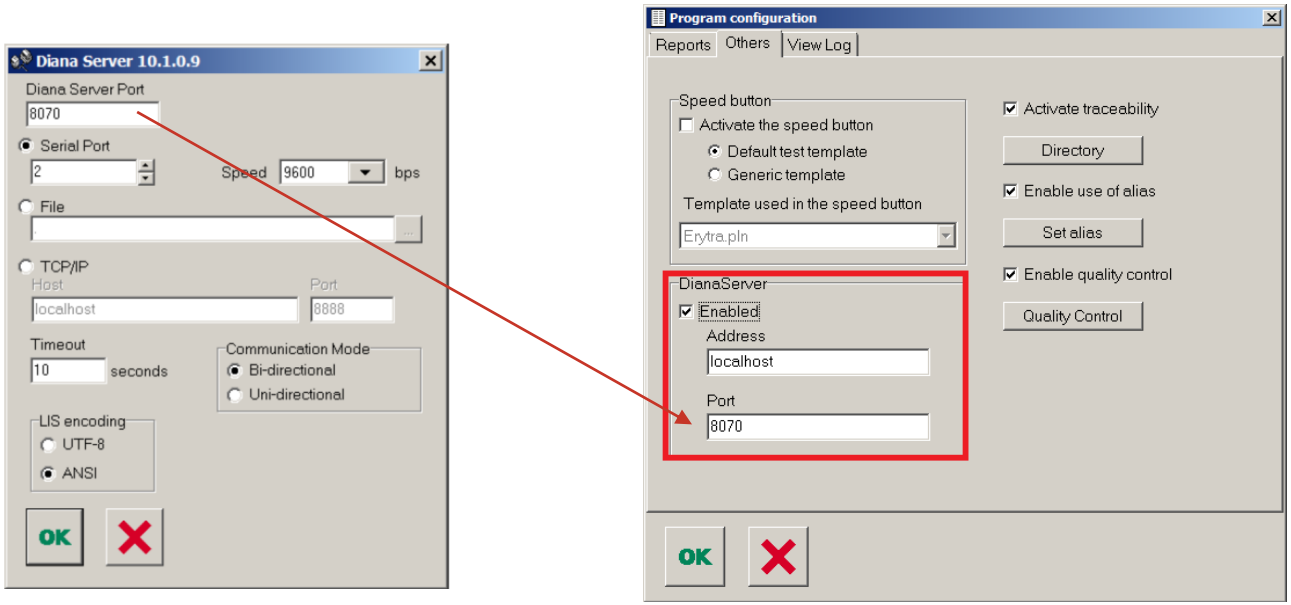
Note: For further information, refer to the Erytra Eflexis® Service Manual or INR/FSB.

GRIFOLS

LIS

LIS configuration: 9. Enable the Diana Server

- Run **Listas.exe** as administrator in E:\Data\Sistema Diana folder.
- Enable Diana Server by ticking checkbox in Others tab.
  - Communication port must match between Diana O and Diana Server.
  - The modification of the Address and Port is an advanced function. Do not modify.



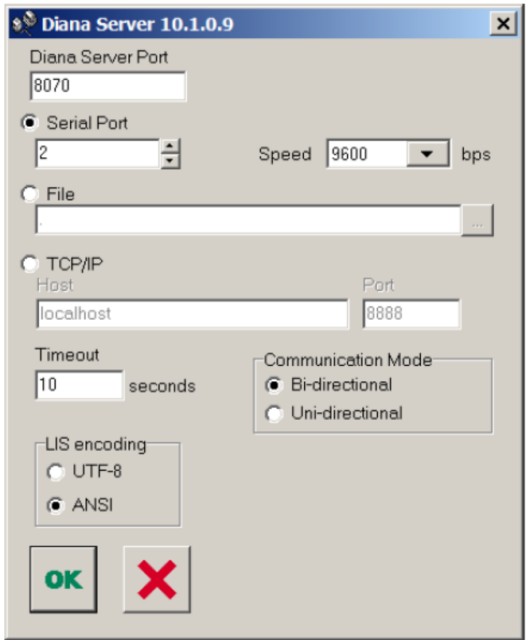
Note: For further information, refer to the Erytra Eflexis® Service Manual or INR/FSB.

GRIFOLS

LIS

LIS configuration: 10. Configure the Diana Server

- **Diana Server Port:** Verify that the port is the same in Diana O (Listas.exe).
- **Communication mode:**
  - Working in **bi-directional mode**, 2 types of Query (not configurable from Diana Server):
    - **Query All:** request for ALL the samples in LIS.
    - **Query by Sample:** individual request per sample present in the instrument.
  - Receiving workload from LIS, tests will start automatically.
  - If sent by LIS, demographic data (Name, Family name, gender and date of birth) are stored in the system and can be displayed in the reports (under configuration). Only for CE market.



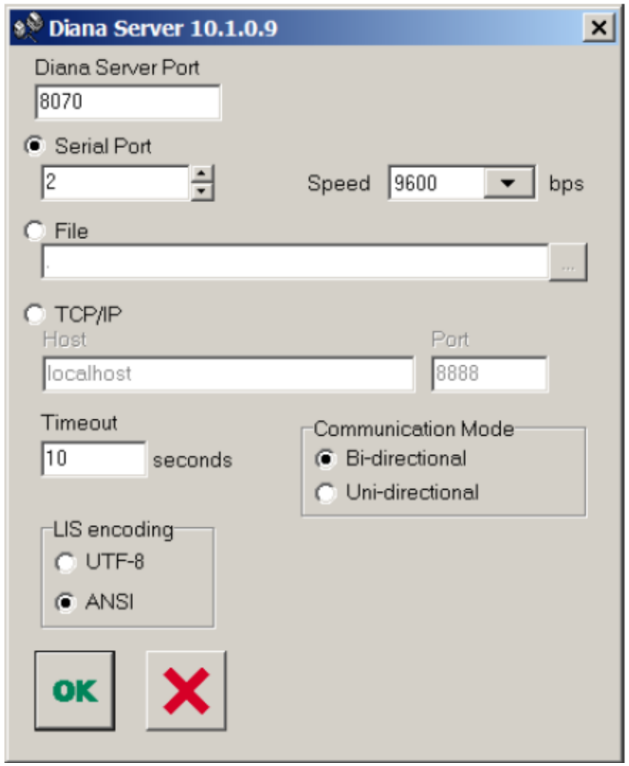
Note: For further information, refer to the Erytra Eflexis® Service Manual or INR/FSB.

GRIFOLS

LIS

LIS configuration: 10. Configure the Diana Server

- **Timeout:** Time that the Diana Server software use to verify that the transmission or reception is performed. After this time, an error is generated.
- **LIS Encoding:** Is an advanced configuration.
  - **UTF-8** is the configuration by default and widely used on the fields .
  - **ANSI** encoding is used for communicating with Host Demo.



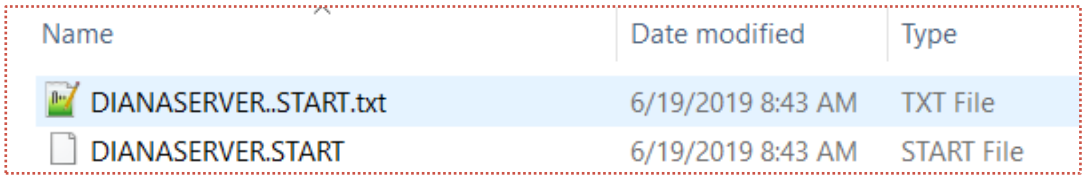
Note: For further information, refer to the Erytra Eflexis® Service Manual or INR/FSB.

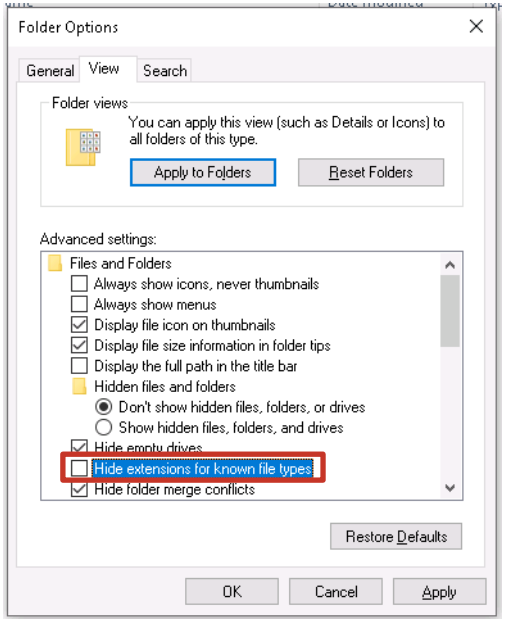
GRIFOLS

LIS

LIS configuration: 11. Automatic start of Diana Server

- Go to E:/Program Files/Startup.
- Go to View and select options.
- Deselect the option “Hide extensions for known file types”.
- Create a file called ***DIANASERVER.START*** **in *E:\Program files\Startup* folder** (not ~~**DIANASERVER.START.txt**~~).





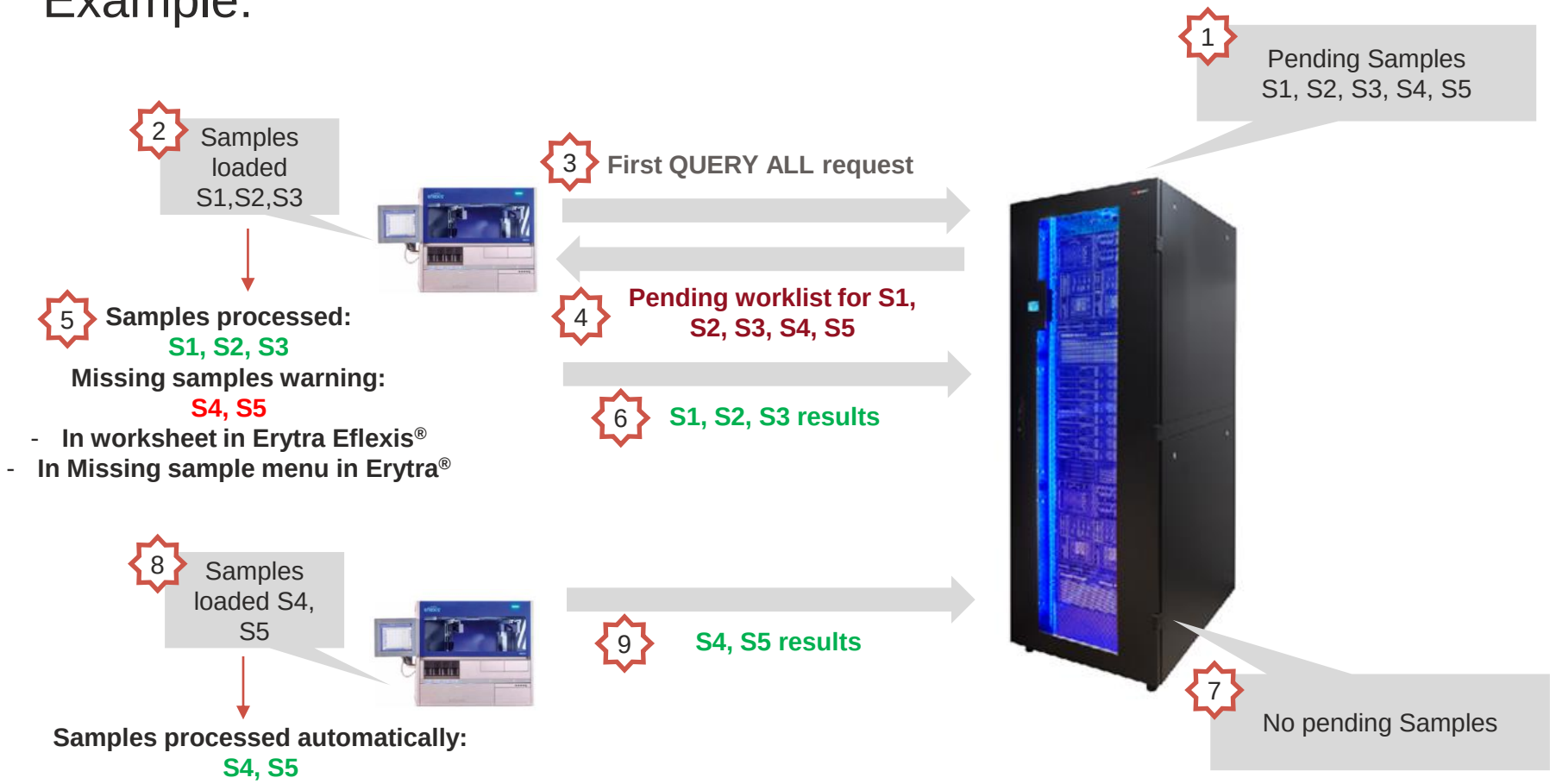
Note: For further information, refer to the Erytra Eflexis® Service Manual or INR/FSB.

GRIFOLS

LIS

LIS Share Mode false

Example:



Note: The above functionality explained is as expected. Other middleware or LIS drivers may run with another features.

GRIFOLS

LIS

LIS Share Mode false

Example (ASTM.log):

1 Pending Samples
S1, S2, S3, S4, S5

3 **First QUERY ALL request**

```
Send: H|\^&|||WADIANA^0000880^ERY||||||P|1|20170614180920
Send: Q|1|ALL||ALL||||||||I|
Send: L|1|N
```

2 Samples loaded
S1,S2,S3



Pending worklist for S1, S2, S3, S4, S5



4

```
OnPetitionEvent: W:0000880:ERY
Receive: H|\^&|||WADIANA^0000880^ERY||||||P|1|20170614180920
Receive: P|1|105116|||King^Alan||19970208|M|
Receive: O|1|S1||^^^DGW01A.gru^Erytra.pln|||||||||||
Receive: P|2|148135|||Anderson^Maika||19660610|M|
Receive: O|1|S2||^^^DGW01A.gru^Erytra.pln|||||||||||
Receive: P|3|178137|||Spoon^Logan||19410905|M|
Receive: O|1|S3||^^^DGW05B.gru^Erytra.pln|||||||||||
Receive: P|4|116997|||Cool^Jackson||19650315|M|
Receive: O|1|S4||^^^DGW01A.gru^Erytra.pln|||||||||||
Receive: P|5|172182|||Pierce^Layla||19761016|F|
Receive: O|1|S5||^^^DGW05B.gru^Erytra.pln|||||||||||
Receive: L|1|N
```

5 **Samples processed:**
S1, S2, S3
Missing samples:
S4, S5

S1, S2, S3 results

6

```
Send: H|\^&|||WADIANA^0000880^ERY||||||P|1|20170920141146
Send: P|1|105116|||King^Alan||19970208|M|
Send: O|1|S1||^^^DGW01A.gru^Erytra.pln|R|20170920133750|||||||#21^031581200145^71760118001018264 9|||||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW01A.gru^Erytra.pln|A^Neg^|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920133750|20170920135412|WADIANA^0000880^ERY|
Send: P|2|148135|||Anderson^Maika||19660610|M|
Send: O|1|S2||^^^DGW01A.gru^Erytra.pln|R|20170920133830|||||||#20^031581200145^021072780553^021172780552^711101180020107770|||||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW01A.gru^Erytra.pln|A^Pos^|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920133830|20170920135455|WADIANA^0000880^ERY|
Send: P|3|178137|||Spoon^Logan||19410905|M|
Send: O|1|S3||^^^DGW05B.gru^Erytra.pln|R|20170920133750|||||||#21^031581200145^71760118001018264 9|||||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW05B.gru^Erytra.pln|^^CEce|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920133750|20170920135412|WADIANA^0000880^ERY|
Send: P|4|116997|||Cool^Jackson||19650315|M|
Send: O|1|S4||^^^DGW01A.gru^Erytra.pln|R|20170920133813|||||||#20^031581200145^021072780553^021172780552^711101180020107772|||||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW01A.gru^Erytra.pln|0^Pos^|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920133813|20170920135434|WADIANA^0000880^ERY|
Send: P|5|172182|||Pierce^Layla||19761016|F|
Send: O|1|S5||^^^DGW05B.gru^Erytra.pln|R|20170920134219|||||||#21^031581200145^71760118001018264 3|||||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW05B.gru^Erytra.pln|^^cece|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920134219|20170920135545|WADIANA^0000880^ERY|
Send: L|1|N
OnPetitionEvent: R:000:ERY
```

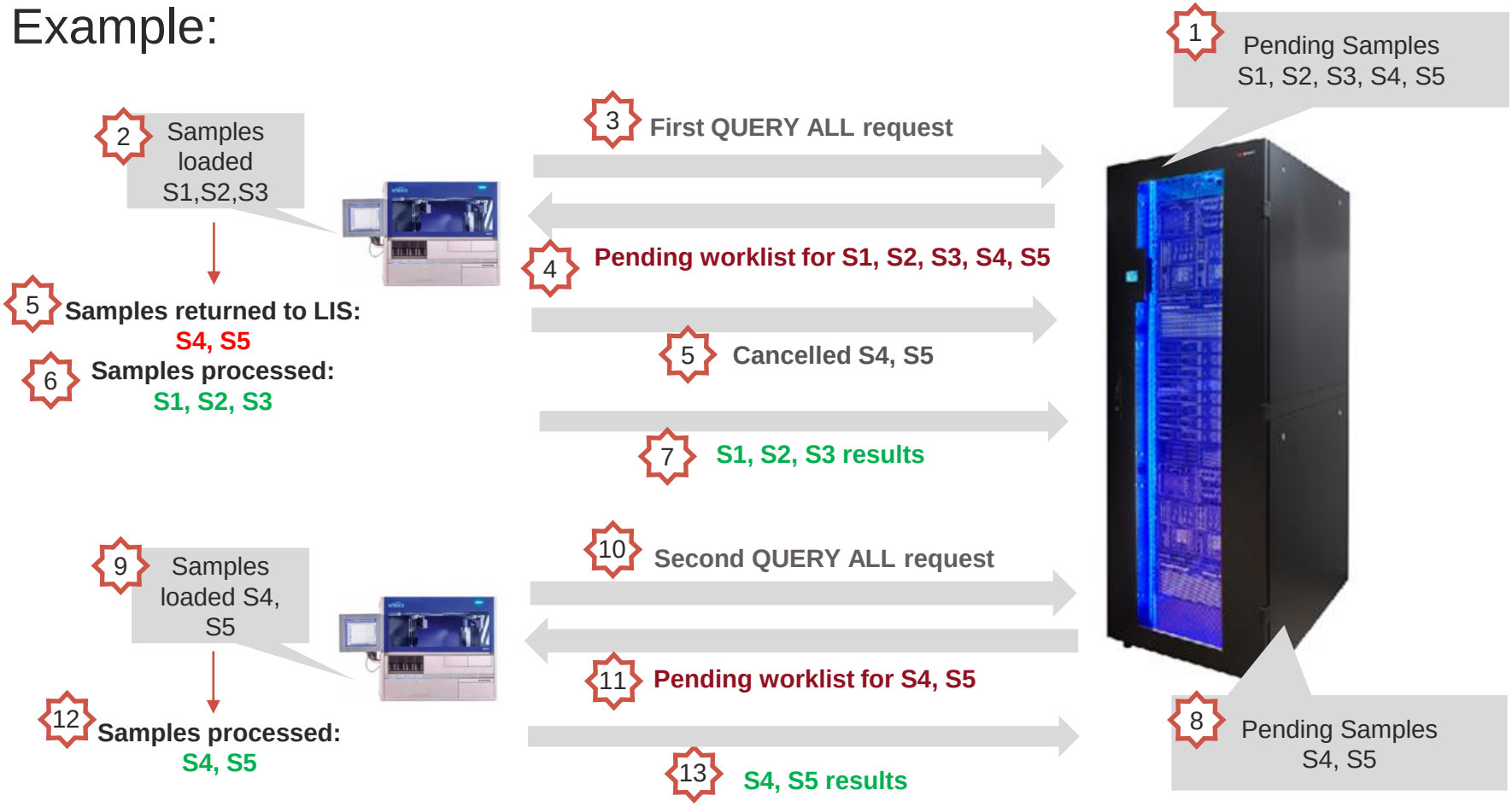
Note: The above functionality explained is as expected. Other middleware or LIS drivers may run with another features.

GRIFOLS

LIS

LIS Share Mode true

Example:



Note: The above functionality explained is as expected. Other middleware or LIS drivers may run with another features.

GRIFOLS

LIS

LIS Share Mode true

Example:

1 Pending Samples
S1, S2, S3, S4, S5

2 Samples loaded
S1,S2,S3



3 **First QUERY ALL request**

```
Send: H|\^&|||WADIANA^0000880^ERY||||||P|1|20170614180920
Send: Q|1|ALL||ALL||||||||I|
Send: L|1|N
```

Pending worklist for S1, S2, S3, S4, S5

4

```
OnPetitionEvent: W:0000880:ERY
Receive: H|\^&|||WADIANA^0000880^ERY||||||P|1|20170614180920
Receive: P|1|105116|||King^Alan||19970208|M|
Receive: O|1|S1||^^^DGW01A.gru^Erytra.pln|||||||||||
Receive: P|2|148135|||Anderson^Maika||19660610|M|
Receive: O|1|S2||^^^DGW01A.gru^Erytra.pln|||||||||||
Receive: P|3|178137|||Spoon^Logan||19410905|M|
Receive: O|1|S3||^^^DGW05B.gru^Erytra.pln|||||||||||
Receive: P|4|116997|||Cool^Jackson||19650315|M|
Receive: O|1|S4||^^^DGW01A.gru^Erytra.pln|||||||||||
Receive: P|5|172182|||Pierce^Layla||19761016|F|
Receive: O|1|S5||^^^DGW05B.gru^Erytra.pln|||||||||||
Receive: L|1|N
```



5 **Samples returned to LIS:**
S4, S5

6 **Samples processed:**
S1, S2, S3

Cancelled S4, S5

5

```
Send: P|4|116997|||Cool^Jackson||19650315|M|
Send: O|1|S4||^^^DGW01A.gru^Erytra.pln|R|||||||||||||||||||WADIANA^0000880^ERY|X|
Send: P|5|172182|||Pierce^Layla||19761016|F|
Send: O|1|S5||^^^DGW05B.gru^Erytra.pln|R|||||||||||||||||||WADIANA^0000880^ERY|X|
Send: L|1|N
```

S1, S2, S3 results

7

```
Send: H|\^&|||WADIANA^0000880^ERY||||||P|1|20170920141146
Send: P|1|105116|||King^Alan||19970208|M|
Send: O|1|S1||^^^DGW01A.gru^Erytra.pln|R|20170920133750||||||#21^031581200145^71760118001018264 9|||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW01A.gru^Erytra.pln|A^Neg^|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920133750|20170920135412|WADIANA^0000880^ERY|
Send: P|2|148135|||Anderson^Maika||19660610|M|
Send: O|1|S2||^^^DGW01A.gru^Erytra.pln|R|20170920133830||||||#20^031581200145^021072780553^021172780552^711101180020107770|||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW01A.gru^Erytra.pln|A^Pos^|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920133830|20170920135455|WADIANA^0000880^ERY|
Send: P|3|178137|||Spoon^Logan||19410905|M|
Send: O|1|S3||^^^DGW05B.gru^Erytra.pln|R|20170920133750||||||#21^031581200145^71760118001018264 9|||||||||WADIANA^0000880^ERY|F|
Send: R|1|^^^DGW05B.gru^Erytra.pln|^^CEce|Bloodgroup^Rh^Rh Phen||N||F||ADMIN|20170920133750|20170920135412|WADIANA^0000880^ERY|
Send: L|1|N
OnPetitionEvent: R:000:ERY
```

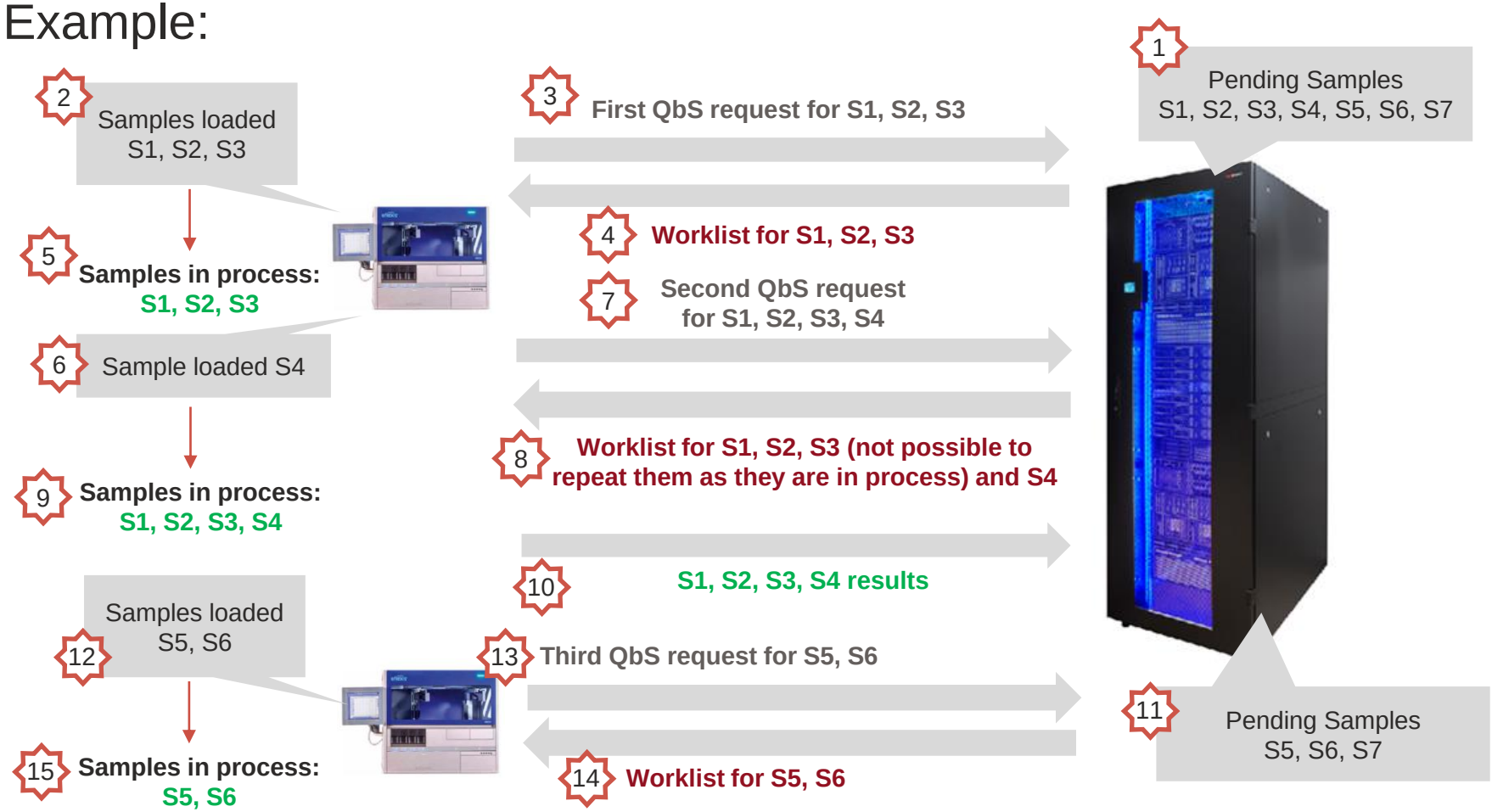
Note: The above functionality explained is as expected. Other middleware or LIS drivers may run with another features.

GRIFOLS

LIS

Query by sample

Example:



Note: The above functionality explained is as expected. Other middleware or LIS drivers may run with another features.

GRIFOLS

LIS

Query by sample

Example:

1 Pending Samples
S1, S2, S3, S4, S5, S6, S7



3 First QbS request for S1, S2, S3

```
Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170920144308
Send: Q|1|^S1^||ALL||||||||I|
Send: Q|2|^S2^||ALL||||||||I|
Send: Q|3|^S3^||ALL||||||||I|
Send: L|1|N
```

2 Samples loaded S1,S2,S3



Worklist for S1, S2, S3

4

```
Receive: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170920144308
Receive: P|1|154485|||White^David||20000726|M|
Receive: O|1|S1||^^^DGW01A.gru^Erytra.pln||||||||||||
Receive: O|2|S1||^^^DGW05B.gru^Erytra.pln||||||||||||
Receive: P|2|159005|||Richardson^Gab||19700415|M|
Receive: O|1|S2||^^^DGW01A.gru^Erytra.pln||||||||||||
Receive: O|2|S2||^^^DGW05B.gru^Erytra.pln||||||||||||
Receive: P|3|107900|||Jones^Jason||20000317|M|
Receive: O|1|S3||^^^DGW01A.gru^Erytra.pln||||||||||||
Receive: O|2|S3||^^^DGW05B.gru^Erytra.pln||||||||||||
Receive: L|1|N
```

5 **Samples processed automatically:**
S1, S2, S3

6 Sample loaded S4

Second QbS request for S1, S2, S3, S4

7

```
Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170920145531
Send: Q|1|^S1^||ALL||||||||I|
Send: Q|2|^S2^||ALL||||||||I|
Send: Q|3|^S3^||ALL||||||||I|
Send: Q|4|^S4^||ALL||||||||I|
Send: L|1|N
```

8

```
Receive: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170920145531
Receive: P|1|154485|||White^David||20000726|M|
Receive: O|1|S1||^^^DGW01A.gru^Erytra.pln||||||||||||
Receive: O|2|S1||^^^DGW05B.gru^Erytra.pln||||||||||||
Receive: P|2|159005|||Richardson^Gab||19700415|M|
Receive: O|1|S2||^^^DGW01A.gru^Erytra.pln||||||||||||
Receive: O|2|S2||^^^DGW05B.gru^Erytra.pln||||||||||||
Receive: P|3|107900|||Jones^Jason||20000317|M|
Receive: O|1|S2||^^^DGW01A.gru^Erytra.pln||||||||||||
Receive: O|2|S3||^^^DGW05B.gru^Erytra.pln||||||||||||
Receive: P|4|159778|||Anderson^Harper||19670321|F|
Receive: O|1|S4||^^^DGW01A.gru^Erytra.pln||||||||||||
Receive: O|2|S4||^^^DGW05B.gru^Erytra.pln||||||||||||
Receive: L|1|N
```

9 **Samples in process:**
S1, S2, S3
Sample processed automatically:
S4

10 S1, S2, S3, S4 results

Worklist for S1, S2, S3 (not possible to repeat them as they are in process) and S4

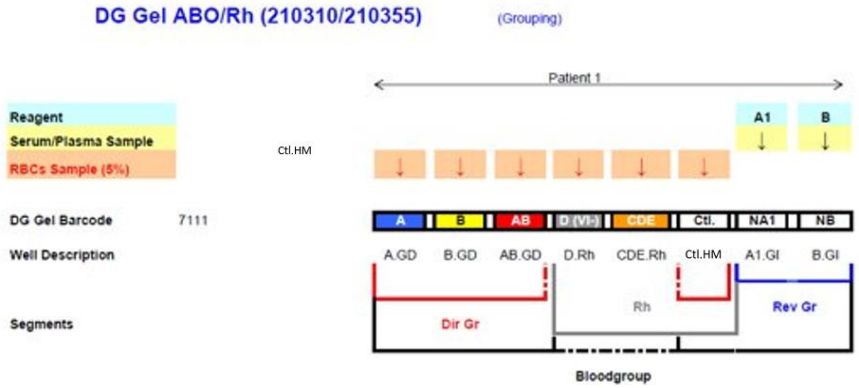
Note: The above functionality explained is as expected. Other middleware or LIS drivers may run with another features.

GRIFOLS

LIS

Plantillas (Templates)

- Plantilla is the Spanish word for Template.
- Is used for sending the information of the results to the LIS through ASTM protocol.
- For instance, the information contained in a whole DG Gel ABO/Rh card is:
  - Wells: A.GD, B.GD, AB.GD, D.Rh, CDE.Rh, CTL.HM, A1.GI and B.GI
  - Segments: Dir Gr., Rev Gr. and Rh
  - Meta-segments: Group and Group Rh



GD: Direct group
GI: Indirect group

Note: Well names might vary between different regions and countries.

GRIFOLS

LIS

Plantillas (Templates)

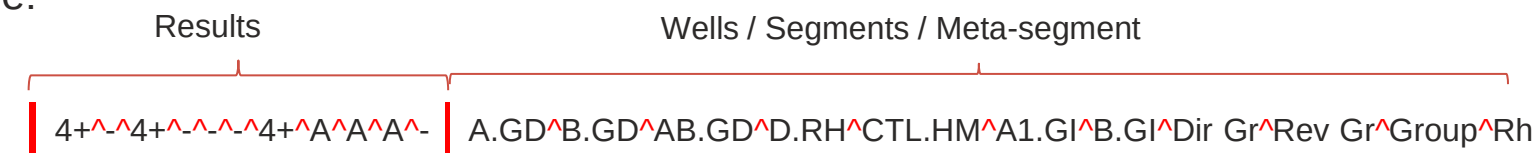
- When the instrument finishes a technique, the result will be stored in a table similar to these:

Results	Wells/Segments
4+	A.GD
-	B.GB
4+	AB.DG
-	D.RH
-	CTL.HM
-	A1.GI
4+	B.GI
A	DIR GR
A	REV GR

GROUP	RH
A	-

- The way that ASTM protocol sends the information depends of the template used.

Example:



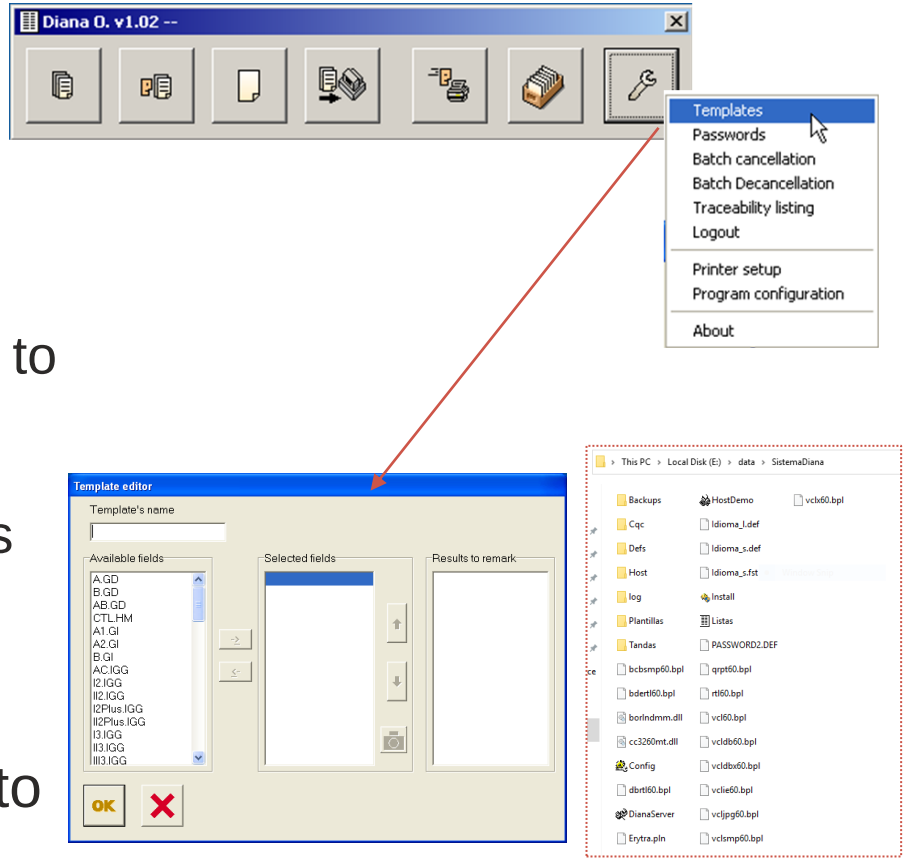
Note: Well names might vary between different regions and countries.

GRIFOLS

LIS

Plantillas (Templates)

- To create a new template:
  - Open Listas.exe found in E:\Data\Sistema Diana folder.
  - Select “Templates” in the Maintenance button.
  - Enter the template’s name.
  - Select the fields that will be sent to LIS.
  - Template will be saved in the folder E:\SistemaDiana\Plantillas
- Before creating a template is important to ask customer and LIS provider what results and how they want to receive them to the LIS.



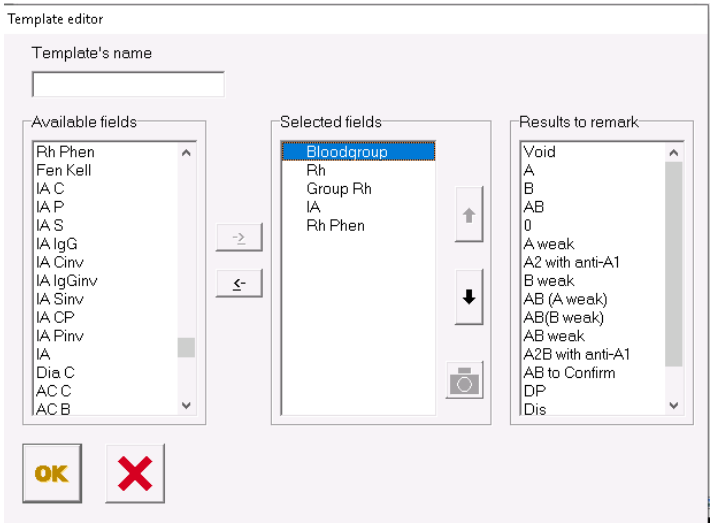
Note: Well names might vary between different regions and countries.

GRIFOLS

LIS

Plantillas (Templates)

- The most common segments used to send the results are:
  - Bloodgroup (Group)
  - Rh
  - Group Rh
  - IA (Antibody screening interpretation)
  - Rh Pheno (C, c, E, e interpretation using traditional nomenclature)



Note: The way to configure the template might change depending on the LIS provider and/or customer needs.

GRIFOLS

LIS

ASTM protocol

- Data is transmitted using ASTM protocol (international standard language used for data transmission).
- The different messages are divided in:
  - **Test request:** From the analyser to the LIS
    - Query all
      - LIS share mode false
      - LIS share mode true – Cancelled samples
    - Query by Sample
  - **Response to the request:** From the LIS to the analyser
    - Query all: Worklist of all the pending samples in the LIS.
    - Query by sample: Worklist of all the samples inside the instrument.
  - **Sending of the results:** From the analyser to the LIS

The records type used are:

- H – Header record
- L – Terminator record
- Q – Query record
- P- Patient Information record
- O –Test Order Record
- R – Result record

GRIFOLS

LIS

ASTM log file

- The log file of the communication between Diana Server and LIS is located in E:\Data\SistemaDiana\Log\DianaASTM.log

DianaASTM.log

```
[5/25/2017 10:13:20 AM]-> ## LOG SIGNATURE FAILED!! ##
[5/25/2017 10:13:20 AM]-> [I] Free Space : LOCAL_DRIVE: 98% (106191232 KB)
[5/25/2017 10:13:20 AM]-> ## START LOG ##
[5/25/2017 10:13:20 AM]-> Check samples.db
[5/25/2017 10:13:20 AM]-> The samples list was empty
[5/25/2017 10:14:34 AM]-> ## END LOG ##
A41DB1E0EC9D190EF9E396B8F172E9B0DB6ED6187366C050D21797E57F3A5C504244BA56208613477C87EEB2AE
12-6-2017  14:41:38 The application boot up
12-6-2017  14:41:50   Application close
**************************************************************
14-6-2017  10:11:36 The application boot up
14-6-2017  10:11:46   Application close

[6/14/2017 5:25:00 PM]-> ## LOG SIGNATURE FAILED!! ##
[6/14/2017 5:25:00 PM]-> [I] Free Space : LOCAL_DRIVE: 97% (104646400 KB)
[6/14/2017 5:25:00 PM]-> ## START LOG ##
[6/14/2017 5:25:00 PM]-> Check samples.db
[6/14/2017 5:25:00 PM]-> The samples list was empty
[6/14/2017 6:08:39 PM]-> Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170614180839
[6/14/2017 6:08:39 PM]-> Send: Q|1|^=E00251711038521^||ALL||||||||I|
[6/14/2017 6:08:39 PM]-> Send: L|1|N
[6/14/2017 6:08:39 PM]-> OnPetitionEvent: W:0000880:ERY
[6/14/2017 6:09:19 PM]-> OnCommIn: Remote Host not ready
[6/14/2017 6:09:20 PM]-> Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170614180920
[6/14/2017 6:09:20 PM]-> Send: Q|1|ALL||ALL||||||||I|
[6/14/2017 6:09:20 PM]-> Send: L|1|N
[6/14/2017 6:09:20 PM]-> OnPetitionEvent: W:0000880:ERY
[6/14/2017 6:09:20 PM]-> Receive: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170614180920
[6/14/2017 6:09:20 PM]-> Receive: P|1|105116|||King^Alan||19970208|M|
[6/14/2017 6:09:20 PM]-> Receive: O|1|S1||^^^DGW01J.gru^Erytra.pln||||||||||||
```


GRIFOLS

LIS

ASTM log file: Test request from the analyser to the LIS

Query all:

[6/14/2017 6:09:20 PM]-> Send: **H**|\^&|||WADIANA^**0000880**^ERY|||||||**P|1|20170614180920**

[6/14/2017 6:09:20 PM]-> Send: **Q|1|ALL||ALL||||||||I|**

[6/14/2017 6:09:20 PM]-> Send: **L**|1|N

H: Record type Header

^000880^ : serial number of the instrument

P|1|20170614180920: Time and date of the transmission

Q|1|ALL||ALL||||||||I|: Query ALL

L: Record type: End of transmission

GRIFOLS

LIS

ASTM log file: Test request from the analyser to the LIS

Query by Sample:

[6/14/2017 6:09:20 PM]-> Send: **H**|\^&|||WADIANA^**0000880**^ERY|||||||**P|1|20170614180920**

[6/14/2017 6:08:39 PM]-> Send: **Q|1|^=E00251711038521^||ALL||||||||**

[6/14/2017 6:08:39 PM]-> Send: **Q|2|^=E00251711031714^||ALL||||||||**

[6/14/2017 6:09:20 PM]-> Send: **L**|1|N

H: Record type Header

^000880^ : serial number of the instrument

P|1|20170614180920: Time and date of the transmission

Q: Query by samples ^=E00251711038521^ and ^=E00251711031714^

L: Record type: End of transmission

GRIFOLS

LIS

ASTM log file: Response to the request from the LIS to the analyser

[6/14/2017 6:09:20 PM]-> Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170614180920
[6/14/2017 6:09:20 PM]-> Send: Q|1|ALL||ALL||||||||I|
[6/14/2017 6:09:20 PM]-> Send: L|1|N
[6/14/2017 6:09:20 PM]-> OnPetitionEvent: W:0000880:ERY

[6/14/2017 6:09:20 PM]-> Receive: **H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170614180920**
[6/14/2017 6:09:20 PM]-> Receive: **P|1|105116**|||**King^Domonic**||**19970208|M**|

[6/14/2017 6:09:20 PM]-> Receive: **O|1**|E00251711035221||^^^DGW01J.gru^Erytra.pln||||||||||||
[6/14/2017 6:09:20 PM]-> Receive: **O|2**|E00251711035221||^^^DGW05B.gru^Erytra.pln||||||||||||
[6/14/2017 6:09:20 PM]-> Receive: **O|3**|E00251711035221||^^^DGW11E.gru^Erytra.pln||||||||||||

[6/14/2017 6:09:20 PM]-> Receive: P|2|148135|||Anderson^Joseph||19660610|M|
P|2|148135 : Patient 2
[6/14/2017 6:09:20 PM]-> Receive: O|1|E00251711035821||^^^**DGW01J.gru^Erytra.pln**||||||||||||
[6/14/2017 6:09:20 PM]-> Receive: P|3|178137|||Hall^Logan||19410905|M|
P|3|178137: Patient 3
[6/14/2017 6:09:20 PM]-> Receive: O|1|E00251711035921||^^^DGW01J.gru^Erytra.pln||||||||||||
[6/14/2017 6:09:20 PM]-> Receive: O|2|**Sample BC^Donor BC**||^^^DGW14J.gru^Xmatch.pln||
[6/14/2017 6:09:20 PM]-> Receive: **L|1|N**

P|1|105116
Patient 1, Hosiptal ID

King^Domonic
Patient name

19970208|M
BHD+Gender

O|1 O|2+ O|3: Three different test for the same sample

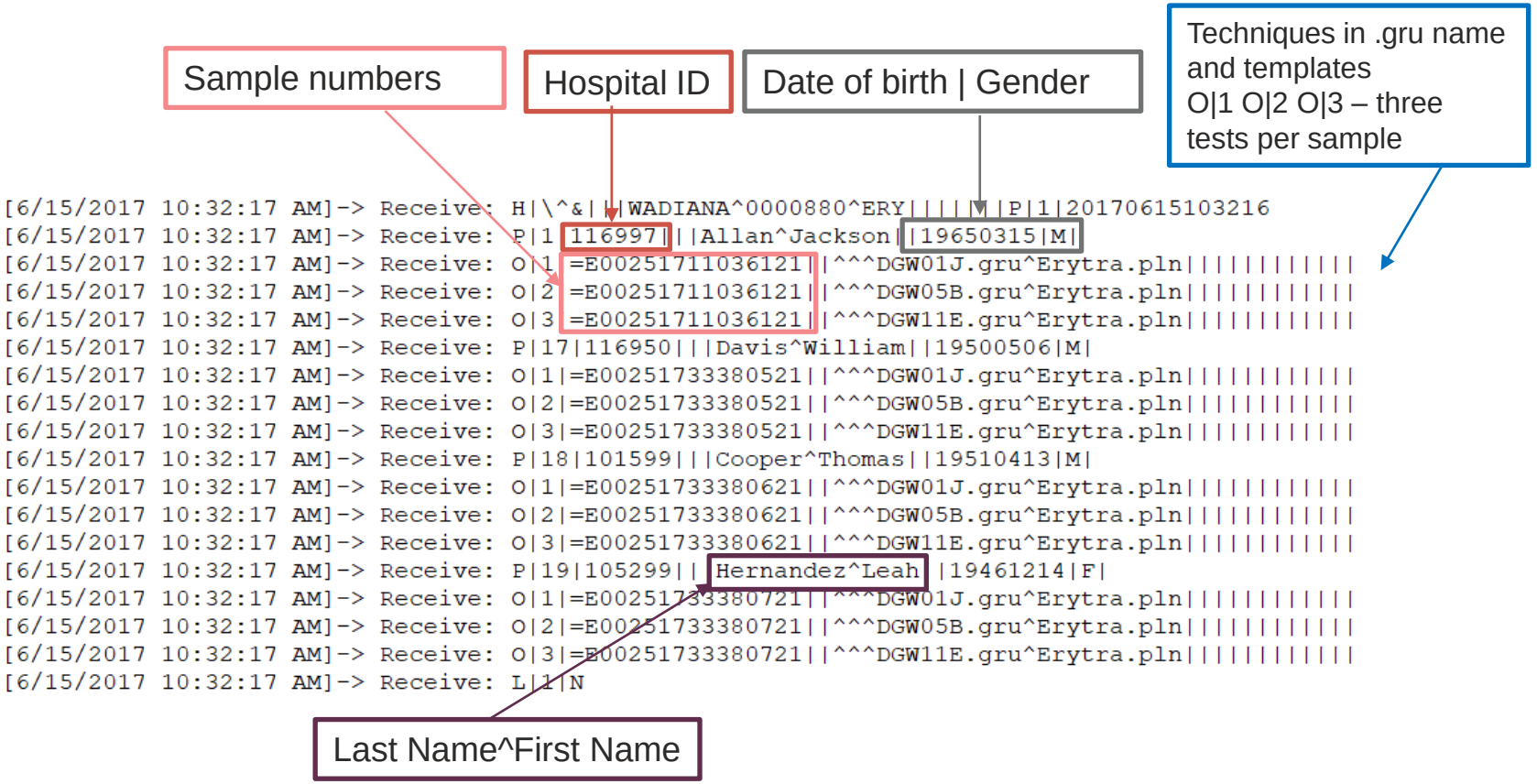
DGW01J.gru^Erytra.pln
Gru and template

Programing a CrossMatch:
Sample BC^Donor BC

GRIFOLS

LIS

ASTM log file: Response to the request from the LIS to the analyser



GRIFOLS

LIS

ASTM log file: Samples cancelled and returned to the LIS

- The cancelled message will be sent in the following cases:
  - LIS SHARE mode is TRUE and the samples are not present into the instrument.
  - Missing samples warning is cancelled by the user.
  - Analyser is switched off before testing the pending samples.

[6/14/2017 6:09:25 PM]-> Send: H|\^&|||WADIANA^0000880^ERY|||||||P|1|20170614180925
[6/14/2017 6:09:25 PM]-> Send: P|1|105116|||King^Domonic||19970208|M|
[6/14/2017 6:09:25 PM]-> Send: O|1|E00251711035221||^^^DGW01J.gru^Erytra.pln|R|||||||||||||||||WADIANA^0000880^ERY**|X|**
[6/14/2017 6:09:25 PM]-> Send: O|2|E00251711035221||^^^DGW05B.gru^Erytra.pln|R|||||||||||||||||WADIANA^0000880^ERY**|X**|
[6/14/2017 6:09:25 PM]-> Send: O|3|E00251711035221||^^^DGW11E.gru^Erytra.pln|R|||||||||||||||||WADIANA^0000880^ERY**|X|**

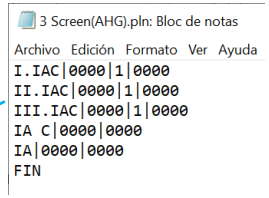
GRIFOLS

LIS

ASTM log file: Sending the results from analyser to the LIS

- Unidirectional communication:
  - The instrument send the result to LIS as result of the run a profile or because the instrument is forwarding an existing result (REPORT^000^ERY)

[7/6/2017 11:26:32 AM]-> Send: H|\^&|||**REPORT^000^ERY**|||||||P|1|20170706112632
[7/6/2017 11:26:32 AM]-> Send: **P|1**|QCS117011170715|||^^^^|
[7/6/2017 11:26:32 AM]-> Send: **O|1**|QCS117011170715||^^^DGW11E.gru^3 Screen(AHG).pln|R|
20170706105041||||||||||||||||||REPORT^000^ERY|F|
[7/6/2017 11:26:32 AM]-> Send: **R|1**|^^^**DGW11E.gru^3 Screen(AHG).pln|3+^3+^-^Pos^Pos**
|I.IAC^II.IAC^III.IAC^IA C^IA||N||F||**ADMIN|20170706105041|20170706111643**|REPORT^000^ERY|
[7/6/2017 11:26:32 AM]-> Send: L|1|N



ADMIN|20170706105041|20170706111643
Operator| Start of transmission | End of transmission

DGW11E.gru = 3 Screen (AHG) profile
3 Screen(AHG).pln = template used to send the results

LIS

ASTM log file: Sending the results from analyser to the LIS

- Bidirectional communication:
  - The instrument send the result responding to a LIS request (WADIANA^0000880^ERY)

Cells I, II and III
2 Coombs cards

[7/6/2017 2:18:06 PM]-> Send: **H|**\^&|||**WADIANA^0000880^ERY**|||||||P|1|20170706141806

[7/6/2017 2:18:06 PM]-> Send: **P|1|**MAS1486|||LastName^FirsName||20121212|F|

[7/6/2017 2:18:06 PM]-> Send: **O|1|**MAS1486||^^^**DGW11E.gru^3 Screen(AHG).pln**|R|20170706134301|||||||**023100280370^023000280371^023200280379^71300920181012792 7940**||||||||||WADIANA^0000880^ERY|F|

[7/6/2017 2:18:06 PM]-> Send: **R|1**|^^^**DGW11E.gru^3 Screen(AHG).pln|3+^3+^-^Pos^Pos**|I.IAC^II.IAC^III.IAC^IA C^IA||N||F||**ADMIN|20170706134301|20170706141027**|WADIANA^0000880^ERY|

M4+: Data modified manually

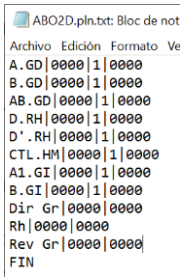
[7/6/2017 2:18:06 PM]-> Send: **P|2**|MAS1539||| LastName^FirsName||20121212|F|

[7/6/2017 2:18:06 PM]-> Send: **O|1**|MAS1539||^^^DGW01J.gru^ABO2D.pln|

R|20170706134130|||||||**031510590156^021100230913^021000230914^716505200090113116**||||||||||WADIANA^0000880^ERY|F|

[7/6/2017 2:18:06 PM]-> Send: **R|1**|^^^**DGW01J.gru^ABO2D.pln|M4+**^-^4+^-^-^-^-^3+^A^Neg^A|A.GD^B.GD^AB.GD^D.RH^D'.RH^CTL.HM^A1.GI^B.GI^Dir Gr^Rh^Rev Gr||N||F||ADMIN|20170706134113|20170706135427|WADIANA^0000880^ERY|

[7/6/2017 2:18:06 PM]-> Send: L|1|N



DGW01J.gru = ABO-Rh (2D) profile
ABO2D.pln = template used to send the results

GRIFOLS

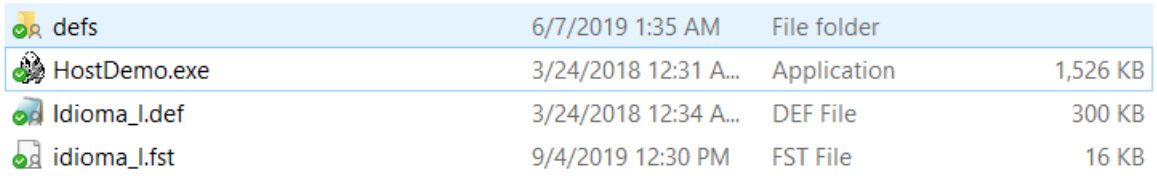
Host demo

GRIFOLS

Host demo

Introduction

- Program simulates a LIS and allows testing of Diana Server configuration.
- Isolates the problem to either Diana Server or Customer LIS.
- Host demo is included in the software by default in E:\Data\SistemaDiana.
- To be run in an external computer, be sure that the following folders are copied in it:



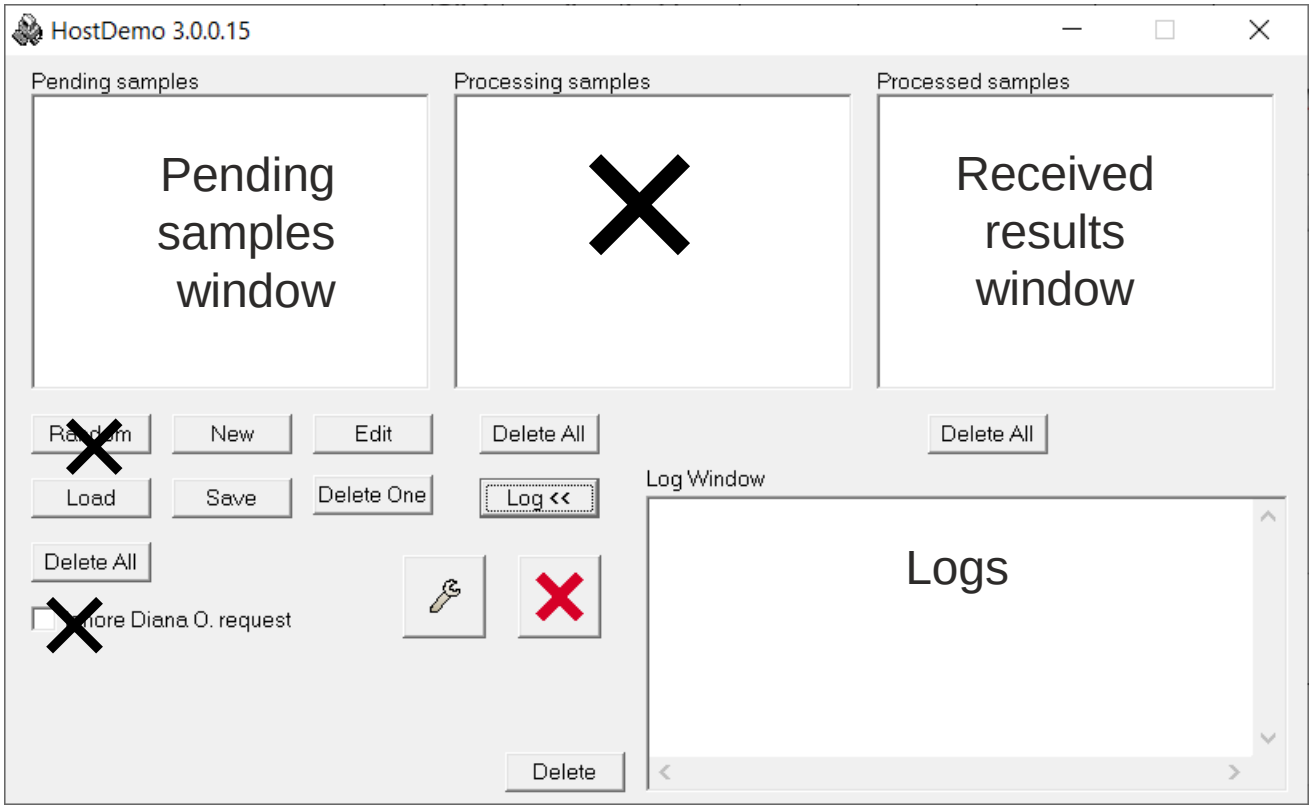
GRIFOLS

Host demo

Configuration

- Open the software and enter the username and password.

✕ Not used



Username: ST
Password: d+g_diana

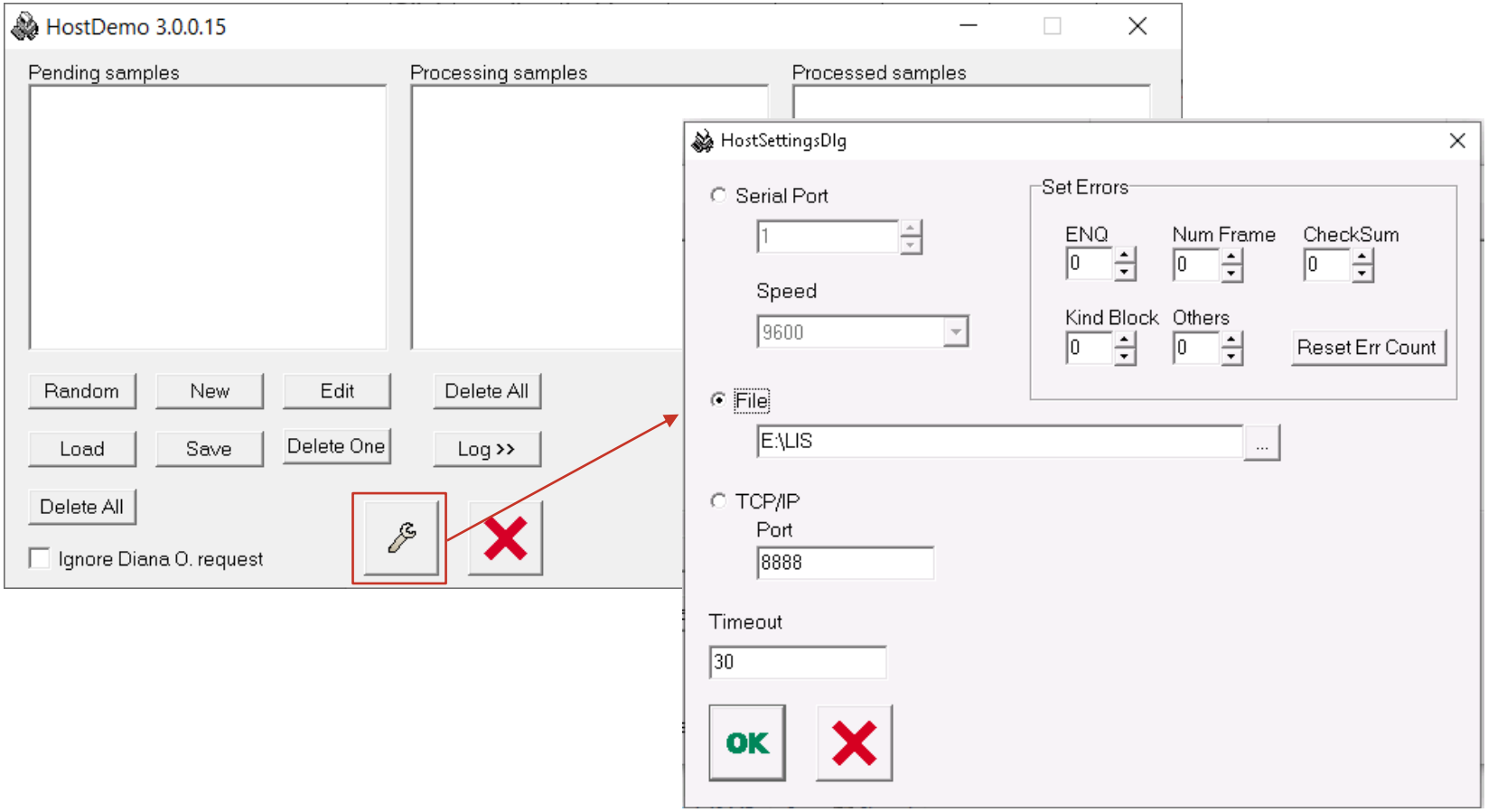
(Case sensitive)

GRIFOLS

Host demo

Configuration

- Select the configuration button.

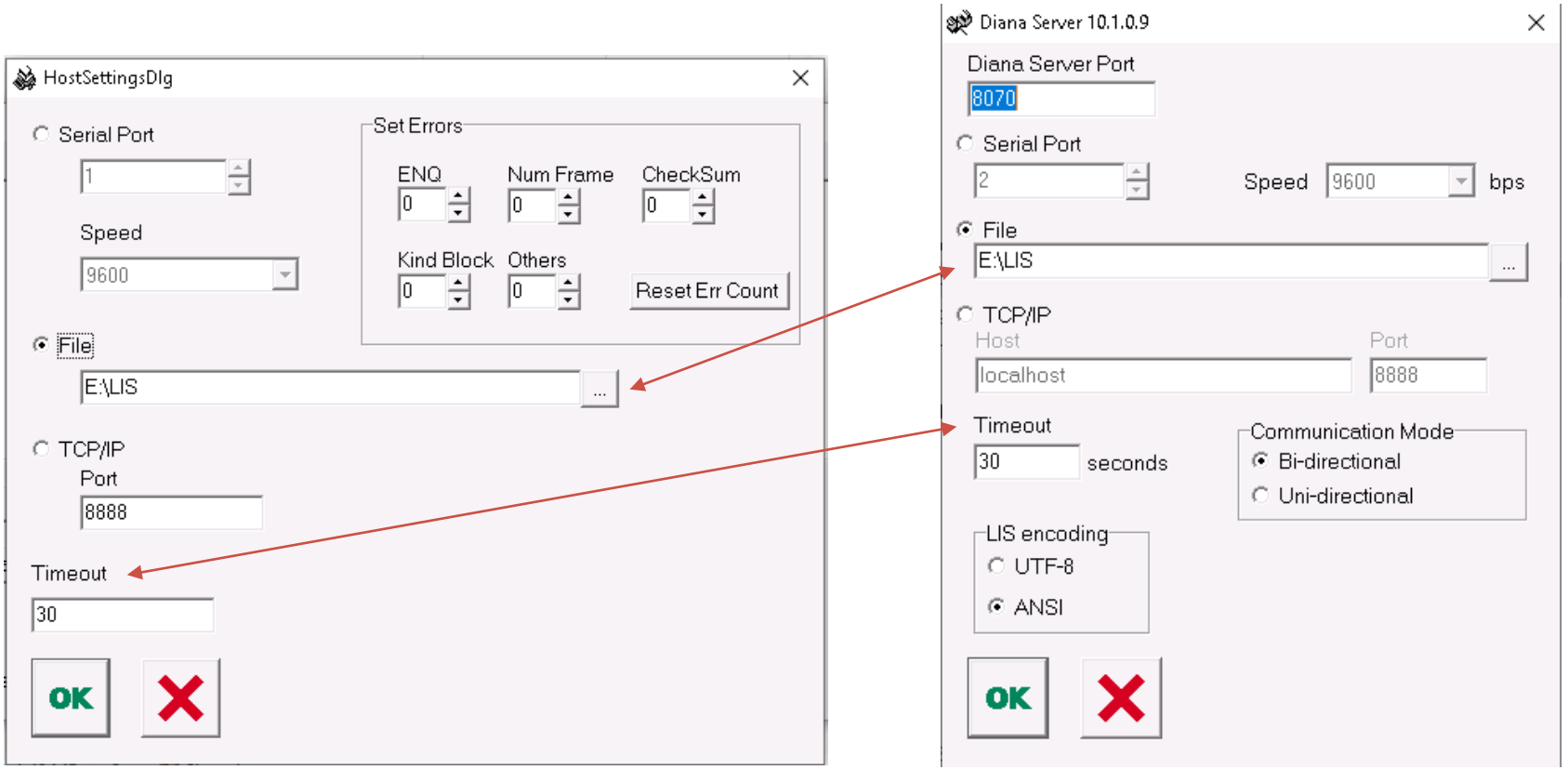


GRIFOLS

Host demo

Configuration

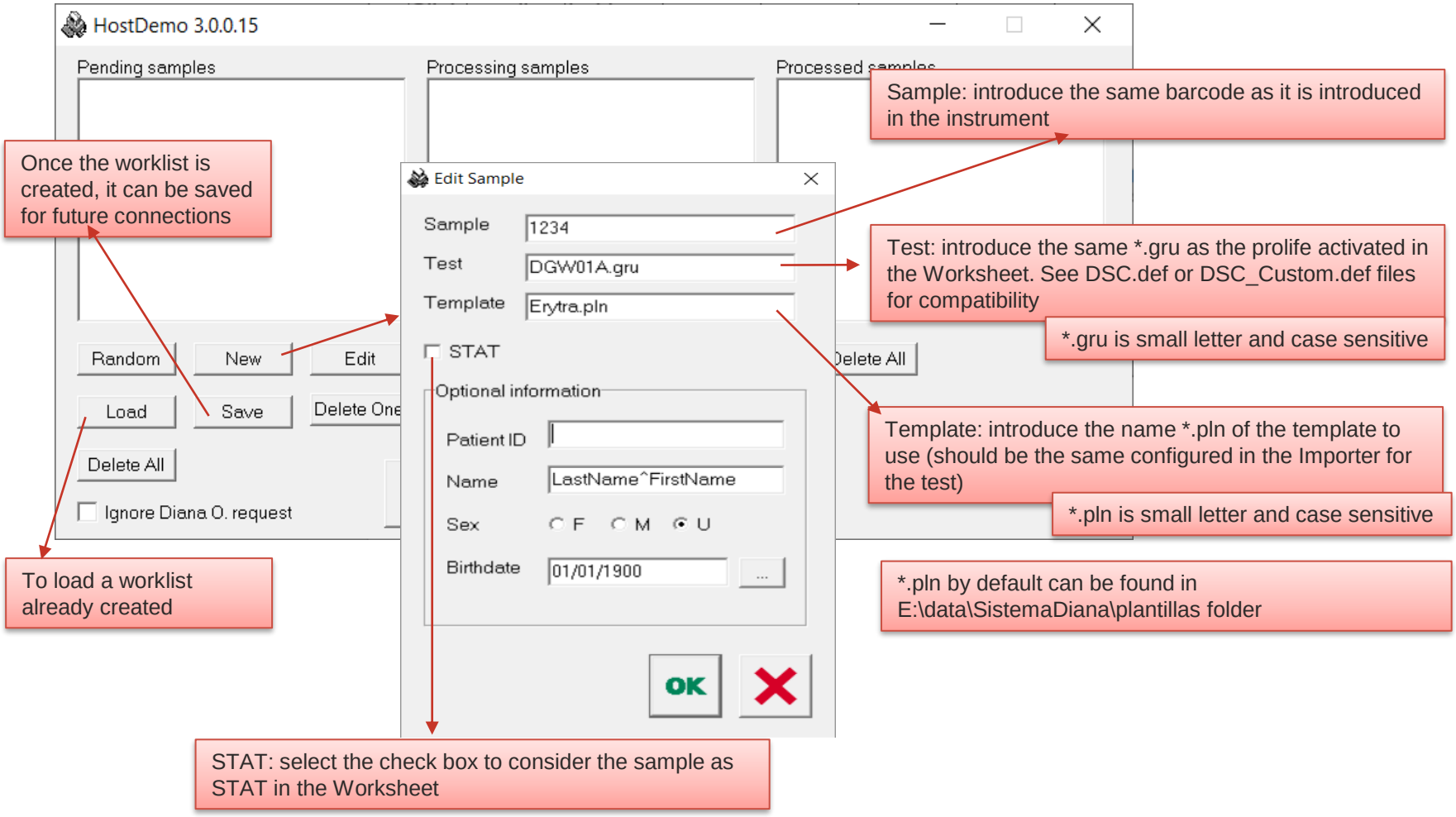
- The HostDemo configuration must be the same as the Diana Server configuration.



GRIFOLS

Host demo

Configuration

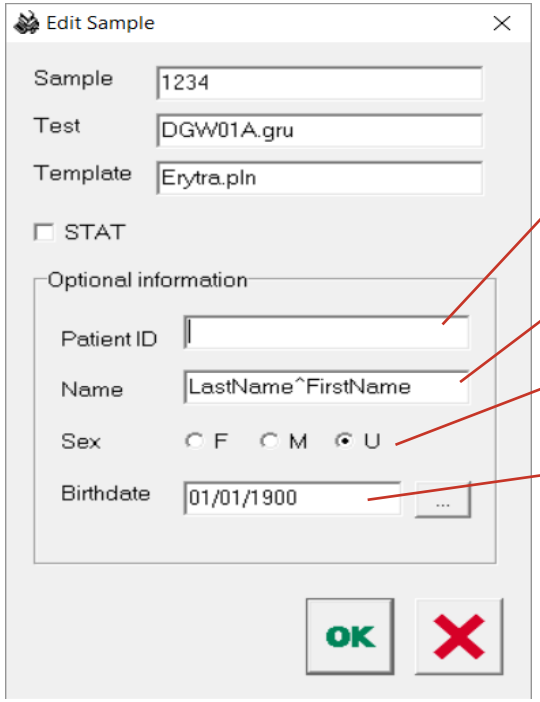


GRIFOLS

Host demo

Configuration

- To test the demographic data transmission, it is possible to configure the following parameters:



Patient ID: patient identification defined by the hospital/laboratory and unique for each patient

Name: name and surname of the patient

Sex: sex of the patient

Birthdate: date of birth of the patient

GRIFOLS

Q & A

GRIFOLS